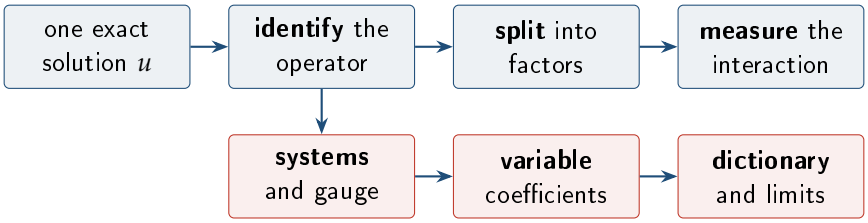
Complete edition · Phases 0–10

Exact Recovery of a Differential Operator from a Single Solution

Identification, factor decomposition, the interaction term, nonlinear equations, systems, variable coefficients, convergence, the analytic encoder and the universal dictionary



Companion volume: *Catalogue of proofs, Phases 0–10*. Every theorem below names its proof as "Dim. A*n*" (key proofs) or "Dim. B*n*". The edition for Phases 0–5 alone is a separate pair of volumes.

October 2026

Abstract

We are given one exact solution $u(t, x)$ of a differential equation that we do not know. What does u remember about its equation? This edition answers the question for the results of Phases 0–10 of the project. The answer has four parts for constant coefficients (Part I) and four more for what lies beyond (Part II).

Identification. For constant coefficients, u determines its operator P exactly when no partial quotient

$$w_i = \left(\frac{P}{P_i}\right)(\partial)\, u$$

is an exponential-polynomial. No finite amount of data can certify this, and a black-box test is impossible. One extra integer, a bound on the exponential-polynomial part of u, makes it a finite computation.

Decomposition. On a simply connected region, u splits into one component per irreducible *factor* of the symbol. The ambiguity has the exact finite dimension $\delta(P)$. The split per additive *term* of the operator is false. The components keep the Sobolev regularity of u exactly when a conditioning number γ_N is positive.

Interaction. For an evolution equation whose N terms commute, the part of the solution due to the interplay of the terms is

$$E_{\text{int}} = \Big[\prod_{n=1}^{N}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n\Big] u_0.$$

One value costs $O(N)$ operations, while the full table of cross terms costs $\Theta(2^N)$. The map $u_0 \mapsto E_{\text{int}}$ is not continuous in Sobolev norms and cannot be regularised.

Nonlinear equations. For Burgers, KdV–Burgers and KdV the same questions have sharp answers. The interaction has a t^2 jet and a closed form before the shock, and size $\Theta(\nu)$ and $\nu|\log \nu|$ after it. Identification holds relative to a dictionary, modulo the differential consequences of the equation. For the N-soliton of KdV the identifying polynomial has degree $(N+2)2^{N-1}$.

Beyond constant coefficients (Part II). *Systems:* a vector solution determines a matrix operator iff every socle of its module has dimension at most one, which holds for every non-singular square system in two or more variables; gauge can never be separated from physics by one field. *Variable coefficients:* a potential $V(x)$ is read from any non-zero solution, a potential $V(t, x)$ can hide in silent time strips, and a huge coefficient class identifies nothing. *Convergence and noise:* identification holds in the limit without a certificate, no learner decides whether a law exists, and noise is handled only at a finite level of the dictionary. *The encoder and the dictionary:* an explicit Volterra transformation removes a known potential; a universal dictionary exists for approximation but never for exact identification, and a misfit has a finite certificate while a fit never does. Phase 10 corrected several statements of Phases 1–3: a continuity no-go that was false as stated, and the failure of density on regions with holes.

How to read this edition

Layout. Every chapter opens with a box with three entries. *Starting from* lists the assumptions in force. *Why* says what problem the chapter solves. *Role* says where its output is used. Inside a chapter the order is always the same: motivation, then intuition, then statement, then a worked example, then consequences.

Boxes. The colours are fixed:

- blue boxes are theorems, propositions and lemmas;
- green boxes are definitions;
- violet bars are worked examples;
- amber boxes give the intuition;
- red boxes are warnings and counterexamples.

Proofs. Proofs are not in this volume. Each statement names its proof in the companion *Catalogue of proofs*, for example Dim. A3. Parts I and II of the catalogue contain the load-bearing proofs of Phases 0–5 (A1–A31) and of Phases 6–10 (A32–A53); Parts III and IV contain all the others.

Trust labels. Every statement carries one label. [proved] means a complete proof that passed an independent review. [proved mod. X] means complete except for a cited published theorem X whose source was checked. [proof written, not audited] means a full proof exists but no independent audit has been done. [from cited source] means a published theorem we rely on. [conditional] means the result depends on an unverified lemma. [checked on examples] means it was only checked on examples. [open] means unsolved. [false] means a counterexample is proved. The reviewers were independent AI agents, not human referees. Grey tags such as (C3-T1) are the identifiers in the project ledger `claims/LEDGER.md`.

Prerequisites. Linear algebra (eigenvalues, Jordan form), calculus in two variables, the Fourier transform, and linear ODEs with constant coefficients. Chapter 2 introduces every algebraic notion from scratch: ideals, annihilators, exponential-polynomials and the Chinese remainder theorem.

Scope. This edition covers Phases 0–10; a separate pair of volumes covers Phases 0–5 alone. In case of a difference the present edition prevails.

Contents

Abstract i

I Phases 0–5: the constant-coefficient theory and its nonlinear extension 1

1 Introduction 2

1.1 The question . . . 2
1.2 Overview of the answers . . . 3
1.3 Notation at a glance . . . 3
1.4 Roadmap . . . 4

2 A primer for readers new to the algebra 6

2.1 Ideals and annihilators, starting from integers . . . 6
2.2 Exponential-polynomials . . . 8
2.3 The Chinese remainder theorem . . . 8
2.4 The companion matrix . . . 9

3 The algebraic language 11

3.1 Derivatives as polynomials . . . 11
3.2 Annihilators and what a solution knows . . . 12
3.3 Exponential-polynomials . . . 14
3.4 One variable: the ODE prototype . . . 15

4 Identification: what one solution determines 18

4.1 The question in algebraic form . . . 18
4.2 The criterion . . . 18
4.3 The intrinsic operator and the spectral form . . . 21
4.4 Refinements . . . 23

5 Finite data and decidability 25

5.1 Three data models . . . 25
5.2 Exponential-polynomial data: a finite threshold . . . 26

5.3 What no finite data can show . . . 27
5.4 One extra integer makes it decidable . . . 28
5.5 A decidable class: holonomic solutions . . . 28
5.6 Variable coefficients: the first limits . . . 29
5.7 Summary . . . 30

6 Decomposition into components **31**
6.1 Which pieces? Terms versus factors . . . 31
6.2 Why factors work: the Chinese remainder theorem . . . 32
6.3 The decomposition theorem . . . 34
6.4 Where the decomposition fails: holes . . . 35
6.5 Explicit projectors . . . 36

7 Regularity of the components **38**
7.1 The pieces can be rougher than the whole . . . 38
7.2 Two components: the chord criterion . . . 39
7.3 Any number of components . . . 40

8 Terms inside a factor: the interaction term **43**
8.1 From the sum of operators to the interaction term . . . 43
8.2 The function class and the exact formula . . . 44
8.3 Cost: $O(N)$ versus 2^N . . . 46
8.4 The flows are not determined without a class . . . 47
8.5 Stability: the interaction cannot be computed stably . . . 47
8.6 Time-dependent perturbation theory . . . 48
8.7 Non-commuting terms . . . 50

9 Nonlinear equations **51**
9.1 The interaction term for nonlinear flows . . . 51
9.2 Before, at and after the shock . . . 52
9.3 Identification relative to a dictionary . . . 54
9.4 Integrable models and scattering . . . 56
9.5 Linear versus nonlinear . . . 56

10 The algorithm **57**

10.1 The pipeline . . . 57
10.2 The steps in detail . . . 57
10.3 Worked example 1: the wave equation . . . 60
10.4 Worked example 2: advection–diffusion . . . 61
10.5 Cost and feasibility . . . 62

II Phases 6–10: beyond constant coefficients 63

11 Systems of equations and gauge 64
11.1 From one equation to a module . . . 64
11.2 When does an identifying solution exist? . . . 65
11.3 Sums of kernels for systems . . . 65
11.4 Dirac in Hamiltonian form . . . 66
11.5 Gauge, magnetic fields, and Lamé . . . 66

12 Variable coefficients 68
12.1 Too much freedom: a huge class identifies nothing . . . 68
12.2 A potential $V(x)$ is read from any non-zero solution . . . 69
12.3 A potential $V(t,x)$: silent strips . . . 69
12.4 Stability and the phase . . . 70
12.5 Other results of this phase . . . 70

13 Convergence, noise and the limits of learning 71
13.1 Identification in the limit . . . 71
13.2 No learner decides whether a law exists . . . 72
13.3 Noisy data . . . 72
13.4 Other results of this phase . . . 73

14 The analytic encoder 74
14.1 What an encoder is . . . 74
14.2 The Volterra transformation operator . . . 74
14.3 What no encoder can do . . . 75
14.4 Other results of this phase . . . 76

15 The universal dictionary and the misfit certificate 77

15.1 What “universal” can mean . . . 77
15.2 What it cannot mean . . . 78
15.3 Fit and misfit . . . 78
15.4 Stopping rules and the limit . . . 79
15.5 Other results of this phase . . . 79

16 What Phase 10 changed **80**
16.1 Statements that turned out false . . . 80
16.2 Problems closed . . . 81
16.3 Textbook examples and formalisation . . . 81

17 The extended algorithm **83**
17.1 Which senses of “identify” . . . 83
17.2 Data models and the contract . . . 84
17.3 The steps . . . 85

18 Limits, open problems and status **87**
18.1 What is impossible . . . 87
18.2 Statements that turned out false . . . 88
18.3 Open or conditional . . . 88
18.4 How reliable the results are . . . 89
18.5 What this edition does not contain . . . 90

19 The complete list of open problems **91**
19.1 Why the analytic part can be closed . . . 91
19.2 Research problems with no known route . . . 92
19.3 Items blocked on sources unavailable offline . . . 93
19.4 Open questions of each research package . . . 94
19.5 Open questions of the extended algorithm . . . 96
19.6 Statuses of the claims used by the algorithm . . . 97

20 Directions for improvement and routes not taken **98**
20.1 Improving what is already there . . . 98
20.2 Routes scouted and parked . . . 99
20.3 The statistical wrapper that was proposed and postponed . . . 100

20.4 What was never done 101
20.5 Honest limits of the whole project 101

A Glossary of classical tools **102**

B Bibliography of the project **103**

Bibliography **109**

Part I

Phases 0–5: the constant-coefficient theory and its nonlinear extension

CHAPTER 1

Introduction

In this chapter	
Starting from	one function $u(t,x)$, known exactly (no noise) on an open region $\Omega \subset \mathbb{R}^2$, which solves a linear equation $P(\partial_t, \partial_x)u = 0$ with constant coefficients. The operator P is *not* given.
Why	to state precisely what we want to learn from u and to give a map of the answers before any detail.
Role	frames the three questions Q1–Q3. Chapters 4–5 answer Q1, Chapters 6–7 answer Q2, Chapter 8 answers Q3. Chapter 9 extends all three to nonlinear equations and Chapter 10 assembles them into an algorithm. Part II (Chapters 11–18) goes beyond constant coefficients and states the limits of the whole project.

1.1 The question

Suppose we observe a physical field $u(t,x)$: a temperature, a wave, a concentration. We know u exactly on some region Ω of the (t,x) plane, and we know that it obeys some linear law. We do not know which law. Three questions arise naturally.

Q1. Identification. Can the operator P be reconstructed from u? When is the answer unique?

Q2. Decomposition. If P is built from pieces, can u be split into the parts that each piece generates?

Q3. Interaction. If the pieces are the *terms* of the operator (diffusion, advection, dispersion, ...), which part of u is due to no single term but to their interplay? Can it be computed exactly and cheaply?

The setting is deliberately analytic. There is no statistical model, no learned weights and no noise. The aim is an *exact* procedure, together with a precise statement of what is impossible.

> **Example 1.1 Three solutions of the heat equation**
>
> Take the heat equation $u_t = Du_{xx}$ with $D > 0$. Consider three of its solutions.
>
> - A single Fourier mode:
>
> $$u_1 = e^{-D\kappa^2 t}\cos(\kappa x).$$

- A sum of two modes:

$$u_2 = e^{-Dt}\cos x + e^{-4Dt}\cos 2x.$$

- The heat kernel, for $t > 0$:

$$u_3 = \frac{1}{\sqrt{t}}\, e^{-x^2/(4Dt)}.$$

All three solve the same equation. We will see that only u_3 determines it. The function u_1 also solves $u_{xx} = -\kappa^2 u$ and $u_t = -D\kappa^2 u$, and from u_1 alone we cannot tell which of these laws is "the" law. The difference is that u_1 and u_2 are finite sums of exponentials, while u_3 is not (Chapter 4, Example 4.2).

1.2 Overview of the answers

Figure 1.1 shows the logical chain, read from left to right. Each box is a task. Each arrow is a theorem that allows the next task to be carried out, under its own hypotheses.

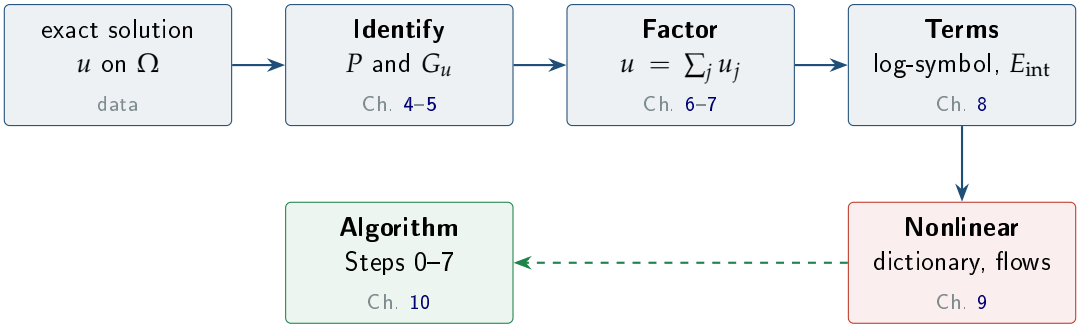


Figure 1.1: Logical structure of the method. The three linear stages (blue) are followed by the nonlinear extension (red). The algorithm (green) chains all of them. A step can only be trusted under the hypotheses of the theorem that precedes it.

Table 1.1 lists the main statements with their status. Every row points to the chapter where the statement is made precise.

1.3 Notation at a glance

Table 1.2 collects the symbols used throughout, with the place where each one is defined. Some letters have different meanings in different chapters; the caption lists them.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P)$ iff no $w_i = (P/P_i)(\partial)u$ is an exponential-polynomial (Ch. 4)	[proved]
Spectral form	$\mathrm{Ann}(u) = (P)$ iff the point spectrum is empty and $G_u \sim P$ (Ch. 4)	[proved]
Finite data	a jet of order $2d-1$ suffices for exponential-polynomials; no finite certificate otherwise; black box impossible; decidable with a bound (Ch. 5)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, ambiguity $\delta(P)$ (Ch. 6)	[proved]
Terms	splitting per additive term is false (Ch. 6)	[proved]
Regularity	components in H^m iff the sum of kernels is closed ($\gamma_N > 0$) and dense (Ch. 7)	[proved]
Interaction	closed product formula; $O(N)$ per value, $\Theta(2^N)$ per table (Ch. 8)	[proved]
Stability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m; no stable regularisation (Ch. 8)	[proved]
Nonlinear	Burgers: jet, closed form before the shock, $\Theta(\nu)$ and $\nu\lvert\log\nu\rvert$ after (Ch. 9)	[proved] / [proved mod. Kružkov]
Dictionary	identification modulo differential consequences; KdV degree $(N+2)2^{N-1}$ (Ch. 9)	[proved] / [proved mod. Novikov]

Table 1.1: Main results, in the order in which they appear.

1.4 Roadmap

The chapters build on each other in the following order.

- **Chapter 2** is a primer. It explains ideals, annihilators, exponential-polynomials, the Chinese remainder theorem and the companion matrix, starting from integers and one-variable ODEs. A reader who already knows these notions can skip it.
- **Chapter 3** fixes the algebraic language: symbols, the annihilator ideal, its gcd splitting and the finite-dimension test for exponential-polynomials. It ends with the one-variable prototype.
- **Chapters 4 and 5** answer Q1. The first gives the exact criterion and the intrinsic operator G_u. The second asks what can be decided from finite data.
- **Chapters 6 and 7** answer Q2. The first proves the factor decomposition and shows why splitting by terms fails. The second asks when the pieces keep the regularity of u.
- **Chapter 8** answers Q3: the exact interaction term, its cost, and its instability.
- **Chapter 9** extends the three questions to nonlinear equations.
- **Chapter 10** turns the theorems into an algorithm with two fully worked examples.
- **Chapters 11–15** (Part II) go beyond constant coefficients: systems and gauge, variable coefficients, convergence and noise, the analytic encoder, and the universal dictionary.
- **Chapter 16** records what Phase 10 changed in the statements of the earlier phases.

Symbol	Meaning	Defined in
(t, x), Ω	time and space; the open region where u is known	Ch. 1
$\partial = (\partial_t, \partial_x)$	partial derivatives	Ch. 3
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	ring of constant-coefficient operators, $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$	Def. 3.1
P, P_i, e_i	the operator (its symbol), its irreducible factors, their multiplicities	Ch. 3
$Q_j = P_j^{m_j}$	pairwise coprime factors used for decomposition	Ch. 6
$V(P)$	characteristic variety: zeros of the symbol in $\mathbb{C}^2$	Def. 3.1
$\mathrm{Ann}(u)$	the ideal of all operators that annihilate u	Def. 3.2
G_u, J_u	$G_u = \gcd \mathrm{Ann}(u)$ and the cofactor ideal: $\mathrm{Ann}(u) = G_u J_u$	Lemma 3.2
EP	exponential-polynomials	Def. 3.3
$\sigma_p(u)$	point spectrum $V(J_u)$	Ch. 4
$\delta(P)$	dimension of the ambiguity of the factor decomposition	Ch. 6
c_F, γ_N	Friedrichs cosine; conditioning number of the sum map	Ch. 7
A_n, $S_n(t)$, Δ_n	terms of an evolution, their flows, increments $S_n - \mathrm{Id}$	Ch. 8
E_{int}	interaction term	Eq. (8.1)
T_n, V, $I(u)$	nonlinear terms; dictionary span; ideal of relations of u	Ch. 9

Table 1.2: Notation guide. Overloaded letters: S is a flow in Chapter 8 and the sum map in Chapter 7; D is d/dx in Section 3.4 and a diffusion constant elsewhere; J denotes ideals.

- **Chapter 17** extends the algorithm to all of the above, with a contract for every step.
- **Chapter 18** states what is impossible, what is open, and how reliable each result is, for the whole project.

CHAPTER 2

A primer for readers new to the algebra

In this chapter	
Starting from	only linear algebra, polynomials and linear ODEs with constant coefficients.
Why	the whole theory speaks the language of rings and ideals. Every notion used later is introduced here from a familiar starting point: integers first, then ODEs in one variable, then two variables.
Role	nothing later depends on reading this chapter, but every later chapter uses its vocabulary. It covers ideals and annihilators (§2.1), exponential-polynomials (§2.2), the Chinese remainder theorem (§2.3) and the companion matrix (§2.4).

2.1 Ideals and annihilators, starting from integers

2.1.1 Integers

The multiples of 6 form the set

$$(6) = \{\dots, -12, -6, 0, 6, 12, \dots\}.$$

This set has two properties. The sum of two multiples of 6 is a multiple of 6. A multiple of 6 times any integer is again a multiple of 6. A subset of a ring with these two properties is called an *ideal*.

Now take the ideal generated by 4 and 6, that is, all combinations $4a + 6b$ with integers a, b:

$$(4, 6) = \{4a + 6b : \ a, b \in \mathbb{Z}\}.$$

Since $2 = 6 - 4$ lies in it, and every element is even, we get $(4, 6) = (2)$. The ideal "remembers" only the greatest common divisor of its generators. In $\mathbb{Z}$ every ideal is generated by one number.

2.1.2 Operators in one variable

Replace integers by differential operators with constant coefficients in one variable x, and replace "is a multiple of 6" by "kills the function u".

Definition 2.1 Annihilator (one variable)

For a function $u(x)$, the annihilator of u is the set of all polynomials Q such that the operator $Q(d/dx)$ kills u:

$$\operatorname{Ann}(u) = \Big\{ Q \in \mathbb{C}[\xi] : \; Q\Big(\frac{d}{dx}\Big)u = 0 \Big\}.$$

It is an ideal. If Q kills u and S is any polynomial, then SQ kills u too, because $S(d/dx)$ applied to 0 is 0. Sums of annihilators are annihilators.

Example 2.2 Annihilators of familiar functions

Write $\partial = d/dx$. The operator $\partial - 1$ kills e^x, and so does every multiple of it. The basic cases are:

$$\begin{aligned} \operatorname{Ann}(e^x) &= (\partial - 1), \\ \operatorname{Ann}(\sin x) &= (\partial^2 + 1), \\ \operatorname{Ann}(xe^x) &= \big((\partial - 1)^2\big), \\ \operatorname{Ann}(e^x + e^{2x}) &= \big((\partial - 1)(\partial - 2)\big). \end{aligned}$$

The function e^{x^2} has annihilator $\{0\}$: no nonzero polynomial in ∂ kills it, because each derivative $\partial^k e^{x^2}$ equals a polynomial of degree exactly k times e^{x^2}.

In one variable every ideal has a single generator, exactly as in $\mathbb{Z}$. The generator of $\operatorname{Ann}(u)$ is the *minimal annihilator* of u: the ODE of lowest order that u satisfies.

2.1.3 Two variables: the new feature

With two variables t and x we have two derivatives. An operator is a polynomial in both. Here an ideal may need *several* generators.

Example 2.3 An ideal with two generators

Let $u = e^{2t+3x}$. Then

$$(\partial_t - 2)u = 0 \qquad \text{and} \qquad (\partial_x - 3)u = 0.$$

Every operator of the form $A\,(\partial_t - 2) + B\,(\partial_x - 3)$ kills u, and these are all of them. So

$$\operatorname{Ann}(u) = (\partial_t - 2,\ \partial_x - 3).$$

No single polynomial generates this ideal: a polynomial in the ideal vanishes at the point $(2, 3)$, and a single polynomial vanishes on a whole curve, not at a single point.

This is the source of every subtlety in the theory. A solution of a PDE has an annihilator that may contain a "curve part" (one polynomial) and a "point part" (finitely many points). Lemma 3.2 of the next chapter makes this precise.

2.2 Exponential-polynomials

An *exponential-polynomial* is a finite sum of polynomials times exponentials:

$$w(t,x) = \sum_{j=1}^{r} p_j(t,x)\, e^{\tau_j t + \xi_j x}.$$

These are exactly the solutions of ODEs with constant coefficients, in each variable. They are the "trivial" solutions of the theory: they live on finitely many points of the characteristic variety.

The key property is a finite-dimension test. Differentiate w as often as you like: you always stay inside a finite-dimensional space.

Example 2.4 Which functions are exponential-polynomials?

Function	In EP?	Span of all derivatives
xe^x	yes	span$\{xe^x,\ e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite: $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$
e^{-x^2}	no	$\partial_x^k e^{-x^2} = (\text{degree } k)\cdot e^{-x^2}$: infinite dimension
$1/(1+x^2)$	no	denominators $(1+x^2)^{k+1}$ grow: infinite dimension
$f(x+t)$, f a Gaussian	no	same as the Gaussian

The test "is the span of the derivatives finite-dimensional?" is Lemma 3.3.

2.3 The Chinese remainder theorem

2.3.1 Integers

Solve the system

$$x \equiv 1 \pmod 2, \qquad x \equiv 2 \pmod 3.$$

The unique solution modulo 6 is $x \equiv 5$. In general, if a and b are coprime, a remainder modulo a and a remainder modulo b determine a unique remainder modulo ab. In the language of rings:

$$\mathbb{Z}/(ab) \;\cong\; \mathbb{Z}/(a) \times \mathbb{Z}/(b).$$

2.3.2 ODEs in one variable

Let $p(\partial) = (\partial - 1)(\partial - 2)$. Its solutions are $u = ae^x + be^{2x}$. The two pieces can be extracted by *projector operators*:

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = a\, e^x,$$

$$u_2 = \frac{\partial - 1}{2 - 1}\, u = b\, e^{2x}.$$

Check the first line: $(\partial - 2)(ae^x + be^{2x}) = -ae^x$, and dividing by $1 - 2 = -1$ gives ae^x. The sum is direct:

$$\ker p(\partial) = \ker(\partial - 1) \oplus \ker(\partial - 2).$$

This works because $\partial - 1$ and $\partial - 2$ are coprime: there are polynomials a, b with $a(\partial - 1) + b(\partial - 2) = 1$ (here $a = 1$, $b = -1$).

2.3.3 Two variables: what changes

For the wave operator $P = \tau^2 - \xi^2 = (\tau - \xi)(\tau + \xi)$ the two factors are coprime, but no combination of them equals 1. Indeed

$$(\tau - \xi,\ \tau + \xi) = (\tau,\ \xi),$$

the ideal of the single point $(0, 0)$ where the two lines meet. The quotient

$$\mathcal{R}/(\tau, \xi) \cong \mathbb{C}$$

has dimension 1. This number is the ambiguity of the splitting: the d'Alembert components $f(x + t)$ and $g(x - t)$ are defined only up to a constant moved from one to the other. Theorem 6.3 says that this finite overlap is the *only* failure of the integer picture.

2.4 The companion matrix

A linear ODE of order m becomes a first-order system for the vector of derivatives. For

$$u'' - 3u' + 2u = 0$$

put $X = (u, u')^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}.$$

The characteristic polynomial of C is $\lambda^2 - 3\lambda + 2$, the symbol of the ODE, with roots 1 and 2. The Lagrange projectors are

$$\Pi_1 = \frac{C - 2\,\mathrm{Id}}{1 - 2}, \qquad \Pi_2 = \frac{C - \mathrm{Id}}{2 - 1},$$

and the matrix exponential splits into modes:

$$e^{xC} = e^{x}\,\Pi_1 + e^{2x}\,\Pi_2.$$

This is the same formula as the projectors of the previous section, written for matrices. It reappears as the Lagrange formula (6.1) for evolution factors.

Takeaway. Four ideas carry the theory. An *ideal* records all equations a function satisfies. Its *gcd* separates a curve part from finitely many points. *Exponential-polynomials* are the functions with finitely many derivatives up to linear dependence. The *Chinese remainder theorem* splits a solution into pieces, up to a finite overlap.

CHAPTER 3

The algebraic language

In this chapter	
Starting from	constant coefficients; functions or distributions on $\Omega \subset \mathbb{R}^2$ with complex values. The vocabulary of Chapter 2.
Why	three notions carry the whole theory. The *symbol* turns derivatives into polynomials. The *annihilator* collects all equations a function satisfies. *Exponential-polynomials* are the trivial solutions.
Role	foundation for every later chapter. It proves the two lemmas that every identification argument uses (Lemmas 3.2 and 3.3) and ends with the one-variable prototype (Section 3.4).

3.1 Derivatives as polynomials

A linear operator with constant coefficients is a polynomial in the two partial derivatives:

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j\, \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials are eigenfunctions of every derivative. Write

$$e_{\tau,\xi}(t,x) = e^{\tau t + \xi x}.$$

Then

$$\partial_t\, e_{\tau,\xi} = \tau\, e_{\tau,\xi}, \qquad \partial_x\, e_{\tau,\xi} = \xi\, e_{\tau,\xi},$$

and therefore, for every operator P,

$$P(\partial)\, e_{\tau,\xi} = P(\tau,\xi)\, e_{\tau,\xi}. \tag{3.1}$$

So applying the operator to an exponential is the same as multiplying by the value of a polynomial. This is the reason why all the algebra below is about polynomials in two variables.

Definition 3.1 Operator ring, symbol, characteristic variety

The ring of constant-coefficient operators is

$$\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \;\cong\; \mathbb{C}[\tau,\xi].$$

The polynomial $P(\tau,\xi)$ is the *symbol* of $P(\partial)$. The *characteristic variety* of P is its zero set

$$V(P) = \{(\tau,\xi) \in \mathbb{C}^2 :\; P(\tau,\xi) = 0\}.$$

By (3.1), the exponential $e_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when (τ, ξ) lies on $V(P)$. Fourier modes are the special case $\xi = i\kappa$ with real frequency κ.



Figure 3.1: Real slices of two characteristic varieties. A single exponential solution is a single point of the variety. A rich, non-exponential solution "fills" the whole curve.

3.2 Annihilators and what a solution knows

A single function is usually killed by many operators. For example, e^x is killed by $\partial_x - 1$, by $\partial_x^2 - 1$, by ∂_t, and by every combination of these.

Definition 3.2 Annihilator

The annihilator of u is the set of all operators that kill u:

$$\operatorname{Ann}(u) = \{Q \in \mathcal{R} : \; Q(\partial)u = 0\}.$$

$\operatorname{Ann}(u)$ is an ideal of $\mathcal{R}$: if $Q(\partial)u = 0$ and $S \in \mathcal{R}$, then

$$(SQ)(\partial)u = S(\partial)\big(Q(\partial)u\big) = 0,$$

and the sum of two annihilators is an annihilator. Three classical facts about $\mathbb{C}[\tau, \xi]$ are used throughout.

Three facts about polynomials in two variables

(F1) Unique factorisation. Every nonzero polynomial factors as

$$P = c \prod_i P_i^{e_i},$$

with P_i irreducible and pairwise non-proportional (the "primes"), $e_i \geq 1$ their multiplicities, and c a constant. The factorisation is unique up to constants, so gcd is defined as for integers.

(F2) Bézout. Two polynomials without a common factor have only finitely many common zeros, at most the product of their degrees. Geometrically: two curves without a common component meet in isolated points.

(F3) Nullstellensatz (Hilbert). If a polynomial q vanishes on the common zero set $V(I)$ of an ideal I, then $q^k \in I$ for some $k \geq 1$. In particular, an ideal without common zeros is the whole ring.

All three are classical [1, 2]. They hold over $\mathbb{C}$; the Nullstellensatz fails over $\mathbb{R}$.

The next lemma says that every ideal splits into a "curve part" (one polynomial) and a residue that vanishes only at finitely many points.

Lemma 3.3 Splitting off the gcd (C3-L2) [proved]

Let $I \subset \mathcal{R}$ be a nonzero ideal and $g = \gcd(I)$. Then

$$I = g \cdot J, \qquad J = \{f/g : \ f \in I\},$$

where J is an ideal whose zero set $V(J)$ is finite.

Dim. C3-L2, Catalogue p. 3

Intuition

Divide every element of I by the common factor g. What is left has no common factor, so by Bézout any two "generic" elements of J meet only at finitely many points.

For the annihilator of a function we use a fixed notation:

$$G_u = \gcd \operatorname{Ann}(u), \qquad \operatorname{Ann}(u) = G_u \cdot J_u.$$

G_u is the "genuinely infinite-dimensional" part of the equation of u. J_u records the finitely many exponentials that happen to be present in u.

Example 3.4 Two extreme cases

A single exponential. For $u = e^x$, seen as a function of (t, x):

$$\operatorname{Ann}(u) = (\tau, \ \xi - 1), \qquad G_u = 1, \qquad J_u = \operatorname{Ann}(u).$$

The ideal is that of the single point $(0, 1)$. The function has no infinite part.

A travelling profile. For $u = f(x + t)$ with f a Gaussian:

$$\operatorname{Ann}(u) = (\tau - \xi), \qquad G_u = \tau - \xi, \qquad J_u = \mathcal{R}.$$

The whole ideal is the curve part; there are no isolated points. This is proved in Chapter 4.

3.3 Exponential-polynomials

Definition 3.5 Exponential-polynomials

EP is the set of finite sums

$$w(t,x) = \sum_{j=1}^{r} p_j(t,x)\, e^{\tau_j t + \xi_j x},$$

with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. We write $\mathrm{EP}(\Omega)$ for their restrictions to Ω.

For a function w, let $\mathcal{R}w$ be the space of all its derivatives and their linear combinations:

$$\mathcal{R}w = \operatorname{span}\{\partial_t^j \partial_x^k w : \ j,k \geq 0\} = \{Q(\partial)w : \ Q \in \mathcal{R}\}.$$

The map $Q \mapsto Q(\partial)w$ goes from $\mathcal{R}$ onto $\mathcal{R}w$ and its kernel is $\operatorname{Ann}(w)$. Hence

$$\mathcal{R}w \ \cong \ \mathcal{R}/\operatorname{Ann}(w).$$

Lemma 3.6 Finite-dimension criterion (C3-L3) [proved]

For a function or distribution w on a connected Ω:

$$w \in \mathrm{EP}(\Omega) \iff \dim_{\mathbb{C}}(\mathcal{R}w) < \infty.$$

In that case $V(\operatorname{Ann} w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.

Dim. C3-L3, Catalogue p. 4

Intuition

($\Rightarrow$) Derivatives of $p\, e^{\tau t + \xi x}$ are polynomials of degree at most $\deg p$ times the same exponential, so they stay in a finite-dimensional space.

($\Leftarrow$) On the finite-dimensional space $\mathcal{R}w$, the derivatives ∂_t and ∂_x act as two commuting matrices. Commuting matrices can be triangularised together. Their joint eigenvalues are the exponents, and their nilpotent parts produce the polynomial prefactors.

Example 3.7 The test on two functions

For $w = xe^x$ the derivatives are $xe^x + e^x$, $xe^x + 2e^x$, and so on: all lie in $\operatorname{span}\{xe^x, e^x\}$, of dimension 2. So $w \in \mathrm{EP}$.

For $w = e^{-x^2}$, the k-th derivative is a polynomial of degree exactly k times e^{-x^2}. These are linearly independent for all k, so $\dim \mathcal{R}w = \infty$ and $w \notin \mathrm{EP}$.

Why it matters. Lemma 3.3 is the computational heart of the method. The question "is w an exponential-polynomial?" is a question of finite-dimensional linear algebra. It is the test used in the identification criterion (Theorem 4.2) and in Step 3 of the algorithm.

3.4 One variable: the ODE prototype

In one variable every object of the later chapters appears with explicit formulas and without the geometry of curves. We consider an ODE $p(D)u = 0$ in the variable x, with $D = d/dx$ in this section only.

Setting

The symbol factors over $\mathbb{C}$ as

$$p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j},$$

with distinct roots λ_j and multiplicities $m_j \geq 1$. The order is $m = \sum_j m_j$. The solution space is

$$\ker p(D) = \Big\{ \sum_j q_j(x)\, e^{\lambda_j x} : \deg q_j < m_j \Big\}.$$

The form of the solutions comes from one computation. For one root,

$$(D - \lambda)\big(q\, e^{\lambda x}\big) = q'\, e^{\lambda x} :$$

applying $D - \lambda$ lowers the degree of q by one. So $(D - \lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $\deg q < m_j$.

Example 3.8 Reading the solution space from the roots

Let $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$. The roots are 1 (multiplicity 2) and -2 (multiplicity 3), and the order is 5. A basis of solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2e^{-2x}.$$

3.4.1 Identification by the Wronskian

Proposition 3.9 ODE identification (R4-T1) [proved]

Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of basis functions actually present in u. It equals p if and only if

$$u,\ u',\ \ldots,\ u^{(m-1)}$$

are linearly independent (nonvanishing Wronskian), that is, iff every root of p appears in u with its full multiplicity.

Dim. R4-T1, Catalogue p. 97

Example 3.10 A missing root

Let $p(\lambda) = (\lambda - 1)(\lambda - 2)$ and $u = 3e^x$. Then $u' = u$, so u and u' are dependent and the minimal annihilator is $\lambda - 1$, of degree 1. The root 2 is missing from u, and u cannot tell p from $\lambda - 1$.

3.4.2 The companion matrix and the spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ and put $X = (u, u', \ldots, u^{(m-1)})^\top$. Then $X' = CX$ with the companion matrix

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC}X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the exponential has the spectral form

$$e^{xC} = \sum_{j=1}^{m} e^{\lambda_j x}\,\Pi_j, \qquad \Pi_j = \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \tag{3.2}$$

where the Π_j are the Lagrange projectors onto the eigenspaces.

Proposition 3.11 Projectors of the companion matrix (II-T2, II-T3) [proved]

For an operator monic in τ with pairwise coprime factors, the projectors onto the factor components are explicit rational functions of ξ. Their poles lie exactly where two factor varieties meet. They are polynomial if and only if the varieties are disjoint, that is, iff $(Q_i, Q_j) = \mathcal{R}$ for all $i \neq j$.

Dim. II-T2, Catalogue p. 100

The companion matrix is not normal in general

An early conjecture said that the companion matrix is normal exactly when $p = s^m - c$ with $|c| = 1$. For $m = 2$ this is [false]: the polynomial $s^2 + s - 1$ is a counterexample. The correct normality criterion for $m = 2$ is (II-P5) [proved], and it is formalised in Lean. Consequence: Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on coefficient vectors.

3.4.3 The jet is a cyclic vector

Proposition 3.12 Cyclic vector (II-C1) [proved]

Let P be monic in τ of degree k. Then $\mathrm{Ann}(u) = (P)$ if and only if the t-jet

$$u,\ \partial_t u,\ \ldots,\ \partial_t^{k-1} u$$

is linearly independent over the ring $\mathbb{C}[\partial_x]$.

Dim. II-C1, Catalogue p. 103

This is the two-variable Wronskian: it replaces "the derivatives are independent" by "the time derivatives are independent with coefficients that are operators in x".

Takeaway. In one variable the polynomial encodes the solution space. Its roots give the exponential rates, the multiplicities give the polynomial prefactors, the companion matrix represents differentiation, and the Lagrange projectors isolate the modes. In two variables the same structure survives. The minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Chapter 4) and the Lagrange projectors become Eq. (6.1) (Chapter 6).

CHAPTER 4

Identification: what one solution determines

In this chapter	
Starting from	$\Omega \subset \mathbb{R}^2$ open and *connected*; $u \in \mathcal{D}'(\Omega)$; constant coefficients; an operator P with $P(\partial)u = 0$ is available (any annihilator, not necessarily the "true" one), with irreducible factorisation $P = c\prod_i P_i^{e_i}$.
Why	before splitting a solution into pieces one must know its equation. We ask exactly when u alone pins down P, and what the intrinsic operator of u is when it does not.
Role	first link of the chain in Figure 1.1. Its output is the pair (G_u, J_u), which every later chapter uses, and the test "$\in$ EP", which Step 3 of the algorithm implements.

4.1 The question in algebraic form

Suppose $P(\partial)u = 0$. Then every multiple SP also kills u, so the ideal (P) is contained in $\mathrm{Ann}(u)$:

$$(P) \subseteq \mathrm{Ann}(u).$$

We say that u *identifies* P when the two coincide. In that case every equation satisfied by u is a consequence of P, and P is determined by u up to a constant factor.

Why can the inclusion be strict? Because u may satisfy extra equations that P knows nothing about. The single Fourier mode of the heat equation is the typical case: it also satisfies $u_{xx} = -\kappa^2 u$.

Connectedness of Ω is necessary. On a disconnected region the two pieces can satisfy different equations, and no criterion on u as a whole can see that.

4.2 The criterion

Terminology

- P is *reduced* if all $e_i = 1$ (no repeated factor).
- P is *monic in* τ if its leading coefficient as a polynomial in τ is a nonzero constant. Then $k = \deg_\tau P$ is its order in time, and the *t-jet* of u is the vector $(u, \partial_t u, \dots, \partial_t^{k-1} u)$.

- For each irreducible factor we form the partial quotients

$$w_i = \left(\frac{P}{P_i}\right)(\partial)\, u, \qquad w_i' = \left(\frac{P}{P_i^{e_i}}\right)(\partial)\, u.$$

  In w_i we remove one copy of P_i; in w_i' we remove the whole power.

Theorem 4.1 Identification criterion (C3-T1) [proved]

Let Ω be connected, $u \in \mathcal{D}'(\Omega)$ and $P(\partial)u = 0$.

(a) $\operatorname{Ann}(u) = (P)$ if and only if $w_i \notin \mathrm{EP}(\Omega)$ for every irreducible factor P_i.

(b) $V(\operatorname{Ann} u) = V(P)$ if and only if $w_i' \notin \mathrm{EP}(\Omega)$ for every i.

(c) If P is reduced, (a) and (b) coincide.

(d) If P is monic in τ, the test can be carried out on the t-jet of u (Proposition 3.4.3).

Dim. C3-T1, Catalogue p. 7

Intuition

Strip one irreducible factor off P and look at what is left: w_i lies in the kernel of P_i.

If w_i is a "generic" member of that kernel, not a finite sum of exponentials, then no smaller operator can kill u.

If w_i is an exponential-polynomial, then u contains a finite-dimensional piece. That piece is also killed by an operator of lower order, and u cannot tell the large equation from the small one.



Figure 4.1: The heat curve $\tau = \xi^2$. Left: a solution that is not an exponential-polynomial (the heat kernel) uses the whole curve, and $\operatorname{Ann}(u) = (P)$. Right: a finite sum of exponentials lives on finitely many points. Then $\operatorname{Ann}(u)$ contains polynomials that vanish at those points but are not multiples of P.

Proposition 4.2 Uniqueness of the identified operator

If $\operatorname{Ann}(u) = (P)$, then every operator Q with $Q(\partial)u = 0$ is a multiple $Q = SP$ with $S \in \mathcal{R}$. In particular P is, up to a nonzero constant, the only annihilator of minimal degree.

Dim. C3-T1, Catalogue p. 7

The proof is one line: $Q \in (P)$ means $Q = SP$, and degrees add, so $\deg Q \geq \deg P$ with equality only when S is a constant.

Example 4.3 The heat equation, three cases

Take $P = \tau - D\xi^2$ with $D > 0$. It is irreducible, so $P/P_1 = 1$ and the criterion reads

$$\mathrm{Ann}(u) = (P) \iff u \notin \mathrm{EP}.$$

One Fourier mode. Let $u = e^{-D\kappa^2 t + i\kappa x}$. Then $u_t = -D\kappa^2 u$ and $u_{xx} = -\kappa^2 u$. So u is also killed by $\partial_x - i\kappa$ and by $\partial_t + D\kappa^2$:

$$\mathrm{Ann}(u) = (\tau + D\kappa^2,\ \xi - i\kappa).$$

This is the ideal of one point of $V(P)$, strictly larger than (P). Indeed P belongs to it:

$$P = (\tau + D\kappa^2) - D\,(\xi - i\kappa)(\xi + i\kappa).$$

A finite superposition. For $u = \sum_{j=1}^{r} c_j e^{-D\kappa_j^2 t + i\kappa_j x}$ with distinct κ_j, $u \in \mathrm{EP}$. Its annihilator is the ideal of the r points $(-D\kappa_j^2,\ i\kappa_j)$. It contains many operators that are not multiples of P.

The heat kernel. Let $u = t^{-1/2} e^{-x^2/(4Dt)}$ for $t > 0$. Its x-derivatives are polynomials of growing degree times the same Gaussian. They span an infinite-dimensional space, so $u \notin \mathrm{EP}$ (Lemma 3.3). Hence $\mathrm{Ann}(u) = (P)$: every constant-coefficient equation satisfied by the heat kernel is a multiple of $\tau - D\xi^2$.

Message. A "poor" solution remembers only finitely many exponentials and cannot tell the heat equation from many others. A "rich" solution remembers the whole equation.

Example 4.4 The wave equation and d'Alembert (C3-T5) [proved]

Let $P = (\tau - c\xi)(\tau + c\xi)$ with real $c \neq 0$, and $u = f(x - ct) + g(x + ct)$. The two partial quotients are

$$w_1 = (\partial_t + c\partial_x)\,u = 2c\,g'(x + ct),$$
$$w_2 = (\partial_t - c\partial_x)\,u = -2c\,f'(x - ct).$$

So the criterion says:

$$\mathrm{Ann}(u) = (P) \iff f \notin \mathrm{EP} \ \text{ and } \ g \notin \mathrm{EP}.$$

A single travelling wave ($g = 0$) never identifies the wave equation: it also satisfies the first-order transport equation. The same theorem covers Klein–Gordon $\tau^2 - \xi^2 + \mu^2$ and the telegraph operator $\tau^2 + \alpha\tau - c^2\xi^2$: both are irreducible conics, so they are identified iff $u \notin \mathrm{EP}$. For the beam $\tau^2 + \xi^4 = (\tau - i\xi^2)(\tau + i\xi^2)$ both Schrödinger parts must be outside EP.

Dim. C3-T5, Catalogue p. 104

Why it matters. Theorem 4.2 turns "does u identify P?" into finitely many membership tests "$\in$ EP". Each test is a finite-dimensional computation (Lemma 3.3). Whether the tests can be carried out from the data one actually has is the subject of Chapter 5.

4.3 The intrinsic operator and the spectral form

Theorem 4.2 says when (P) is the whole annihilator. When it is not, what does u determine? The answer uses the two pieces of $\mathrm{Ann}(u)$ given by Lemma 3.2.

Point spectrum

Recall $\mathrm{Ann}(u) = G_u\, J_u$ with $G_u = \gcd \mathrm{Ann}(u)$. The *point spectrum* of u is the finite set

$$\sigma_p(u) = V(J_u) \subset \mathbb{C}^2.$$

Its points are the exponents (τ_0, ξ_0) of the exponentials $e^{\tau_0 t + \xi_0 x}$ that are "present" in u. The word spectrum is meant in the algebraic sense, not as the spectrum of an operator on a Hilbert space. We write $G_u \sim P$ when the two agree up to a nonzero constant.

Theorem 4.5 Spectral characterisation (P4-T2) [proved]

Let P be an annihilator of u. Then

$$\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset \text{ and } G_u \sim P.$$

Moreover the exponential-polynomial part of the derivative space is the finite quotient

$$\mathcal{R}u \cap \mathrm{EP} \;\cong\; \mathcal{R}/J_u.$$

Dim. P4-T2, Catalogue p. 10

Example 4.6 Reading the spectrum

One exponential. For $u = e^{\tau_0 t + \xi_0 x}$: $\mathrm{Ann}(u) = (\tau - \tau_0,\ \xi - \xi_0)$, so $G_u = 1$ and $\sigma_p(u) = \{(\tau_0, \xi_0)\}$. A single exponential has no infinite part and a one-point spectrum.

Two exponentials. For $u = e^{\tau_1 t + \xi_1 x} + e^{\tau_2 t + \xi_2 x}$ with distinct exponents: $G_u = 1$ and $\sigma_p(u)$ has two points.

A rich solution. If $\mathrm{Ann}(u) = (P)$ then $\sigma_p(u) = \emptyset$ and $G_u \sim P$.

The operator G_u is the correct "operator intrinsic to u". It depends only on u, while P depends on which annihilator was chosen. The next theorem computes it from any annihilator.

Theorem 4.7 Intrinsic operator (C3-T6) [proved]

Let $P = c\prod_i P_i^{e_i}$ be any annihilator of u. For each i let

$$k_i = \max\left\{k \le e_i : \left(\frac{P}{P_i^k}\right)(\partial)u \in \mathrm{EP}\right\}$$

(with $k_i = 0$ if no such $k \ge 1$ exists), and $f_i = e_i - k_i$. Then

$$G_u \sim \prod_i P_i^{f_i},$$

and $\mathrm{Ann}(u)$ is principal if and only if $G_u(\partial)u = 0$. Computing G_u takes at most $\sum_i e_i$ tests "$\in$ EP".

Dim. C3-T6, Catalogue p. 9

Example 4.8 A repeated factor

Let G be the heat kernel ($D = 1$), $P = (\tau - \xi^2)^3$ and $u = t\,G$. Then:

$$\left(\frac{P}{P_1}\right)(\partial)u = (\partial_t - \partial_x^2)^2(tG) = 0 \in \mathrm{EP},$$

$$\left(\frac{P}{P_1^2}\right)(\partial)u = (\partial_t - \partial_x^2)(tG) = G \notin \mathrm{EP}.$$

So $k_1 = 1$, $f_1 = 2$, and

$$G_u = (\tau - \xi^2)^2, \qquad \mathrm{Ann}(u) = (G_u).$$

The function tG identifies the *square* of the heat operator, not its cube.

Why G_u and not the variety

A tempting stronger statement replaces G_u by the variety $V(\mathrm{Ann}\,u)$. It is correct only for reduced P, because zero sets do not see multiplicities.

Example. Let $P = \tau^2$ and $u = e^{x^2}$. Then $u_{tt} = 0$. No nonzero polynomial in ∂_x alone kills u, so

$$\mathrm{Ann}(u) = (\tau).$$

Here $\sigma_p(u) = \emptyset$, but $G_u = \tau \not\sim P$. So $\mathrm{Ann}(u) \neq (P)$, while $V(\mathrm{Ann}\,u) = \{\tau = 0\} = V(P)$. The variety alone would wrongly say that u identifies τ^2.

4.4 Refinements

4.4.1 Evolutions: identification inside a finite class

For an evolution $u_t = a(\partial_x)u$ the symbol $P = \tau - a(\xi)$ has degree one in τ, so it is irreducible. The criterion says: $\mathrm{Ann}(u) = (P)$ iff $u \notin \mathrm{EP}$. When u *is* an exponential-polynomial, we can still ask whether a is recovered among polynomials of degree at most n.

Theorem 4.9 Evolutions (C3-T4) [proved]

Let $P = \tau - a(\xi)$ and $P(\partial)u = 0$ on a connected Ω.

(a) $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}(\Omega)$.

(b) If $u = \sum_k q_k\, e^{\tau_k t + \xi_k x}$ with distinct exponents and nonzero q_k, put $\delta_k = \deg_x q_k$ and

$$D = \sum_k (\delta_k + 1).$$

If $\deg a \le n$, then among all b of degree at most n, the evolutions $\tau - b$ that kill u are exactly

$$b \in a + \prod_k (\xi - \xi_k)^{\delta_k + 1} \cdot \mathbb{C}[\xi]_{\le n - D}.$$

So a is identified in degree $\le n$ if and only if $D \ge n + 1$.

Dim. C3-T4, Catalogue p. 104

This is Hermite interpolation: a polynomial of degree n is fixed by $n+1$ conditions, and each mode $q_k\, e^{\tau_k t + \xi_k x}$ supplies $\delta_k + 1$ of them.

Example 4.10 Advection–diffusion from two modes

Take $u_t = -cu_x + Du_{xx}$, so $a(\xi) = -c\xi + D\xi^2$ has degree $n = 2$. Two simple modes give $D = 2 < 3$: the pair (c, D) is not identified, because one free parameter remains. Three distinct modes give $D = 3$: both coefficients are fixed.

4.4.2 A naive transfer that fails

On a periodic domain an earlier result said: if the Fourier expansion of u has infinitely many frequencies on every factor curve, then $\mathrm{Ann}(u) = (P)$ (R4-T5). On a *bounded* domain this transfer is false, because the expansion is not unique.

Counterexample 4.11 The spectrum of an expansion is not intrinsic (C3-C1)

Let $P = \tau + \xi$ and $\Omega = (0,1)^2$. On $(-\pi, \pi)$ the Fourier series

$$e^s = \sum_{k \in \mathbb{Z}} c_k\, e^{iks}, \qquad c_k = \frac{\sinh(\pi)(-1)^k}{\pi(1 - ik)} \neq 0,$$

converges in L^2. So $u = e^{x-t} = \sum_k c_k\, e^{ik(x-t)}$ has infinitely many exponents on the line $\tau + \xi = 0$. Yet

$$\mathrm{Ann}(u) = (\tau + 1,\ \xi - 1) \supsetneq (\tau + \xi).$$

Dim. C3-C1, Catalogue p. 105

The criterion must therefore be stated on u itself, as in Theorem 4.2, not on an expansion of u.

4.4.3 Symmetries and local data

Proposition 4.12 Translation symmetry (C3-P3) [proof written, not audited]

Suppose u is invariant under translations along a direction, $\ell(\partial)u = 0$ with $\ell = \alpha\tau + \beta\xi \neq 0$, and $P(\partial)u = 0$ with P nonconstant. If $P \not\sim \ell$, then $\mathrm{Ann}(u) \neq (P)$. If moreover $\ell \nmid P$, then $u \in \mathrm{EP}$.

Dim. C3-P3, Catalogue p. 106

So a symmetry with constant characteristic destroys identification, and it is a special case of the criterion. Symmetries with variable coefficients are invisible in $\mathbb{C}[\partial]$: the heat kernel is scale invariant, $(2t\partial_t + x\partial_x + 1)G = 0$, yet $\mathrm{Ann}(G) = (\tau - \xi^2)$. Identifiability depends on the ring of coefficients.

Proposition 4.13 Local data (C3-P5) [proof written, not audited]

Identification can be lost on a smaller region. Let $P = \tau + \xi$, $F(s) = e^s + \varphi(s)$ with φ a smooth bump supported in $(2,3)$, and $u = F(x-t)$. On $(0,1)\times(-5,5)$, $u \notin \mathrm{EP}$ and $\mathrm{Ann}(u) = (\tau + \xi)$. On the smaller square $(0,1)^2$, $u = e^{x-t}$ and the annihilator is the ideal of a point.

For elliptic P, every solution is real-analytic, and then $\mathrm{Ann}(u|_{\Omega'}) = \mathrm{Ann}(u)$ for every connected $\Omega' \subset \Omega$.

Dim. C3-P5, Catalogue p. 107

Takeaway. A solution identifies its equation exactly when, after removing any one irreducible factor, what is left is not a finite sum of exponentials. In general, u identifies only its intrinsic operator G_u. The difference between P and G_u is a finite set of exponentials, the point spectrum $\sigma_p(u)$.

CHAPTER 5

Finite data and decidability

In this chapter	
Starting from	the criterion of Chapter 4: $\mathrm{Ann}(u) = (P)$ iff no w_i is an exponential-polynomial. It is a statement about the whole function u.
Why	in practice we hold finitely many numbers: a jet at a point, samples, or a formula. We ask which data suffice, which never suffice, and when the test "$\in$ EP" becomes an algorithm.
Role	tells Step 3 of the algorithm what it can and cannot promise. The answer depends on the data model: positive for exponential-polynomials, negative for a black box, positive again with one extra integer.

5.1 Three data models

The reconstruction is exact, so the input must be of an exact kind. Three models are analysed throughout (C3-D1).

M1 A closed formula for u, for example a finite sum of exponentials and Gaussians.

M2 The jet of u at a point p_0: all partial derivatives up to some order, given as exact numbers. In the "black-box" version M2$^+$ we may also ask a *zero-test oracle* whether $Q(\partial)u = 0$ for a given Q.

M3 u real-analytic on a connected Ω, described by finitely many linear equations with polynomial coefficients (a *holonomic* description) with algebraic numbers as coefficients.

Words used in this chapter

Jet of order k	the list of all partial derivatives of u of order $\le k$ at p_0: its Taylor coefficients.
Zero-test oracle	a procedure that says whether a given expression built from u is identically zero.
Holonomic	described by linear differential equations with polynomial coefficients whose solution space is finite-dimensional.
Weyl algebra	A_1 or A_2: the ring of differential operators with polynomial coefficients.

5.2 Exponential-polynomial data: a finite threshold

When u itself is an exponential-polynomial, it never identifies its equation in the whole ring: its annihilator has finitely many zeros, while $V(P)$ is a curve (C3-C2). What we can ask is whether u identifies P *among operators of degree at most* n.

Theorem 5.1 Bézout threshold (C3-T2) [proved]

Let $u = \sum_k c_k\, e^{\langle z_k,(t,x)\rangle}$ be a sum of simple modes with distinct exponents $z_k \in V(P)$, P reduced, and $n \geq \deg P$. If every factor curve carries enough exponents,

$$\big|\{z_k\} \cap V(P_i)\big| > n \cdot \deg P_i \qquad \text{for every } i,$$

then the annihilators of degree at most n are exactly the multiples of P:

$$\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = P \cdot \mathcal{R}_{\leq n-\deg P}.$$

The threshold cannot be lowered uniformly: for $P = \tau - \xi^2$ and every n there are $2n$ exponents on the parabola that do not suffice.

Dim. C3-T2, Catalogue p. 12

Intuition

An operator Q of degree n that kills u must vanish at every exponent z_k. If P_i does not divide Q, the two curves meet in at most $n \deg P_i$ points (Bézout). More points than that force P_i to divide Q.

Example 5.2 The parabola needs $2n+1$ points

Let $P = \tau - \xi^2$, so $\deg P = 2$. Take the $2n$ exponents

$$Z_n = \{(j^2, \pm j) : \ j = 1, \dots, n\} \subset V(P).$$

The operator $Q_n = \prod_{j=1}^{n} (\tau - j^2)$ has degree n, vanishes on Z_n, and is not a multiple of $\tau - \xi^2$ because it does not depend on ξ. So $2n$ points are not enough. With $2n+1$ points the theorem applies.

The next theorem says how many derivatives at a single point are needed.

Theorem 5.3 Jet threshold $2d-1$ (C3-T3) [proved]

Let $0 \neq u \in \mathrm{EP}(\Omega)$ on a connected Ω, and let $d = \dim \mathcal{R}u$. Then:

(i) the operators of degree at most n that kill u are determined by the jet of u of order $n + d - 1$ at any point;

(ii) $\mathrm{Ann}(u)$ is generated in degree at most d, so it is determined by the jet of order $2d - 1$;

(iii) the order $2d - 1$ is optimal in the worst case, already in one variable: for $u = \sum_{j=1}^{d} e^{jx}$ the jet of order $2d - 2$ does not determine $\mathrm{Ann}(u)$.

An upper bound $d' \geq d$ may replace d everywhere.

Dim. C3-T3, Catalogue p. 12

Example 5.4 One exponential, $d = 1$

For $u = e^{2x}$, $d = 1$ and the threshold is $2d - 1 = 1$. The values $u(0) = 1$ and $u'(0) = 2$ give $u' = 2u$, so $\mathrm{Ann}(u)$ contains $\xi - 2$, and τ comes from the absence of t. The value $u(0)$ alone (order 0) cannot distinguish e^{2x} from e^{3x}.

5.3 What no finite data can show

For a solution outside EP, the situation reverses.

Theorem 5.5 No finite certificate (C3-P1) [proved] / [from cited source]

Let Ω be connected, P nonconstant and $u \in \ker P$. For every order K there is an exponential-polynomial $v \in \ker P$ with the same jet of order K at p_0 as u, and $\mathrm{Ann}(v) \neq (P)$.

On convex Ω the same holds for any finite family of continuous linear functionals (Malgrange's density theorem) [from cited source]. On a non-convex region a single functional can certify.

Dim. C3-P1, Catalogue p. 14

Intuition

Every finite jet is matched by a finite sum of exponentials taken from the curve $V(P)$. Finitely many numbers cannot distinguish u from that sum, and the sum never identifies anything. So an algorithm that sees finitely many numbers can only *refute* identification, never confirm it.

The black-box version is sharper.

Theorem 5.6 Black-box impossibility (P4-T5) [proved]

In the model $\mathrm{M2}^+$ (exact jets plus a zero-test oracle), for every nonconstant P there is a solution u with $\mathrm{Ann}(u) = (P)$ such that the following holds. For every jet order k_0 and every finite set of zero tests, some exponential-polynomial $v \in \ker P$ gives the same jet and the same answers, while $\mathrm{Ann}(v) \neq (P)$. Hence no algorithm decides "$\mathrm{Ann}(u) = (P)$" correctly on all inputs.

Dim. P4-T5, Catalogue p. 16

5.4 One extra integer makes it decidable

Theorem 5.7 Decidability with a bound (P4-T4) [proved]

In the model M2$^+$:

(i) for every finite-dimensional space of operators $\mathcal{L}$, the annihilators of u in $\mathcal{L}$ are computable;

(ii) "$\mathrm{Ann}(u) \neq (P)$" is semi-decidable: an algorithm stops with "yes" exactly when it is true;

(iii) if an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ is given, "$\mathrm{Ann}(u) = (P)$" is decidable. With

$$N = B + \deg P - 1,$$

one has $\mathrm{Ann}(u) = (P)$ iff $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq N} = (P) \cap \mathcal{R}_{\leq N}$.

The bound also makes G_u computable.

Dim. P4-T4, Catalogue p. 15

Intuition

If identification fails, some w_i is an exponential-polynomial of dimension at most B. Its annihilator has a generator of degree at most B that is not a multiple of P_i. Multiplying by P/P_i gives a witness of degree at most $B + \deg P - 1$. So it is enough to look for witnesses up to that degree, and that is a finite linear-algebra problem.

Is the bound B the least data that works? The minimal sufficient datum is "a degree D such that, if identification fails, a witness of degree $\leq D$ exists" (OP-ALG-T3) [proof written, not audited]. The bound B is strictly stronger on some regions when P has at least two irreducible factors counted with multiplicity. Whether it is strictly stronger on every region is open [open] (OP-ALG-Q2).

5.5 A decidable class: holonomic solutions

Theorem 5.8 Holonomic model (OP-ALG-T1, P4-T6) [proved mod. (AC)]

In the model M3 (real-analytic u with a holonomic description over algebraic numbers), assume (AC): jets at an algebraic point are computable in a computable field. Then

- "$Q(\partial)u \in \mathrm{EP}$?" is decidable for every Q;
- "$\mathrm{Ann}(u) = (P)$?" and "$V(\mathrm{Ann}\, u) = V(P)$?" are decidable, and G_u is computable;
- a bound B and a finite set of generators of $\mathrm{Ann}(u)$ are computable.

Dim. OP-ALG-T1, Catalogue p. 108

Example 5.9 The heat kernel plus an exponential

Let $u = G + e^{t+x}$, with G the heat kernel. The slice $t = 1$ is $x \mapsto G(1,x) + e^{1+x}$, a function with a holonomic description. The algorithm finds that the slice is not an exponential-polynomial, and concludes $\mathrm{Ann}(u) = (\tau - \xi^2)$. For the slice of G alone the bound is $B = 0$: the heat kernel identifies the heat equation without further computation.

5.6 Variable coefficients: the first limits

With coefficients that depend on x, the ring of operators becomes the Weyl algebra. Two of its features break the constant-coefficient picture.

Proposition 5.10 Restriction does not identify (P4-C1) [proof written, not audited]

For every nonzero smooth u of one variable, the action of the Weyl algebra A_1 on the module $A_1 \cdot u$ is faithful. Yet for $u = e^x$ and the class $\mathcal{L} = \mathrm{span}\{1, \partial\}$, the operator $\partial - 1$ kills u. So "the restriction is injective" does not detect identification. The correct object is the evaluation map $L \mapsto Lu$, whose kernel is $\mathrm{Ann}(u) \cap \mathcal{L}$.

Dim. P4-C1, Catalogue p. 108

Theorem 5.11 Saturated criterion (OP-ALG-T4) [proof written, not audited]

Let P be monic in ∂_t of degree m, with coefficients in a ring $\mathfrak{A}$ of operators in x, and $Pu = 0$. Then u identifies P if and only if the t-jet $u, \partial_t u, \ldots, \partial_t^{m-1} u$ is free over $\mathfrak{A}$: no nontrivial relation $\sum_{j<m} b_j \partial_t^j u = 0$ with $b_j \in \mathfrak{A}$.

Dim. OP-ALG-T4, Catalogue p. 110

In the commutative case this is Proposition 3.4.3. With variable coefficients the reduction of the test to the finitely many quotients P/P_i fails: an explicit counterexample with $\theta = x\partial_x$ exists (OP-ALG-C1) [proof written, not audited]. In the full Weyl algebra A_2, a holonomic solution never identifies its equation (OP-ALG-P5) [proof written, not audited].

5.7 Summary

Data	What can be done	Meaning for the algorithm	Status
$u \in \mathrm{EP}$	jet of order $2d-1$ determines $\mathrm{Ann}(u)$; Bézout threshold for modes	a finite computation with a guarantee	[proved]
$u \notin \mathrm{EP}$, finite data	no finite certificate	output "not refuted", never "identified"	[proved] / [from cited source]
black box	"$\in$ EP?" undecidable	extra hypothesis needed	[proved]
bound B given	decidable, test degree $B + \deg P - 1$	stops with a certain verdict	[proved]
holonomic, algebraic	decidable	exact symbolic route	[proved mod. (AC)]
variable coefficients	restriction never identifies; jet freeness criterion	restrict to a finite class	[proof written, not audited]

Table 5.1: When the identification test can be carried out, and what each case means for Step 3 of the algorithm.

Takeaway. Identification is an infinite statement about u. It cannot be verified from finitely many numbers, and no black box can decide it. One extra integer, a bound on how much of u is exponential-polynomial, turns it into a finite computation.

CHAPTER 6

Decomposition into components

In this chapter

Starting from	the operator is known and factored: $P = c\prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime. For the main theorem Ω is open, connected and simply connected (more precisely $H^1_{dR}(\Omega) = 0$).
Why	once P is known, we want to split u into the parts generated by each piece of P, so that the pieces can be studied separately.
Role	second link of the chain. The split exists for *factors* of the symbol, with a finite and exactly known ambiguity. It fails for additive *terms*, and that failure is what Chapter 8 measures.

6.1 Which pieces? Terms versus factors

An operator can be broken up in two different ways. Take the heat operator

$$L = \partial_t - D\partial_x^2.$$

Its *terms* are $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$: they are added. Its symbol $\tau - D\xi^2$ is irreducible, so it has only one *factor*. For the wave operator the situation is the opposite: $\tau^2 - \xi^2$ has two terms and also two factors, $\tau - \xi$ and $\tau + \xi$. The first question is which kind of piece gives a decomposition of the solution.

6.1.1 Terms do not give components

Proposition 6.1 Splitting by terms fails (R2-T3) [proved]

Let $L = L_1 + L_2$ with constant coefficients. If $u = a + b$ with $L_1 a = 0$ and $L_2 b = 0$ solves $Lu = 0$, then u is also killed by $L_1 L_2$:

$$\ker L \cap (\ker L_1 + \ker L_2) \;\subseteq\; \ker(L,\ L_1 L_2).$$

The common zero set of L and $L_1 L_2$ is $V(L_1) \cap V(L_2)$. When this set is finite, only a finite-dimensional space of exponential-polynomial solutions can be split by terms.

Dim. R2-T3, Catalogue p. 19

Intuition

The two operators commute, so

$$L_1 L_2\, u = L_2(L_1 a) + L_1(L_2 b) = 0.$$

A solution split by terms satisfies one extra equation. A generic solution does not.

Example 6.2 The heat equation cannot be split by terms

Take $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$. The kernels are

$$\ker L_1 = \{f(x)\}, \qquad \ker L_2 = \{a(t) + b(t)\, x\}.$$

Every element $f(x) + a(t) + b(t)x$ of the sum is killed by $\partial_t \partial_x^2$. A heat solution is not. For $u = e^{-D\kappa^2 t} \cos \kappa x$:

$$\partial_x^2 u = -\kappa^2 u, \qquad \partial_t \partial_x^2 u = D\kappa^4 u \neq 0.$$

So a generic heat solution is *not* a sum of a solution of $u_t = 0$ and a solution of $u_{xx} = 0$.

The exceptions are few. The function $u = \frac{1}{2}x^2 + Dt$ solves the heat equation and lies in $\ker L_1 + \ker L_2$, but it is a polynomial: the exceptional space is finite-dimensional.

6.1.2 Factors do give components

Example 6.3 The wave equation and d'Alembert

Let $c > 0$ and

$$P = \tau^2 - c^2\xi^2 = (\tau - c\xi)(\tau + c\xi) = Q_1 Q_2.$$

A direct check gives

$$(\partial_t - c\partial_x)\, f(x + ct) = 0, \qquad (\partial_t + c\partial_x)\, g(x - ct) = 0.$$

So $\ker Q_1 = \{f(x+ct)\}$ (profiles moving left) and $\ker Q_2 = \{g(x-ct)\}$ (profiles moving right). D'Alembert's formula says that every solution is a sum of the two:

$$\ker P = \ker Q_1 + \ker Q_2.$$

The pieces are unique up to a constant: $f \mapsto f + k$, $g \mapsto g - k$. This ambiguity has dimension 1, and it equals $\dim \mathcal{R}/(Q_1, Q_2)$:

$$Q_1 + Q_2 = 2\tau, \qquad Q_2 - Q_1 = 2c\xi \quad \Longrightarrow \quad (Q_1, Q_2) = (\tau, \xi), \qquad \mathcal{R}/(\tau, \xi) \cong \mathbb{C}.$$

6.2 Why factors work: the Chinese remainder theorem

One variable. If two polynomials Q_1, Q_2 in one variable are coprime, there are polynomials a, b with

$$a\, Q_1 + b\, Q_2 = 1.$$

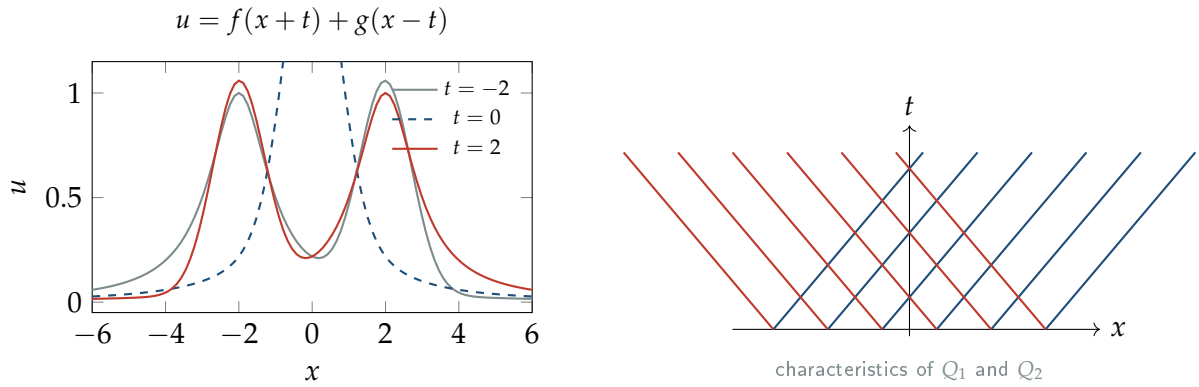


Figure 6.1: Left: the d'Alembert solution with $f(s) = 1/(1+s^2)$ moving left and $g(s) = e^{-s^2}$ moving right ($c = 1$). Right: the two families of characteristic lines, one per factor. Each factor of the symbol carries one rigidly translating component.

Then every u with $Q_1 Q_2 u = 0$ splits explicitly as

$$u = \underbrace{b\, Q_2\, u}_{\in \ker Q_1} + \underbrace{a\, Q_1\, u}_{\in \ker Q_2},$$

and the sum of the two kernels is direct.

Two variables. "Coprime" now means only that the common zeros are finitely many (Bézout), not that $(Q_1, Q_2) = \mathcal{R}$. The finite quotient $\mathcal{R}/(Q_1, Q_2)$ measures the failure of the one-variable picture.

Lemma 6.4 Finite quotient (C12-L3) [proved]

Let $f, g, h \in \mathcal{R} \setminus \{0\}$.

(a) If f, g are coprime, the ideal (f, g) contains a nonzero polynomial in ξ alone and a nonzero polynomial in τ alone. Hence $\dim \mathcal{R}/(f, g) < \infty$.

(b) If f is coprime to g and to h, then

$$\dim \mathcal{R}/(f, gh) = \dim \mathcal{R}/(f, g) + \dim \mathcal{R}/(f, h).$$

(c) $(f, g) = \mathcal{R}$ if and only if $V(f) \cap V(g) = \emptyset$.

Dim. C12-L3, Catalogue p. 21

The geometric meaning of $\dim \mathcal{R}/(Q_i, Q_j)$ is the number of crossing points of the curves $V(Q_i)$ and $V(Q_j)$, counted with intersection multiplicity.

6.3 The decomposition theorem

Theorem 6.5 Factor decomposition (C12-T1, C12-T1′) [proved]

Let $\Omega \subset \mathbb{R}^2$ be open, connected, with $H^1_{dR}(\Omega) = 0$ (for example simply connected: a rectangle, a convex set, a slit disc). Let $\mathcal{A}$ be $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $P = c\prod_{j=1}^{s} Q_j$ with pairwise coprime Q_j. Then:

(i) every solution splits:

$$\ker_{\mathcal{A}} P = \ker_{\mathcal{A}} Q_1 + \cdots + \ker_{\mathcal{A}} Q_s;$$

(ii) the ambiguity (the s-tuples of components with sum 0) has dimension exactly

$$\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j),$$

and consists of exponential-polynomials;

(iii) the splitting is unique if and only if $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

Dim. C12-T1p, Catalogue p. 23

Intuition

At a crossing point z of two factor curves, the exponential $e^{\langle z,(t,x)\rangle}$ belongs to both kernels. It can be moved freely from one component to the other. The number of such free moves, counted with multiplicity, is $\delta(P)$.

Algebraically the proof uses the exact sequence

$$0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$$

and shows that applying $\mathrm{Hom}(\,\cdot\,, \mathcal{A})$ keeps it exact. The only possible obstruction lives on the finite-dimensional module $\mathcal{R}/(Q_1, Q_2)$, and it vanishes when Ω has no holes (Lemma 6.3).

Example 6.6 The ambiguity on four operators

Operator	Factors	$\delta(P)$	Ambiguity
wave $\tau^2 - \xi^2$	$\tau - \xi$, $\tau + \xi$	1	constants
beam $\tau^2 + \xi^4$	$\tau - i\xi^2$, $\tau + i\xi^2$	2	span$\{1, x\}$
heat–transport	$\tau - \xi^2$, $\tau - \xi$	2	span$\{1,\ e^{t+x}\}$
parallel transports	$\tau - \xi$, $\tau - \xi - 1$	0	none (unique)

For heat–transport the curves $\tau = \xi^2$ and $\tau = \xi$ meet at $(0,0)$ and $(1,1)$, so the shared exponentials are 1 and e^{t+x}. For the two parallel transports the lines $\tau = \xi$ and $\tau = \xi + 1$ never meet, and the splitting is unique.

The key analytic tool is a version of the Poincaré lemma "closed forms are exact" with an exponential weight.

Lemma 6.7 Twisted Poincaré lemma (C12-L1′, L2′) [proved]

Let Ω be connected with $H^1_{dR}(\Omega) = 0$, and $z = (z_1, z_2) \in \mathbb{C}^2$. Write $D_t = \partial_t - z_1$ and $D_x = \partial_x - z_2$. If $\varphi, \psi \in \mathcal{A}$ satisfy the compatibility condition

$$D_x \varphi = D_t \psi,$$

then there is $w \in \mathcal{A}$ with $D_t w = \varphi$ and $D_x w = \psi$. Consequently, for every finite-dimensional $\mathcal{R}$-module M, the obstruction group $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A})$ vanishes.

Dim. C12-L1p, Catalogue p. 22

Multiplying by $e^{-z_1 t - z_2 x}$ turns the twisted derivatives into ordinary ones, and the lemma becomes the classical statement that a closed 1-form on a simply connected region is exact.

6.4 Where the decomposition fails: holes

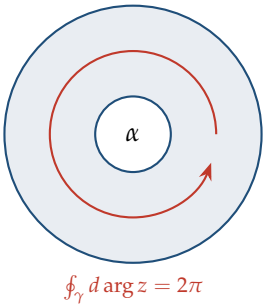
Proposition 6.8 Holes break it (C12-P10) [proved]

Let Ω contain a closed curve γ that winds around a point $\alpha \notin \Omega$. For the Laplacian

$$P = \tau^2 + \xi^2 = (\tau - i\xi)(\tau + i\xi),$$

the harmonic function $u = \log|(t, x) - \alpha|^2$ is *not* a sum of a solution of $\partial_t - i\partial_x$ and a solution of $\partial_t + i\partial_x$.

Dim. C12-P10, Catalogue p. 25



$u = \log|z - \alpha|^2$ is harmonic on the annulus, but $u \neq h(z) + \overline{g(z)}$ with h, g single-valued holomorphic: the logarithm has monodromy.

Figure 6.2: The Laplacian factors into two Cauchy–Riemann operators, but on an annulus the factor decomposition fails. The obstruction is topological: $H^1_{dR}(\Omega) \neq 0$.

For a fixed P the exact condition is cohomological.

Theorem 6.9 Cohomological criterion (OPF-T4) [proof written, not audited]

Let Ω be connected, with first Betti number $b_1(\Omega)$. Let $\mathcal{N}$ be the space of s-tuples of global exponential-polynomial solutions with sum zero (dimension $\delta(P)$). The cokernel of the sum map

$$\Sigma_\Omega : \bigoplus_j \ker_{\mathcal{A}(\Omega)} Q_j \to \ker_{\mathcal{A}(\Omega)} P$$

is the kernel of an explicit map $H^1(\Omega;\mathbb{C}) \otimes \mathcal{N} \to \bigoplus_j H^1(\Omega;\mathcal{K}_j)$. So the decomposition holds on Ω iff that map is injective, and

$$\dim \operatorname{coker} \Sigma_\Omega \le \delta(P)\, b_1(\Omega).$$

Dim. OPF-T4, Catalogue p. 118

Example 6.10 Unique splittings survive holes

If $\delta(P) = 0$ (disjoint factor curves), the bound gives $\operatorname{coker} \Sigma_\Omega = 0$ on *every* connected region, with or without holes. The two parallel transports $\tau - \xi$ and $\tau - \xi - 1$ split on an annulus as well.

6.5 Explicit projectors

For evolution factors the components are given by a formula. Let

$$P = \prod_{j=1}^{N} (\tau - a_j(\xi))$$

with distinct polynomials a_j. A component is an eigenfunction of ∂_t:

$$\partial_t u_j = a_j(\partial_x)\, u_j.$$

Apply $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$. Every u_l with $l \neq j$ is killed, because the product contains its own factor. The component u_j is multiplied by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Dividing gives the Lagrange formula

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\, u. \tag{6.1}$$

In Fourier variables the right-hand side is a multiplier that is rational in ξ, with poles where two a_k coincide. This is Eq. (3.2) with the companion matrix replaced by ∂_t.

Example 6.11 Two transports

Let $a_1(\xi) = \xi$ and $a_2(\xi) = \xi + 1$. The formula gives

$$u_1 = \frac{\partial_t - \partial_x - 1}{-1}\, u, \qquad u_2 = \frac{\partial_t - \partial_x}{1}\, u.$$

These are differential operators, with no poles, because $a_1 - a_2 = -1$ never vanishes: the two lines are disjoint and the splitting is unique.

Proposition 6.12 Projectors and their poles (II-T3, II-T4) [proved]

The components are obtained by *polynomial* operators if and only if the factor curves are pairwise disjoint, $(Q_i, Q_j) = \mathcal{R}$. Only factors, not terms, admit projectors that act on one value of u at a time.

Dim. II-T3, Catalogue p. 101

Takeaway. A "component" of u belongs to a *factor* of the symbol, not to a term of the operator. On a region without holes the factor split always exists, and the overlap between components is finite-dimensional and measured exactly by $\delta(P)$. The split by terms fails except on a finite-dimensional space.

CHAPTER 7

Regularity of the components

In this chapter

Starting from	the factor decomposition of Theorem 6.3; Ω bounded; u in a Sobolev space $H^m(\Omega)$ (square-integrable derivatives up to order m).
Why	the decomposition is algebraic. To use the components numerically we need them to be as regular as u itself, and to depend continuously on u.
Role	completes the answer to Q2. It gives the exact condition (a conditioning number $\gamma_N > 0$), shows that it can fail, and that the geometry of Ω against the characteristics decides it. Step 5 of the algorithm checks it.

7.1 The pieces can be rougher than the whole

Counterexample 7.1 Loss of a derivative (C12-C1)

The statement "if $u \in H^m(\Omega)$ then every component $u_j \in H^m(\Omega)$" is false. Let $\Omega = (0,1)^2$ and

$$f = \tau - \xi, \qquad g = \tau - \xi - 1.$$

For every m there is a smooth $u \in H^m(\Omega)$ with $f(\partial)g(\partial)u = 0$ whose unique components $u_1 \in \ker f$, $u_2 \in \ker g$ are *not* in H^m. Exactly one derivative is lost: $u_2 = f(\partial)u \in H^{m-1}$.

Dim. C12-C1, Catalogue p. 26

Intuition

In the coordinates $s = x + t$ along the characteristics, both factors are first-order ODEs along the same lines, with solutions 1 and e^t. On a long chord these two functions are clearly different. Near a corner of the square the chords become arbitrarily short, and on a short chord 1 and e^t are almost collinear. Separating a function into these two shapes then amplifies it by a factor that blows up as the chord shrinks (Figure 7.1).

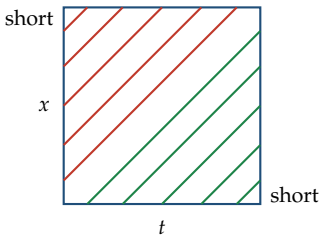


Figure 7.1: The unit square and the characteristic chords of $\partial_t + \partial_x$. Chords near the corners $(0,1)$ and $(1,0)$ are arbitrarily short. On a short chord the two shapes of the components are nearly collinear, and the splitting is ill-conditioned.

7.2 Two components: the chord criterion

For two transports in the same direction, the geometry decides everything.

Theorem 7.2 Chord criterion (C12-T3) [proved]

Let Ω be bounded, open and convex, $c \in \mathbb{R}$, $d_1 \neq d_2$, and

$$f = \partial_t + c\,\partial_x - d_1, \qquad g = \partial_t + c\,\partial_x - d_2.$$

Let $\ell(s)$ be the length of the chord of Ω along the characteristic line $x - ct = s$, and $\mathcal{H}_i = \ker_{L^2} f_i$. The following are equivalent:

(a) the chords do not shrink: $\inf_s \ell(s) > 0$;

(b) every $u \in L^2$ with $fgu = 0$ has components in L^2, with

$$\|u_1\|^2 + \|u_2\|^2 \leq C\,\|u\|^2;$$

(c) the sum $\mathcal{H}_1 + \mathcal{H}_2$ is closed in L^2;

(d) the Friedrichs cosine of the two spaces is smaller than one, $c_F(\mathcal{H}_1, \mathcal{H}_2) < 1$.

Dim. C12-T3, Catalogue p. 27

Example 7.3 Rectangles

On a rectangle $I \times J$ with $c \neq 0$, the chords in direction $(1, c)$ shrink at two corners. So the global L^2 decomposition always fails ($c_F = 1$). For $c = 0$ the chords are vertical, all of length $|I|$, and the decomposition holds.

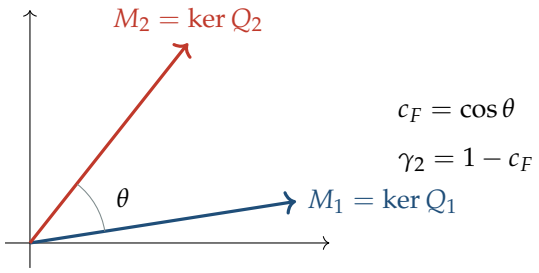


Figure 7.2: The Friedrichs angle θ between two component spaces. A larger angle means a better-conditioned recovery of the pieces from their sum. As $\theta \to 0$ the recovery becomes unstable.

7.3 Any number of components

For N components the right quantity is the conditioning of the sum map

$$S : \bigoplus_{j=1}^{N} \ker_{H^m} Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

Definition 7.4 Conditioning number γ_N

Let $(\ker S)^\perp$ be the tuples orthogonal to the ambiguity of Theorem 6.3. Then

$$\gamma_N = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|_{H^m}^2}{\|v\|^2}, \qquad \|v\|^2 = \sum_j \|u_j\|_{H^m}^2.$$

Equivalently, γ_N is the bottom of the spectrum of S^*S on $(\ker S)^\perp$: the square of the smallest nonzero singular value of S.

Intuition

γ_N is the worst ratio between the size of the sum and the size of the pieces. If $\gamma_N > 0$, the pieces cannot nearly cancel each other:

$$\|v\| \le \gamma_N^{-1/2} \, \|Sv\|,$$

and recovering them from u is stable. If $\gamma_N = 0$, there are large pieces with a tiny sum, and the recovery is unstable. It is exactly the smallest singular value of a least-squares problem.

Theorem 7.5 Regularity criterion (C12-P7) [proved]

(a) Every $u \in \ker_{H^m} P$ has components in H^m if and only if the sum $\sum_j \ker_{H^m} Q_j$ is *closed* and *dense* in $\ker_{H^m} P$.

(b) The sum is closed if and only if $\gamma_N > 0$. Then the minimal-energy split satisfies, optimally,

$$\sum_j \|u_j\|_{H^m}^2 \le \gamma_N^{-1} \, \|u\|_{H^m}^2.$$

(c) For $N = 2$, $\gamma_2 = 1 - c_F$, the Friedrichs cosine of the two spaces after removing their intersection.

(d) For $N \geq 3$ the pairwise cosines do not decide. There are three closed subspaces of a Hilbert space with trivial pairwise intersections and closed pairwise sums ($c_F \leq \frac{1}{2}$), whose triple sum is not closed.

Dim. C12-P7, Catalogue p. 29

The condition has two parts. Closedness is the conditioning above. Density is a separate question, and on convex regions it is always true.

Theorem 7.6 Density on convex regions (OPF5B-T1) [proved mod. Hörmander, Stein]

Let $\Omega \subset \mathbb{R}^2$ be bounded and convex. For every operator P, every coprime factorisation and every $m \geq 0$, the sum $\sum_j \ker_{H^m} Q_j$ is dense in $\ker_{H^m} P$.

Dim. OPF5B-T1, Catalogue p. 30

So on a convex region only $\gamma_N > 0$ remains to be checked.

Added in the complete edition (Phase 10)

Convexity cannot simply be dropped: on a region with a hole the sum of kernels is *not* dense in general (Proposition 16.1, the Laplacian on an annulus), and the extension of this theorem to non-convex regions is false. For simply connected Lipschitz regions density was proved in Phase 10 modulo two cited facts (Chapter 16).

Example 7.7 Cases where the H^m decomposition holds (C12-P4) [proved]

$P = \tau\xi$ on a rectangle. The components are a function of t and a function of x. After removing the constants their H^m inner product vanishes, so $c_F = 0$ and $\gamma_2 = 1$.

The wave equation on a square. In characteristic coordinates $s = x + t$, $r = x - t$ the square becomes a rhombus. It can be covered by five pieces on each of which one component is controlled by u.

Heat and transport on a rectangle (OPR2-T1) [proof written, not audited]. For $P = (\tau - \xi^2)(\tau + c\xi)$ the sum is dense in every H^m, so the decomposition holds exactly when $c_F < 1$.

The same operator can behave well on one region and badly on another: regularity depends on the *geometry of* Ω *against the characteristics*, not on P alone.

Dim. C12-P4, Catalogue p. 117

Proposition 7.8 Real transports, $m = 0$ (OPR2-T2) [proof written, not audited]

Let Ω be convex and bounded, and $Q_j = v_j \cdot \nabla - d_j$ real transports, grouped into classes of parallel directions. Assume that of any two classes at least one has chords bounded below. Then $\sum_j \ker_{L^2} Q_j$ is closed if and only if every class with at least two factors has chords bounded below. In particular, for such transports, closed pairwise sums imply a closed triple sum.

Dim. OPR2-T2, Catalogue p. 121

Three or more components: what is still open

Whether the abstract counterexample of Theorem 7.3(d) can occur for *differential* factors is open in general [open] (C12-Q2). Proposition 7.3 rules it out for real transports in L^2.

Takeaway. Algebraically the split always exists (Theorem 6.3). Analytically the components are as regular as u exactly when the component spaces are separated by a positive angle, $\gamma_N > 0$; density is automatic on convex regions. Everything delicate happens at the boundary of Ω (shrinking chords) or in its topology (holes).

CHAPTER 8

Terms inside a factor: the interaction term

In this chapter	
Starting from	an evolution factor $u_t = Au$ with $A = A_1 + \cdots + A_N$, the *terms* (advection, diffusion, dispersion, …), each a polynomial $a_n(\partial_x)$; flows $S_n(t) = e^{tA_n}$; a bounded rectangle Ω and a base time t_0. The terms commute, except in Section 8.7.
Why	Proposition 6.1.1 shows that splitting a solution by terms is never exact. Here we compute exactly what is left over, the interaction term E_{int}, how much it costs, and whether it can be computed stably.
Role	third link of the chain, answering Q3. The answer is positive in a function class of exponential type (an exact product formula, cost $O(N)$ per value) and negative in Sobolev spaces (no continuity, no stable regularisation).

8.1 From the sum of operators to the interaction term

The solution of $u_t = Au$ is given by the flow (semigroup) of A:

$$u(t) = e^{tA}u_0 = S(t)\,u_0.$$

If term n acted alone, it would move the datum to $S_n(t)u_0 = e^{tA_n}u_0$, a change of $S_n(t)u_0 - u_0$.

The naive guess is that the terms act independently, so that the total change is the sum of the individual changes. It is not: the flow of a sum is not the sum of the flows. For advection ($A_1 = -c\partial_x$, a translation) and diffusion ($A_2 = D\partial_x^2$, a smoothing), the solution is translated *and* smoothed. This joint action is not "only translated" plus "only smoothed".

Definition 8.1 Interaction term

With $S_n(t) = e^{(t-t_0)A_n}$ the flow of term n alone,

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{8.1}$$

The multiplicative variant compares the true flow with the product of the single flows:

$$E_{\text{int}}^{\text{mult}}(t) = e^{(t-t_0)A} - \prod_{n=1}^{N} e^{(t-t_0)A_n}.$$

We write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.

8.2 The function class and the exact formula

Before computing, one must decide on which functions the flows of the single terms are defined. This is not a formality. For the heat equation there are nonzero solutions with zero initial datum (Tychonoff), so "the diffusion flow of u_0" is not unique without a class.

Definition 8.2 The class $\mathfrak{B}$ of exponential type (P1-D1)

For $\rho > 0$, $B_\rho(J)$ is the space of smooth f on the interval J with

$$\|f\|_{B_\rho}^2 = \sum_{j\geq 0} \rho^{-2j}\, \|f^{(j)}\|_{L^2(J)}^2 < \infty.$$

$\mathfrak{B}_\rho(\Omega)$ is the analogous space in (t,x), and $\mathfrak{B} = \bigcup_\rho \mathfrak{B}_\rho$. Every exponential-polynomial belongs to $\mathfrak{B}$, and on B_ρ the derivative is bounded: $\|\partial_x\| \leq \rho$.

On $\mathfrak{B}$ every polynomial in ∂_x is a bounded operator, its exponential is an absolutely convergent series, and two such exponentials commute (P1-L1, P1-L2) [proved]. So the algebra of exponentials of numbers applies to operators.

Theorem 8.3 Closed form of the interaction (P1-T2) [proved]

Let $u \in \mathfrak{B}_\rho(\Omega)$ solve $(\partial_t - a(\partial_x))u = 0$ with $a = \sum_n a_n$, and put

$$\Delta_n = e^{(t-t_0)a_n(\partial_x)} - \mathrm{Id}.$$

Then the single flows are unique in $\mathfrak{B}$, $E_{\mathrm{int}}^{\mathrm{mult}} \equiv 0$, and

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \Big[\prod_{n=1}^{N}(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_{n=1}^{N}\Delta_n\Big]u_0 = \sum_{|S|\geq 2}\prod_{n\in S}\Delta_n\, u_0. \tag{8.2}$$

For a datum $u_0 = \sum_k c_k e^{\zeta_k x}$, each mode is multiplied by the number

$$e_{\geq 2}(z) = \prod_n (1+z_n) - 1 - \sum_n z_n, \qquad z_n = e^{(t-t_0)a_n(\zeta_k)} - 1.$$

Dim. P1-T2, Catalogue p. 34

Intuition

Expand the product $\prod_n(\mathrm{Id}+\Delta_n)$. It gives one monomial per subset S of terms. The empty subset gives Id, the singletons give the individual changes Δ_n, and everything else is interaction. So the interaction is the sum of products of *at least two* increments: it appears only when at least two terms act together.

For two terms the formula is simply

$$E_{\mathrm{int}} = \Delta_1\Delta_2\, u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})\, u_0.$$

In each Fourier mode, the interaction is the product of the two single-term responses.

Example 8.4 Advection–diffusion with a Gaussian datum

Take $u_t = -cu_x + Du_{xx}$ and $u_0 = g(x;\sigma^2)$, where

$$g(x;v) = \frac{e^{-x^2/2v}}{\sqrt{2\pi v}}.$$

The two flows are a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x-ct;\sigma^2), \qquad S_2 u_0 = g(x;s^2), \qquad u = g(x-ct;s^2),$$

with $s^2 = \sigma^2 + 2Dt$. The interaction term is

$$E_{\text{int}} = g(x-ct;s^2) - g(x-ct;\sigma^2) - g(x;s^2) + g(x;\sigma^2).$$

Figure 8.1(a) shows that it is comparable in size to u itself: it is not a small correction. In Fourier variables

$$\frac{|\widehat{E_{\text{int}}}(\kappa)|}{|\hat{u}_0(\kappa)|} = \left|e^{-ic\kappa t} - 1\right| \cdot \left|e^{-D\kappa^2 t} - 1\right| \le 2.$$

It vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$, where the translation returns to the identity.

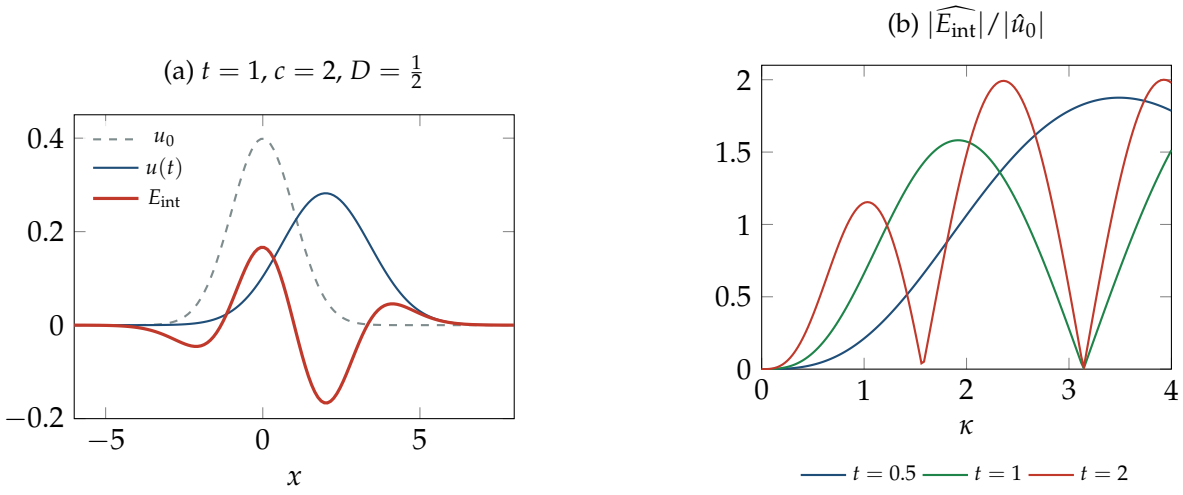


Figure 8.1: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$). (a) The datum, the solution and the interaction term. (b) The Fourier ratio of the interaction to the datum, for three times.

Example 8.5 A stationary state is pure interaction

For advection–diffusion, a stationary solution satisfies $-cu_x + Du_{xx} = 0$, for example $u = e^{cx/D}$. The full flow leaves it unchanged, so $u(t) - u_0 = 0$. The single flows do change it, and the interaction term exactly cancels their changes. In this mode

$$E_{\text{int}} = -4\sinh^2\left(\frac{\theta}{2}\right) u, \qquad \theta = \frac{c^2(t-t_0)}{D}.$$

Equilibrium is the case where all the dynamics is interaction.

8.3 Cost: $O(N)$ versus 2^N

Expanding (8.2) produces one monomial per subset with at least two elements: $2^N - N - 1$ cross terms (Figure 8.2). Writing the *complete table* of cross terms is therefore exponential. Computing one *value* is cheap, because the product is never expanded.

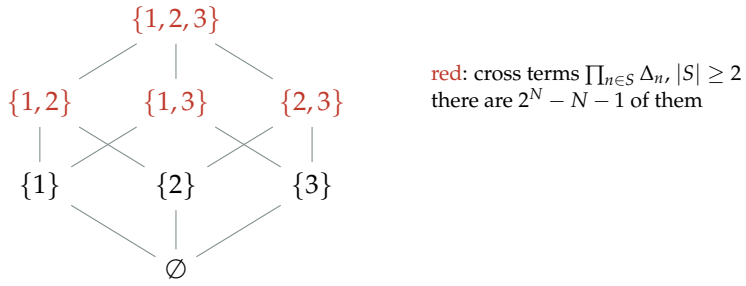


Figure 8.2: The subset lattice for $N = 3$ terms. Each node with at least two elements is one cross term of the interaction.

Theorem 8.6 Cost of the interaction (P1-T3) [proved]

Let the datum be a sum of M exponentials with polynomial prefactors of degree at most L, and let $d_n = \deg a_n$.

(a) One value of E_{int} costs

$$O\Big(M\big[N(L+1)^2 + (L+1)\textstyle\sum_n (d_n+1)\big]\Big);$$

for simple exponentials ($L = 0$) this is $O\big(M(N + \sum_n d_n)\big)$: linear in the number of terms.

(b) All aggregates by order, $E^{(k)} = \sum_{|S|=k}\prod_{n\in S}\Delta_n u_0$ for $2 \leq k \leq N$, cost $O(MN^2(L+1)^2)$ plus the same term in d_n.

(c) The full table of cross terms has $M(2^N - N - 1)$ entries and costs $\Theta\big(M\,2^N(L+1)^2\big)$.

Dim. P1-T3, Catalogue p. 35

Example 8.7 Counting operations for one mode

For one simple mode $e^{\zeta x}$ and N terms, compute the N numbers $z_n = e^{(t-t_0)a_n(\zeta)} - 1$, then the product $\prod_n (1+z_n)$ with $N-1$ multiplications, then subtract $1 + \sum_n z_n$. That is about $3N$ operations. Listing the $2^N - N - 1$ products $\prod_{n\in S} z_n$ one by one costs 2^N times more for large N: already for $N = 20$, about a million.

8.4 The flows are not determined without a class

Proposition 8.8 Non-uniqueness of single flows (P1-P1(a)–(d)) [proved]

On a bounded rectangle, without a function class:

(a) for the transport term $a_n = \xi$, the flow of a datum is any smooth extension of it: infinitely many choices;

(b) for the diffusion term $a_n = \xi^2$, there are nonzero heat solutions with zero datum (Tychonoff) [from cited source];

(c) so E_{int} is not a function of u alone; inside $\mathfrak{B}$ it is unique.

Dim. P1-P1, Catalogue p. 127

8.5 Stability: the interaction cannot be computed stably

Theorem 8.9 No continuous extension to Sobolev spaces (P1-P1(e)) [false]

Let $Q = \tau + c\xi - D\xi^2$ with $c, D > 0$ (advection–diffusion), terms $a_1 = -c\xi$ and $a_2 = D\xi^2$, and any base time t_0. For every m,

$$\sup \left\{ \frac{\|E_{\text{int}}\|_{L^2(\Omega)}}{\|u\|_{H^m(\Omega)}} : u \in \text{EP},\ Qu = 0,\ u \neq 0 \right\} = \infty.$$

So $u \mapsto E_{\text{int}}$ does not extend continuously from exponential-type data to H^m, although exponential-type data are dense. The same holds for every evolution with at least two non-proportional terms; for proportional terms the interaction is continuous (OPA-P3) [proved].

Dim. P1-P1e, Catalogue p. 37

Intuition

On a bounded interval, a real exponential $e^{\zeta x}$ with large ζ is a legitimate datum. The diffusion flow alone amplifies it by $e^{D\zeta^2 t}$. The true solution is only changed by the factor $e^{(D\zeta^2 - c\zeta)t}$ times the datum norm. The interaction term must cancel the gap, and the gap grows faster than any Sobolev norm of the datum.

This instability has three consequences for any algorithm that wants to be stable.

Theorem 8.10 No regularisation removes the interaction (P2-T1, P2-T2, P2-T4) [proved]

(a) **Any penalty leaves the same residual.** For any regulariser (an ℓ^1 sparsity penalty, ℓ^2, total variation, ...) the residual of the best fit converges to the distance of u from the closed span of the per-term pieces.

(b) **Inside an evolution factor that residual is large.** If no single term exhausts the symbol, the part of u outside the span of the per-term kernels satisfies

$$\|E^{\perp}(u)\|_{H^m} \;\geq\; \frac{\|g(\partial_x)u\|_{H^{m-k}}}{C}, \qquad g = \prod_n (a - a_n).$$

Only a finite-dimensional space of exponential-polynomials has a per-term representation. For advection–diffusion it is

$$\operatorname{span}\{1,\ x - ct,\ (x - ct)^2 + 2Dt\}.$$

(c) **No uniformly Lipschitz regularisation exists.** Any family of Lipschitz maps converging pointwise to an unbounded map has Lipschitz constants tending to infinity. Every exact selection of components has Lipschitz constant at least $\gamma_N^{-1/2}$.

Dim. P2-T2, Catalogue p. 38

Added in the complete edition (Phase 10)

The instability above is proved for advection–diffusion. The general statement "for every operator with at least two non-proportional terms" (P1-Q6) is false as written: with constant terms, for example reaction–diffusion, the interaction is continuous from H^1. The correct dichotomy is by the number of terms of degree ≥ 1 (Theorem 16.1, Chapter 16).

The one continuous split found

A *balance* decomposition of u writes each term's contribution as the time integral of that term applied to the true solution:

$$u^{(i)}(t) = \int_{t_0}^{t} a_i(\partial_x)\, u(s)\, \mathrm{d}s.$$

It is exact ($u - u_0 = \sum_i u^{(i)}$) and continuous in H^m (P3-T6) [proof written, not audited]. But it needs an external principle to decide that this is "the" contribution of term i: it uses the full solution, not the flow of the term alone.

8.6 Time-dependent perturbation theory

The product formula needs commuting terms. With non-commuting terms, or with boundary conditions, there is no closed formula, but E_{int} is still defined by (8.1). Perturbation theory

expands the flow in powers of the interaction. Split $A = A_1 + B$, where A_1 is the principal term and B the rest. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k\ge 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s. \end{aligned} \tag{8.3}$$

$S^{(k)}$ collects the histories in which B acts k times, at intermediate times, with the flow of A_1 in between. This is the interaction picture of quantum mechanics [10].

Theorem 8.11 Dyson–Phillips converges for every time (P3-T1) [proved]

Let $A_1 = a(\partial_x)$ with symbol of degree q, and $B = \sum_{k\le r} b_k(x)\partial_x^k$ with bounded coefficients and $r < q$. The symbol condition

$$\operatorname{Re} a(i\kappa) \le \omega - \delta|\kappa|^q \qquad (\delta > 0)$$

holds if and only if q is even and the leading coefficient has negative real part (a dissipative operator of even order). Under it, the Dyson–Phillips series converges in operator norm, uniformly on every bounded time interval, and its sum is the semigroup generated by $A_1 + B$. The same holds with Dirichlet conditions on an interval.

Dim. P3-T1, Catalogue p. 40

So E_{int} is well defined and can be approximated by truncating the series even without a closed formula.

Proposition 8.12 Interaction created by a boundary (P1-P6, P3-T3, OPA-T2, OPF5A-T5) [proved]

On an interval with Dirichlet conditions, advection and diffusion no longer commute: the boundary creates an $E_{\mathrm{int}}^{\mathrm{mult}}$ already at first order (P1-P6). For small times its size is governed by how the datum touches the boundary. If k is the order of contact (the first nonvanishing derivative of u_0 at the endpoints), then

$$\|E_{\mathrm{int}}^{\mathrm{mult}}(t)\| = \Theta\big(t^{k/2+3/4}\big) \qquad (t \to 0).$$

For compatible data the leading constant is explicit and proportional to

$$|c|\,\frac{\tilde N_k}{k!}\,D^{k/2-1/4}\sqrt{u_0^{(k)}(0)^2 + u_0^{(k)}(L)^2}.$$

Dim. OPF5A-T5, Catalogue p. 131

The smoother the datum is against the boundary, the faster the boundary interaction becomes negligible as $t \to 0$.

8.7 Non-commuting terms

If two terms do not commute, the flow of their sum is not the product of the flows. By the Baker–Campbell–Hausdorff formula,

$$e^{tA_1}e^{tA_2} = \exp\left(t(A_1+A_2) + \frac{t^2}{2}[A_1,A_2] + O(t^3)\right).$$

The commutators generate a Lie algebra, and its type decides what can be done.

Theorem 8.13 Lie-algebra classification (P1-T4, P1-P5, P1-T5) [proved]

(a) **Finite and solvable.** An exact formula exists, with $d = \dim \mathfrak{g}$ entire scalar functions (Wei–Norman). Example: $\partial_x^2 + x$, and the Ornstein–Uhlenbeck operator.

(b) **Finite, not solvable** (for example $\mathfrak{sl}_2$). A closed form exists locally in time; it can fail globally.

(c) **Infinite adjoint orbit.** If X and all the iterated commutators $\mathrm{ad}_X^k Y$ are linearly independent (for example $X = x^a\partial_x$, $Y = \partial_x$ with $a \geq 3$), then no finite product of flows of the two terms, with times analytic in t and in the coupling ε and with the flows of Y switched off at $\varepsilon = 0$, reproduces $e^{t(X+\varepsilon Y)}$. Only series expansions exist.

Dim. P1-T5, Catalogue p. 42

Example 8.14 Which potentials allow Wei–Norman

For $i\partial_t = -\partial_x^2 + V(x)$, the operators ∂_x^2 and x^2 generate the finite Lie algebra $\mathfrak{sl}_2$: the harmonic oscillator has a closed-form evolution. With $V = x^3$, repeated commutators produce polynomials of ever higher degree and the algebra is infinite-dimensional. But the hypothesis of part (c) fails there, so whether a finite product of flows is impossible for ∂_x^2 and x^k with $k \geq 3$ is still [open] (P1-Q3). Part (c) applies to the transport with polynomial speed $u_t = (x^a + \varepsilon)u_x$, $a \geq 3$, which nevertheless has an explicit solution by characteristics: the theorem excludes only a factorisation into flows of the terms.

Other ways commutativity is lost

- **Repeated factors.** For $(\tau - a)^e$ the "multiplicity term" is a nilpotent shift. It commutes with the other terms, its flow is polynomial in time (secular growth), and $E_{\mathrm{int}}^{\mathrm{mult}} = 0$ (OPF-T2) [proof written, not audited].
- **Second-order evolutions** (wave, Klein–Gordon). The companion blocks do not commute, so the first-order calculus fails; a Gauss factorisation works mode by mode (II-T7, P1-P4) [proved].

Takeaway. For commuting terms the interaction is explicit and cheap to evaluate, but it can be neither removed nor computed stably in Sobolev spaces. That instability is not a defect of the formula: it is a property of the question. For non-commuting terms even the explicit form is lost, except when the terms generate a finite solvable Lie algebra.

CHAPTER 9

Nonlinear equations

In this chapter

Starting from	a scalar evolution in one space variable, $u_t = \sum_n T_n(u)$, each term T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; a smooth datum u_0; the exact solution is known. Benchmarks: viscous Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS as integrable controls.
Why	for a nonlinear equation the single-term flows do not commute and superposition fails. "Component per term" needs a new definition, and identification needs a new frame (a dictionary). We give both and compute them where everything is explicit.
Role	fourth link of the chain. It reuses the earlier chapters through the jet expansion and the annihilator idea. Its results are weaker and relative, but sharp on the benchmark models.

9.1 The interaction term for nonlinear flows

Let S_n^t be the flow of $u_t = T_n(u)$ alone, which is now a nonlinear map. Definition (8.1) still makes sense:

$$E_{\text{int}}(t) = u(t) - u_0 - \sum_n \big(S_n^t u_0 - u_0\big).$$

There is no product formula, because the flows do not commute. But the Taylor expansion in time is explicit.

Theorem 9.1 Jet of the nonlinear interaction (S4T-T1) [proved]

For smooth data,

$$E_{\text{int}}(t) = \frac{t^2}{2} \sum_{n \neq m} T_n'[u_0]\big(T_m(u_0)\big) + O(t^3),$$

where $T_n'[u_0](v)$ is the derivative (Fréchet derivative) of T_n at u_0 in the direction v.

Dim. S4T-T1, Catalogue p. 45

Intuition

To first order in t every flow moves u_0 by $t\,T_n(u_0)$, and these moves add exactly: there is no interaction at order t. At order t^2 the full flow sees each term react to the motion caused by the others. That reaction is the derivative of T_n in the direction of $T_m(u_0)$.

Example 9.2 Burgers

For $u_t = -uu_x + \nu u_{xx}$ take $T_1(u) = -uu_x$ and $T_2(u) = \nu u_{xx}$. The derivatives are

$$T_1'[u](v) = -(vu_x + uv_x), \qquad T_2'[u](v) = \nu v_{xx}.$$

Substituting the other term:

$$T_1'[u_0]\big(T_2(u_0)\big) = -\nu\left(u_0''u_0' + u_0u_0'''\right),$$
$$T_2'[u_0]\big(T_1(u_0)\big) = -\nu\left(3u_0'u_0'' + u_0u_0'''\right).$$

Half the sum, times t^2, gives

$$E_{\text{int}}(t) = -\nu\left(u_0u_0''' + 2u_0'u_0''\right)t^2 + O(t^3).$$

9.2 Before, at and after the shock

The flow of the single term $-uu_x$ (inviscid Burgers) forms shocks: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t\sin\xi,$$

and since

$$\frac{\partial x}{\partial \xi} = 1 + t\cos\xi$$

first vanishes at $\xi = \pi$, the breaking time is $t^* = 1$ (Figure 9.1).

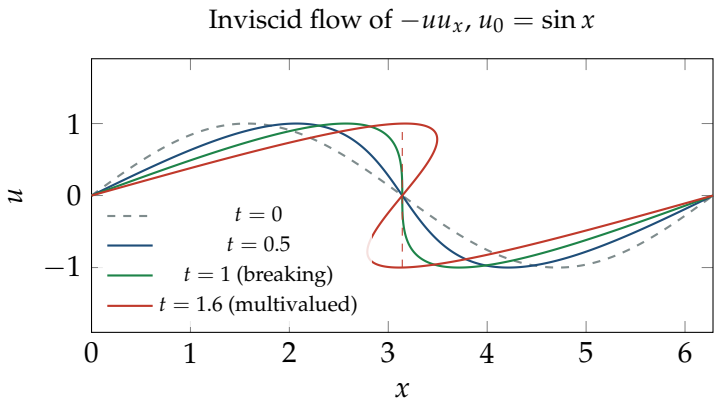


Figure 9.1: At $t = 1$ the profile has a vertical tangent and the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$. This breaking is what makes the interaction term discontinuous after the shock.

The interaction changes qualitatively with time, and each regime needs different tools.

9.2.1 Before the shock: a closed form

Theorem 9.3 Closed form before the shock (S4T-T2, S4T-P3) [proved]

For Burgers with $t < t^*$, the interaction term has a closed form at first order in the viscosity, with remainder

$$O(\nu^2 \varepsilon^2 t^3),$$

where ε measures the size of the datum. The closed form blows up at the breaking time t^*.

Dim. S4T-T2, Catalogue p. 46

The tool is the Cole–Hopf substitution

$$u = -2\nu \frac{\varphi_x}{\varphi},$$

which turns Burgers into the heat equation $\varphi_t = \nu \varphi_{xx}$ [13]. It plays the role of an "exact linearisation". For KdV and NLS the inverse scattering transform plays the same role.

The analyticity radius of the Burgers solution with datum U_0 is at least $2/\mathrm{osc}(U_0)$, and this bound is optimal (F4L-P3) [proved].

9.2.2 After the shock: sizes in L^1

The flow of $-uu_x$ alone forms a shock, so E_{int} has jumps whose size does not depend on ν (S4T-T3) [proved mod. [F1]]. This is forced by the definition. A function with a jump is not in H^1, so Sobolev norms are no longer the right measure. Pointwise values are unreliable, but the *measure* of the discrepancy is well behaved. The meaningful sizes are in L^1.

Theorem 9.4 Interaction after the shock (OPA-T1) [proved mod. Kruzkov]

At a fixed time $t > t^*$, under a nondegeneracy condition (H_t) that holds for example for $u_0 = \sin x$,

$$\|E_{\mathrm{int}}(t)\|_{L^1} = \Theta(\nu) \qquad (\nu \to 0).$$

Each shock of strength ρ contributes

$$4\nu \log\left(2\cosh(\rho/2)\right)$$

to the leading term.

Dim. OPA-T1, Catalogue p. 48

Theorem 9.5 Uniform rate across the shock time (OPF5A-T1, OPF5A-T2) [proved mod. Kruzkov]

Uniformly in time, near the shock time,

$$\|E_{\mathrm{int}}(t)\|_{L^1} \approx 2\nu \log \frac{1}{\max(|t - t^*|, \nu)} + O(\nu),$$

so that

$$\sup_{t \le T} \|E_{\text{int}}(t)\|_{L^1} = \nu |\log \nu| + O(\nu),$$

with leading coefficient exactly 1. The constant at the shock time is explicit (it involves the Pearcey constant 1.188654…). For data with several shock births the uniform rate is [conditional] on a localisation lemma.

Dim. OPF5A-T2, Catalogue p. 50

Proposition 9.6 The splitting error is larger (OPF5D-T6) [proved]

The error of the splitting scheme "advection, then diffusion" is

$$\|S(t)u_0 - S_2^t S_1^t u_0\|_{L^1} = \Theta\big(\sqrt{\nu}\big),$$

larger than the interaction term itself. Treating the terms one after the other costs more than ignoring their interplay.

Dim. OPF5D-T6, Catalogue p. 140

Regime	What happens	Tools and results
before, $t < t^*$	smooth; characteristics do not cross	jet (Thm. 9.1), closed form (Thm. 9.2.1), Cole–Hopf
at breaking, $t = t^*$	the gradient blows up	smoothness estimates fail; the closed form blows up
after, $t > t^*$	the single-term flow has a jump	L^1: $\Theta(\nu)$, uniform $\nu\lvert\log\nu\rvert$, splitting $\Theta(\sqrt{\nu})$

Table 9.1: The three regimes of the Burgers interaction term.

9.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$: there is no ring of operators. Instead one fixes a finite list of candidates.

Dictionary setting

- A *dictionary* $\mathcal{D}$ is a finite set of monomials in the jet $(u, u_x, u_{xx}, \dots)$ of degree at most d and order at most k, for example $\{u,\ u_x,\ uu_x,\ u_{xx},\ u_{xxx}\}$. $V = \operatorname{span}\mathcal{D}$.
- $I(u)$ is the set of all differential polynomial relations that u satisfies.
- $[F]$ is the differential ideal generated by the equation F: F, its derivatives $\partial_x F, \partial_t F, \dots$, and their combinations.
- $\mathcal{J}_u$ is a jet-level obstruction defined from u, and ψ maps the dictionary into the jet space.

Theorem 9.7 Dictionary criterion (S4ID-T1) [proved]

For smooth u on any Ω there is an exact sequence

$$0 \to [F] \cap V \to I(u) \cap V \to \mathcal{J}_u \cap \psi(V) \to 0.$$

In particular

$$I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

Dim. S4ID-T1, Catalogue p. 52

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*". The literal statement $I(u) \cap V = \operatorname{span}\{F\}$ is false in complete dictionaries, because they contain $\partial_x F$ (S4ID-P1) [proved].

Theorem 9.8 Burgers with N shocks (S4ID-T2, OP-T7) [proved]

For the N-shock solutions of Burgers, u identifies its equation in the evolutionary dictionary with d unknown terms if and only if $d \le N$.

Dim. S4ID-T2, Catalogue p. 53

Example 9.9 One shock, two unknowns

The travelling front $u = -\tanh(x/(2\nu))$ is a 1-shock solution. It satisfies Burgers, but also the ODE that every travelling wave satisfies. With $d = 2$ unknown coefficients (in front of uu_x and u_{xx}) and $N = 1$, the theorem says u does not identify the pair. It fixes only their ratio.

Theorem 9.10 KdV N-solitons (S4ID-T3, OPF5C-T2) [proved mod. Novikov 1974]

For the N-soliton solution of KdV, the annihilator in the jet variables is principal, generated by one polynomial Π_N of total degree

$$\deg \Pi_N = (N+2)\, 2^{N-1}.$$

For generic $N = 2$ this degree is 8. The solution identifies KdV in a dictionary of degree d if and only if $d < (N+2)2^{N-1}$.

Dim. OPF5C-T2, Catalogue p. 54

Proposition 9.11 A closed form does not identify (S4ID-T4) [proved]

If u has a closed form, it lives on a smaller variety and satisfies extra relations: it does not identify its equation in a complete dictionary.

Dim. S4ID-T4, Catalogue p. 143

This is the nonlinear analogue of "exponential-polynomials never identify".

9.4 Integrable models and scattering

- **Hirota N-solitons.** The interaction is pairwise and localised: it is determined by pairs of solitons, integrates to 0, and costs $O(N^3)$ (INT-P5, INT-P6) [proved].
- **Scattering identification.** For reflectionless KdV, Novikov's centraliser characterises the potential (INT-T1, OPF5D) [proved]. For NLS with multiplicity, identification by the scattering data holds; without multiplicity it is [false]. The Lax and zero-curvature relations hold for all n (OP-P17, OP-P18) [proved].
- **KdV–Burgers.** The jet is proved. Before the shock the dispersive correction has remainder $O\big((\nu + |\delta|)^2\big)$ (OPR2-T5) [proved mod. local well-posedness].
- **Fisher–KPP.** Only a jet-level statement with an estimate that is not uniform in time (S4T-P5) [proved]. No sharp analogue of the Burgers asymptotics is claimed.

Open in the nonlinear part

For analytic KdV data with nonzero reflection, is there an autonomous ODE in x? Three obstructions are proved, among them the cubic relation of Airy [open] (INT-Q1). The exact minimal syzygy degrees $\delta_N(K)$ are known in nine computer-assisted cases; their general formulas are conjectures [open] (OPF5C-Q1).

9.5 Linear versus nonlinear

	Linear (Chapters 3–8)	Nonlinear (this chapter)
Candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$	a finite dictionary $V = \operatorname{span} \mathcal{D}$
Relations of u	the ideal $\operatorname{Ann}(u)$	the differential ideal $I(u)$
What P generates	the principal ideal (P)	the differential ideal $[F]$
Identification	$\operatorname{Ann}(u) = (P)$	$I(u) \cap V = [F] \cap V$
What escapes	the point spectrum $V(J_u)$	the obstruction $\mathcal{J}_u \cap \psi(V)$
Solutions that never identify	exponential-polynomials	closed forms
Interaction	product formula, $O(N)$	t^2 jet; $\Theta(\nu)$, $\nu\lvert\log\nu\rvert$ after shocks

Table 9.2: The linear and nonlinear problems side by side.

Takeaway. Nonlinearity changes the objects (single-term flows with shocks, dictionaries instead of rings) but not the question. Where the formula is closed (before the shock, solitons) the answer is explicit. Across a shock the interaction does not vanish with the viscosity: it shrinks like $\nu\lvert\log\nu\rvert$.

CHAPTER 10

The algorithm

In this chapter	
Starting from	an input in one of the data models M1–M3 (Chapter 5) and, for each step, the hypotheses of the theorem that justifies it.
Why	to turn the theorems into one procedure with explicit inputs, outputs and guarantees.
Role	operational summary of Chapters 4–9. Table 10.1 lists the steps, Figure 10.1 gives the control flow, and two worked examples make every step concrete.

10.1 The pipeline

Step	Task	Result relied upon	Output
0	Input	data models M1–M3	exact u
1	Annihilator	Gram matrix of a dictionary	a candidate P
2	Factor	unique factorisation (F1)	factors P_i, multiplicities e_i
3	EP test	Lemma 3.3, Thm. 4.2, Ch. 5	a verdict per factor
4	Reduce	Thms. 4.3 and 4.3	G_u, J_u, certificate
5	Decompose	Thm. 6.3, Eq. (6.1), Thm. 7.3	components u_j, ambiguity $\delta(P)$, check γ_N
6	Terms	Thms. 8.2, 8.3, 8.7	log-symbol, E_{int}
7	Nonlinear branch	Thms. 9.3, 9.1	relative identification, flow-based E_{int}

Table 10.1: The steps of the algorithm: what each step does, which result justifies it, and what it returns.

10.2 The steps in detail

Step 1: obtain an annihilator

Let $\Phi(u) = (T_1 u, \dots, T_M u)^\top$ be the vector of M candidate terms applied to u, and

$$\mathbb{G}(u) = \int_\Omega \Phi\, \Phi^*\, \mathrm{d}\mu$$

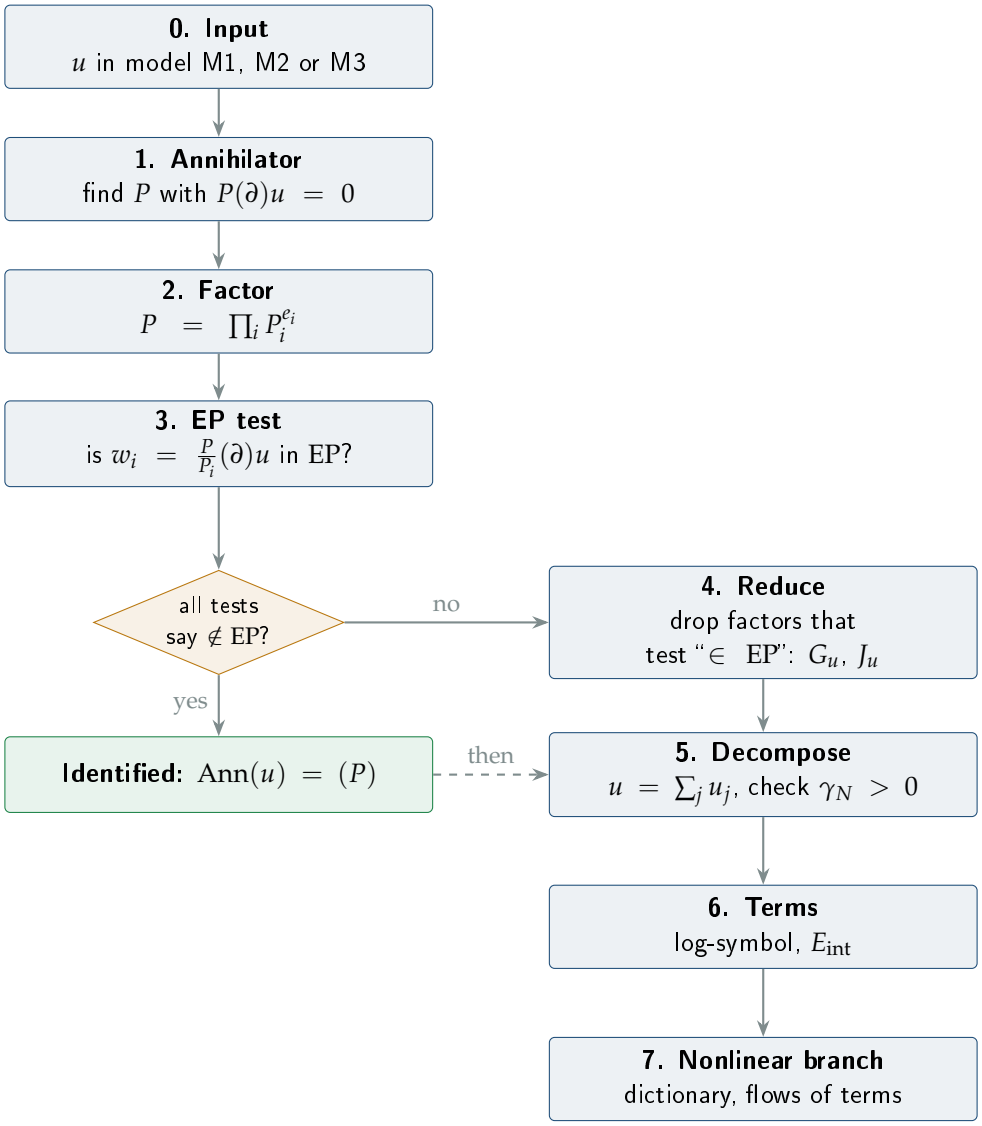


Figure 10.1: The algorithm as a flowchart. The left column certifies what the operator is. If every factor test says "not an exponential-polynomial", the operator is identified. Otherwise Step 4 reduces it to G_u. The right column extracts structure from u once the operator is known.

its Gram matrix (positive semidefinite). A relation $\lambda \cdot \Phi = 0$ holds if and only if $\lambda \in \ker \mathbb{G}(u)$. The computation is exact (rational or algebraic numbers, Gaussian elimination): no learning, no statistics. If $\ker \mathbb{G}$ has dimension larger than one, u satisfies several relations, which is itself information about G_u and J_u.

Step 2: factor

Factor $P(\tau, \xi)$ into irreducibles with multiplicities. This is a standard computer-algebra operation over the algebraic numbers.

Step 3: exponential-polynomial test

For each i compute $w_i = (P/P_i)(\partial)u$ and decide whether $\dim \mathcal{R}w_i < \infty$. How depends on the data model:

- M1: a syntactic check (is the closed form a finite sum of polynomials times exponentials?);
- M3: a holonomic computation, decidable modulo (AC) (Theorem 5.5);
- M2: undecidable from the jet alone (Theorem 5.3); with a bound B, compare annihilators up to degree $B + \deg P - 1$ (Theorem 5.4).

Step 4: reduce to G_u

Drop from P each factor whose test returned "EP", with the multiplicity bookkeeping of Theorem 4.3: at most $\sum_i e_i$ tests in total. The output is G_u; the dropped part is J_u, whose finite zero set lists the exponentials contained in u. The certificate reads either "$\mathrm{Ann}(u) = (P)$" or "u identifies only G_u and contains these exponentials".

Step 5: decompose

For evolution factors $\tau - a_j(\xi)$ apply the Lagrange formula (6.1). For general coprime factors use the construction of Theorem 6.3; the components are determined up to the $\delta(P)$-dimensional exponential-polynomial space, fixed by a normalisation. Before using the components in H^m, check $\gamma_N > 0$ (Theorem 7.3).

Step 6: terms and interaction

Inside an evolution factor with symbol $a = \sum_n a_n$, the terms are read from the log-symbol, which is linear in time:

$$\log \hat{u}(t, \kappa) = \log \hat{u}_0(\kappa) + t \sum_n a_n(i\kappa).$$

Then evaluate E_{int} by the product formula (8.2) at cost $O(N)$ per mode. For non-commuting terms, classify the generated Lie algebra (Theorem 8.7).

Step 7: nonlinear case

Replace Step 3 by the dictionary criterion (Theorem 9.3) and Step 6 by the flow-based interaction term and its jet (Theorem 9.1).

10.3 Worked example 1: the wave equation

We follow Steps 1–5 on the wave equation with $c = 1$ and the solution

$$u(t,x) = \underbrace{\frac{1}{1+(x+t)^2}}_{f(x+t)} + \underbrace{e^{-(x-t)^2}}_{g(x-t)}.$$

Example 10.1 Step 1: annihilator

Take the dictionary $\{u_{tt}, u_{xx}, u_{tx}, u_t, u_x, u\}$. The kernel of the Gram matrix is the line spanned by $u_{tt} - u_{xx} = 0$. Output:

$$P = \tau^2 - \xi^2.$$

Example 10.2 Step 2: factor

$$P = Q_1 Q_2, \qquad Q_1 = \tau - \xi, \qquad Q_2 = \tau + \xi.$$

Two irreducible, simple factors.

Example 10.3 Step 3: EP tests

Remove Q_1. We test $w_1 = (\partial_t + \partial_x)u$. The two parts behave differently:

$$(\partial_t + \partial_x) f(x+t) = 2f'(x+t), \qquad (\partial_t + \partial_x) g(x-t) = 0.$$

With $f(s) = 1/(1+s^2)$:

$$w_1 = -\frac{4(x+t)}{\left(1+(x+t)^2\right)^2}.$$

This is rational, not a finite sum of polynomials times exponentials: $w_1 \notin \mathrm{EP}$.

Remove Q_2. Symmetrically, with $g(s) = e^{-s^2}$:

$$w_2 = (\partial_t - \partial_x)u = -2g'(x-t) = 4(x-t)\, e^{-(x-t)^2}.$$

A Gaussian times a polynomial is not an exponential-polynomial: $w_2 \notin \mathrm{EP}$.

Example 10.4 Step 4: reduce

Both tests say "not EP", so nothing is dropped. The point spectrum is empty, and the wave equation is identified:

$$\mathrm{Ann}(u) = (\tau^2 - \xi^2).$$

Example 10.5 Step 5: decompose

$(Q_1, Q_2) = (\tau, \xi)$, so $\delta(P) = 1$. The components are $u_1 = f(x+t)$ and $u_2 = g(x-t)$, determined up to a constant moved from one to the other. Whether the components of every solution stay in H^m on a given region is decided by Theorem 7.3; on a square it holds (Example after Theorem 7.3).

10.4 Worked example 2: advection–diffusion

Now take the solution of Example 8.2,

$$u = g(x - ct;\ \sigma^2 + 2Dt), \qquad c = 2,\ D = \tfrac{1}{2},\ \sigma^2 = 1.$$

Example 10.6 Steps 1–4

The Gram matrix of $\{u_t, u_x, u_{xx}, u\}$ has kernel spanned by $u_t + 2u_x - \frac{1}{2}u_{xx}$, so

$$P = \tau + 2\xi - \tfrac{1}{2}\xi^2.$$

It is irreducible (degree one in τ). The test is on u itself: a Gaussian is not an exponential-polynomial, so $\mathrm{Ann}(u) = (P)$ and the equation is identified.

Example 10.7 Step 5

There is a single factor: no decomposition is needed.

Example 10.8 Step 6: terms and interaction

The log-symbol is

$$\log \hat{u}(t,\kappa) = \log \hat{u}_0(\kappa) + t\big(-2i\kappa - \tfrac{1}{2}\kappa^2\big).$$

Its two coefficients are the two terms: advection with speed 2 and diffusion with constant $\frac{1}{2}$. The interaction term follows from (8.2), at cost $O(2)$ per Fourier mode:

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \big(e^{-2i\kappa t} - 1\big)\big(e^{-\kappa^2 t/2} - 1\big)\,\hat{u}_0(\kappa).$$

Figure 8.1 plots it.

10.5 Cost and feasibility

Component	Cost or feasibility	Source
annihilator from a dictionary	Gaussian elimination, $O(M^3)$	Step 1
factorisation	standard computer algebra	Step 2
EP test, model M1	syntactic	Step 3
EP test, model M3	holonomic, decidable mod (AC)	Thm. 5.5
EP test, black box	impossible without a bound	Thm. 5.3
EP test with bound B	linear algebra in degree $B + \deg P - 1$	Thm. 5.4
components (evolution)	one rational multiplier per factor	Eq. (6.1)
E_{int}: value / by order / table	$O(N)$ / $O(N^2)$ / $\Theta(2^N)$	Thm. 8.3

Table 10.2: Cost and feasibility of the components of the algorithm.

Takeaway. Every step of the algorithm is a theorem with explicit hypotheses. The algorithm is exact where the theorems are positive (Steps 2, 4, 5 and 6 on exponential-type data), and it reports honestly where they are negative (Step 3 on finite data, Step 6 in Sobolev spaces).

Part II

Phases 6–10: beyond constant coefficients

CHAPTER 11

Systems of equations and gauge

In this chapter	
Starting from	a vector field $u = (u_1, \dots, u_k)$ on a connected open set $\Omega \subset \mathbb{R}^n$ that solves a system $P(\partial)u = 0$ with a *matrix* of constant coefficients $P \in R^{l\times k}$, $R = \mathbb{C}[\zeta_1, \dots, \zeta_n]$. The operator P is not given.
Why	the equations of physics that are not scalar (Dirac, Maxwell, elasticity) are systems; Phases 1–5 treated one scalar equation. We ask the same three questions: can one solution determine the system, when can a sum of kernels be split, and what does a change of gauge do to what can be known?
Role	extends Chapters 4–6 from R to matrices over R. It supplies Step 3b of the extended algorithm (Chapter 17). The proofs are Dim. A32–A35 and B73–B78.

11.1 From one equation to a module

For a scalar equation the annihilator $\mathrm{Ann}(u)$ was an ideal of R. For a system the object that replaces it is a *module*. The rows of P generate a submodule $U_P \subseteq R^{1\times k}$, and the quotient

$$M_P = R^{1\times k}/U_P$$

is the module of the system: solutions of P are exactly the homomorphisms $M_P \to \mathcal{A}$ (Malgrange–Oberst duality). A vector u *identifies* P when its annihilator $\mathrm{Ann}(u) = \{r : r(\partial)u = 0\}$ equals U_P, i.e. when the map $M_P \to \mathcal{A}$, $[r] \mapsto r(\partial)u$, is injective.

Intuition

Think of M_P as the list of everything the system says, up to consequences. A single function u must carry all of it in an injective way. The obstruction is local: at a point λ the solutions of the system that are pure exponentials $e^{\lambda\cdot x}$ form one line, but M_P may contain a plane of "independent" behaviours at that point. A plane cannot be embedded in a line.

11.2 When does an identifying solution exist?

Define the *socle* of M at a maximal ideal $\mathfrak{m}_\lambda$ as the set of elements of M killed by $\mathfrak{m}_\lambda$; it is the space of exponential behaviours $e^{\lambda\cdot x}$ the module contains.

Theorem 11.1 Existence of an identifying solution (SYS-EX) [proved]

Let $\Omega \subseteq \mathbb{R}^n$ be non-empty, open and connected. A finitely generated module M embeds in $C^\infty(\Omega)$ (or $\mathcal{D}'(\Omega)$) if and only if $\dim_{\mathbb{C}} \operatorname{soc}_{\mathfrak{m}}(M) \le 1$ for every maximal ideal $\mathfrak{m}$. In particular a system P has an identifying solution if and only if $\dim \operatorname{Hom}_R(R/\mathfrak{m}, M_P) \le 1$ for every maximal $\mathfrak{m}$. Dim. SYS-EX, Catalogue p. 57

Example 11.2 Two ODE systems

For $n = 1$ let $P = \operatorname{diag}(s-1, s-1)$ (two copies of $u' = u$). The socle at $s = 1$ has dimension 2: no function u ever identifies it. For $P = \begin{pmatrix} s-1 & 1 \\ 0 & s-2 \end{pmatrix}$ all socles have dimension ≤ 1, and $u = (ae^x - be^{2x}, be^{2x})$ with $ab \neq 0$ identifies it. In general an ODE system of rank r with invariant factors $e_1 \mid \cdots \mid e_r$ has an identifying solution iff $e_1 = \cdots = e_{r-1} = 1$ (SYS-T0) (known).

Theorem 11.3 Square non-singular systems in two or more variables (SYS-C1) [proved]

If $n \ge 2$ and $P \in R^{k\times k}$ has $\det P \neq 0$, every socle of M_P is zero, so on every connected Ω some $u \in \operatorname{Sol}(P)$ identifies P. The associated primes of M_P are the prime factors of $\det P$. Dim. SYS-C1, Catalogue p. 59

Dirac in $3+1$ dimensions, the matrix Klein–Gordon operator and the Lamé system of elasticity are all of this type. They always have an identifying field.

Where this is not new

For $\Omega = \mathbb{R}^n$ the existence criterion is known (Oberst, Pommaret, Macaulay: the number of trajectories needed to generate a system). The new point here is only that Ω may be any connected open set, including one where $C^\infty(\Omega)$ is not an injective module.

11.3 Sums of kernels for systems

For a scalar $P = Q_1 Q_2$ we proved $\ker P = \ker Q_1 + \ker Q_2$ on a simply connected region (Theorem 6.3). For systems the analogue is stated with the *least common left multiple*: if U_1, U_2 are the row modules of two sub-systems and $N = R^{1\times k}/(U_1 + U_2)$, then $\operatorname{Sol}(Q_1) + \operatorname{Sol}(Q_2) = \operatorname{Sol}(P)$ requires $U_1 \cap U_2 = U_P$.

Theorem 11.4 Sum of kernels (SYS-S1) [proved]

On every open Ω, $\operatorname{Sol}(Q_1) + \operatorname{Sol}(Q_2) = \operatorname{Sol}(P)$ implies $U_1 \cap U_2 = U_P$. Conversely, if $U_1 \cap U_2 = U_P$ and $\operatorname{Ext}^1(N, \mathcal{A}) = 0$ the equality holds; this is the case when Ω is convex,

and when N has finite length and $H^1_{dR}(\Omega) = 0$. The intersection $\mathrm{Sol}(Q_1) \cap \mathrm{Sol}(Q_2)$ is finite-dimensional iff N has finite length, with dimension $\delta_{\mathrm{sys}} = \ell(N)$. For one scalar equation in two variables this is Theorem 6.3. Dim. SYS-S1, Catalogue p. 61

Counterexample 11.5 Simple connectivity does not suffice for infinite-length quotients

[false] There are a simply connected open $\Omega \subset \mathbb{R}^2$ (a "U"-shaped domain) and $Q_1 = (1, 0)$, $Q_2 = (1, \partial_x)$ with $U_1 \cap U_2 = U_P$ but $\mathrm{Sol}(Q_1) + \mathrm{Sol}(Q_2) \subsetneq \mathrm{Sol}(P)$, in C^∞ and in $\mathcal{D}'$. The obstruction is the failure of P-convexity of Ω, not its topology. Dim. SYS-S1, Catalogue p. 61

11.4 Dirac in Hamiltonian form

Write the free or perturbed Dirac equation as $L = I\partial_t + \sum_k \alpha^k \partial_k + B$ with arbitrary constant 4×4 matrices α^k, B (no Clifford relations are assumed).

Theorem 11.6 Dirac coefficients are read exactly (H-DIRAC) [proved]

If $\psi \in \mathrm{Sol}(L_0)$ identifies L_0 and L is another operator of this form with $L\psi = 0$, then $L = L_0$: the matrices α^k and B are determined by ψ exactly, with no ambiguity. An identifying ψ always exists. For massive Dirac in $3+1$ dimensions $M_D \cong (R/\mathfrak{q})^2$ with $\mathfrak{q} = (p_0^2 - |p|^2 - m^2)$, and ψ identifies D iff ψ_3, ψ_4 are independent over $R/\mathfrak{q}$; in $1+1$ dimensions ψ identifies D_2 iff $\psi_2 \notin \mathrm{EP}$. Dim. H-DIRAC, Catalogue p. 63

Example 11.7 Where the Dirac coefficients come from

Normalising the coefficient of ∂_t to the identity removes the ambiguity $D \sim VD$ of the covariant form $\gamma^\mu \partial_\mu$. In $1+1$ dimensions the identification test is just the one of Chapter 4: the second component must not be an exponential-polynomial.

11.5 Gauge, magnetic fields, and Lamé

Proposition 11.8 Gauge covariance and what invariant data reveal (G-L0, G-T2) [proved]

Under $A \mapsto A + d\chi$, $u \mapsto e^{i\chi}u$ the covariant operators transform covariantly, so the identification tests are gauge-invariant. If the observation is invariant under the phase, a reconstruction of the potential A recovers at most its gauge class $[A]$: $A' \sim A$ iff $dA' = dA$ and the holonomies agree (Aharonov–Bohm). On a simply connected region, $A' \sim A$ iff $dA' = dA$. Dim. G-T2, Catalogue p. 152

Proposition 11.9 Magnetic Schrödinger: one field does not identify (A, V) (SYS-N3) [proved]

For $n \geq 2$ and a nowhere-zero solution u the pairs (A', V') compatible with u form an infinite-dimensional affine space, parametrised by divergence-free fields k; their magnetic fields differ, so they are not gauge-equivalent, even in the non-stationary case. The

obstruction is a count: one complex scalar equation against $n+1$ real unknowns. Dim. SYS-N3, Catalogue p. 153

Proposition 11.10 Lamé in space (LAME-3) [proved]

For $\rho, \mu \neq 0$, $\lambda+\mu \neq 0$, $\lambda+2\mu \neq 0$, the system has $\det P = q_P q_S^2$ with Bass multiplicities 1 and 2; an identifying displacement field exists, and u identifies it iff $\operatorname{div} u$ satisfies no equation besides the wave equation of speed $\sqrt{(\lambda+2\mu)/\rho}$ and the two transverse polarisations of $\operatorname{rot} u$ satisfy no extra relation. A pure P-wave ($\operatorname{rot} u = 0$) never identifies. Dim. LAME-3, Catalogue p. 151

Proposition 11.11 Dirac $1+1$ as an evolution: no uniform product (SYS-E1, E2, E3) [proved]

Writing $\psi_t = (X + mY)\psi$ with X the transport and Y the mass term, the operators $X, \operatorname{ad}_X^k Y$ are linearly independent, so no finite product of transport and mass flows with ξ-independent times equals $e^{t(X+mY)}$ uniformly in the mass. The Lie algebra they generate is infinite-dimensional. An explicit product (Euler angles) exists, with times that are not polynomial in ξ and with non-local Fourier multipliers. Dim. SYS-E, Catalogue p. 150

Proposition 11.12 Systems from finite data (SYS-R1) [proved]

From exponential-polynomial data one can identify only systems with finite-dimensional solution space; for an ODE system the identifying solution exists iff $e_1 = \cdots = e_{r-1} = 1$. Dim. SYS-R1, Catalogue p. 149

Why it matters. For a system the unknown is a module and the datum is one vector function. The only obstruction to identification is the socle, a purely algebraic and local count, and it vanishes for every non-singular square system in $n \geq 2$ variables. What a single field can never do is separate gauge from physics.

CHAPTER 12

Variable coefficients

In this chapter

Starting from	a rectangle $\Omega = I \times J \subset \mathbb{R}^2$ and a solution u of $iu_t = -u_{xx} + V\,u$, i.e. $L_0 u = Vu$ with $L_0 = i\partial_t + \partial_x^2$, where the potential $V(x)$ or $V(t,x)$ is unknown. Solutions are *strong* (class SF): $u \in C^{1,2}(\Omega)$ and the equation holds almost everywhere.
Why	in physics the coefficients are rarely constant: a Schrödinger particle feels a potential. Phases 1–5 identified constant coefficients. Here the unknown is a function, and the first question is whether a single solution can reveal it at all.
Role	extends Chapter 4 from $\mathbb{C}$ to function coefficients. It supplies Step 3c of the extended algorithm (Chapter 17). The proofs are Dim. A36–A40 and B78.

12.1 Too much freedom: a huge class identifies nothing

Suppose we allow coefficients in any algebra $\mathcal{K}$ of functions closed under differentiation (all smooth functions, all analytic functions). Then even one solution that never vanishes carries no information beyond itself.

Theorem 12.1 Rank-one annihilator (VAR-N1, VAR-T1(a)) [proved]

Let $u \in C^\infty(\Omega)$ have no zeros, with $a = u_t/u$ and $b = u_x/u$ in $\mathcal{K}$. Then the annihilator of u among differential operators with coefficients in $\mathcal{K}$ is generated by $\partial_t - a$ and $\partial_x - b$, and the module $\mathcal{K}[\partial]u$ is isomorphic to $\mathcal{K}$ itself. For an affine family $L_\theta = L_0 + \sum \theta_k T_k$ the compatible parameters are $\theta_0 + \ker \Phi_u$, so θ is identified iff $\Phi_u(\theta) = \sum \theta_k T_k u$ is injective.
Dim. VAR-N1, Catalogue p. 65

Intuition

If the coefficient class is closed under differentiation, a zero-free u can write each of its own derivatives as "a coefficient times u", so every law is a law about u itself and nothing more. To identify anything the class of coefficients must be restricted in advance.

Zeros matter

With zeros the statement fails: $u = t - x^2$ near 0 satisfies no law $\partial_t + a\partial_x + \beta$ with smooth coefficients, because at the origin it would give $1 = 0$.

12.2 A potential $V(x)$ is read from any non-zero solution

For a time-independent potential the situation is the opposite: nothing can hide.

Theorem 12.2 Identification of $V(x)$ (VAR-T1(b)) [proved]

Let $u \not\equiv 0$ be a strong solution with potential $V \in L^\infty_{\mathrm{loc}}(J)$, and $X_0 = \{x : u(t,x) = 0\ \forall t\}$. Then X_0 is closed and discrete; if $L_0 u = W(x)u$ for another W, then $W = V$ almost everywhere; and

$$V(x) = \frac{iu_t + u_{xx}}{u} \qquad \text{at every } (t,x) \text{ with } u(t,x) \neq 0.$$

Dim. VAR-T1b, Catalogue p. 66

Intuition

If u vanished along infinitely many vertical lines accumulating at x_0, cutting u off to the right of x_0 would give another strong solution that vanishes on a half-strip. Schrödinger waves have infinite speed, and a Carleman estimate (a weighted energy inequality) shows that a solution vanishing on a strip vanishes everywhere. So u cannot have such lines, and the formula $V = (iu_t + u_{xx})/u$ holds on a set of full measure.

Example 12.3 The harmonic oscillator

For $V = x^2$ and the Hermite functions $h_0, \dots, h_3$ the set X_0 is empty or $\{0\}$, and $(iu_t + u_{xx})/u$ returns x^2 exactly. The same formula recovers the Pöschl–Teller and Airy potentials.

12.3 A potential $V(t,x)$: silent strips

Theorem 12.4 Time-dependent potentials (VAR-T1(c)) [proved]

For $V \in C(\Omega)$, the potentials compatible with u are $V + \{W : W = 0 \text{ on } \Omega \setminus Z(u)\}$, where $Z(u) = \{u = 0\}$; so V is identified iff $\operatorname{int} Z(u) = \varnothing$. If $\operatorname{int} Z(u) \neq \varnothing$ there is an open interval $T' \subset I$ with $u = 0$ on $T' \times J$ (a *silent strip*). Silent strips occur: Tychonoff's series gives a solution vanishing for $t \leq 0$ and non-zero for $t > 0$. Hence "every $u \not\equiv 0$ identifies $V(t,x)$" is false. Dim. VAR-T1c, Catalogue p. 68

Counterexample 12.5 Data that switch on from the boundary

[false] On a bounded domain without boundary conditions, a solution can be identically zero up to a time and non-zero afterwards (Tychonoff's series). While it is silent every potential supported in the silent time is compatible with it. The failure is in the time direction: unique continuation holds in x but not in t.

12.4 Stability and the phase

Theorem 12.6 Stability of the reconstruction (VAR-P3) [proved]

On a rectangle K with $|u_1| \geq \delta > 0$: $\|V_1 - V_2\|_\infty \leq \delta^{-1}(1 + \|V_2\|_\infty)\|u_1 - u_2\|_{C^{1,2}(K)}$; the map is not continuous from C^1 to L^∞ (an explicit oscillation); and under an a priori bound M on $\|e_{tt}\|_\infty, \|e_{xxx}\|_\infty$ one has a Hölder modulus with sharp exponent $1/3$. It is a modulus of continuity, not a certificate. Dim. VAR-P3, Catalogue p. 70

Theorem 12.7 Unobserved phase: V modulo constants (VAR-P4) [proved]

If only the class $\{e^{i\theta(t)}u\}$ is observed, the compatible potentials are exactly $\{V + c : c \in \mathbb{R}\}$. For $P(\partial_t, L_q)u = 0$ with P unknown, (P, q) is defined only modulo $q \to q + c$, $P(\tau, \ell) \to P(\tau, \ell - c)$: a redundancy of the parametrisation. Dim. VAR-P4, Catalogue p. 71

Proposition 12.8 Dirac $1+1$ with a potential (VAR-P5) [proved]

Finite speed of propagation leaves regions that the data never visit: for $m = V = 0$ a travelling bump $\psi = f(x - t)v_+$ vanishes for $x \leq 0.5$ and $x \geq 1.7$, so every potential supported there is compatible. The matrix potential is determined at x_0 iff $\operatorname{span}_t\{\psi(t, x_0)\} = \mathbb{C}^2$. Dim. VAR-P5, Catalogue p. 154

12.5 Other results of this phase

- **Algebra of differential operators with polynomial coefficients** (Weyl algebra): for a D-finite u the annihilator is a principal left ideal and u identifies iff its derivatives are independent over K; Airy is irreducible and every solution identifies; the oscillator is reducible exactly for $E \in \pm(2\mathbb{N} + 1)$ (VAR-T2a, VAR-T6, VAR-E1, VAR-E2).
- **Dynamical Lie algebras**: $\dim \mathfrak{g}_V < \infty$ iff $V \in \operatorname{span}\{1, x, x^2\}$; the modified Pöschl–Teller potential has an explicit evolution through a Darboux chain (VAR-P8).
- **Non-autonomous splitting**: $E^{\text{mult}} = \int\int [A_2(s_1), A_1(s_2)] + O(t^3)$; commuting at equal times is not enough (VAR-P9). Dyson–Phillips converges for bounded time-dependent perturbations (VAR-P11); Floquet logarithms are bounded but not differential (VAR-P12).

These are list cards of the catalogue (Part IV); the closures made in Phase 10 are in Chapter 16.

> **Why it matters.** A function-valued coefficient is identified by one solution exactly when the solution cannot hide: for $V(x)$ it cannot (infinite speed and unique continuation); for $V(t, x)$ it can fall silent; for a huge class of coefficients, nothing is identified. Stability is of order two and the phase costs one constant.

CHAPTER 13

Convergence, noise and the limits of learning

In this chapter	
Starting from	one analytic function u on a connected open set $\Omega \subset \mathbb{R}^n$, observed through its Taylor jet at a point p_0 (the datum at step K is $j_K u$), or through a noisy copy $\tilde{u}$ with $\|\tilde{u} - u\| \leq \delta$ in a stated norm.
Why	Chapters 4 and 5 proved that no finite amount of data certifies identification. A real experiment gives more and more data: does the identified operator converge, how fast, and what can a noisy measurement still say?
Role	supplies Steps 1–4 and 7 of the extended algorithm (Chapter 17). The proofs are Dim. A41–A43.

13.1 Identification in the limit

At step K the rule computes, for each degree $m \leq K$, the space $N_{m,K}$ of operators of degree $\leq m$ whose action on u has vanishing Taylor coefficients up to order $K - m$. These spaces decrease with K and their limit is $\mathrm{Ann}(u) \cap R_{\leq m}$. The *minimal-degree rule* outputs the gcd of $N_{m^*(K),K}$ at the smallest degree $m^*(K)$ where it is non-zero.

Theorem 13.1 Identification in the limit (CONV-L1, CONV-T1) [proved]

If $\mathrm{Ann}(u) \neq 0$ with minimal degree d, the rule outputs $g_d = \gcd(\mathrm{Ann}(u) \cap R_{\leq d})$ from some step on; if $\mathrm{Ann}(u) = (P)$ it outputs P, without knowing $\deg P$; if $\mathrm{Ann}(u) = 0$ the degree $m^*(K) \to \infty$. The sequence of outputs takes at most $(d+1)(d+2)/2$ values. But:

- the step K^* after which the output is correct is not bounded by any function of d;
- no function of the jet of order K is a correct certificate: any finite jet is shared by a polynomial, whose annihilator is a different ideal.

Dim. CONV-T1, Catalogue p. 73

Example 13.2 Convergence at an unpredictable time

For $u = 1 + x^{N+2}e^{x^2}$ one has $\mathrm{Ann}(u) = (\tau)$. The rule outputs "no law of degree 1" for $K \leq N+1$ and τ from $K = N+2$, for every N: nothing in the first $N+1$ derivatives announces the law.

13.2 No learner decides whether a law exists

Theorem 13.3 Analytic Gold theorem (CONV-T2, CONV-T2′) [proved]

For any sequence of maps φ_K from jets of order $\leq K$ to $\{\text{law}, \text{no law}\}$ (no computability or measurability assumed) there is a real entire function u on which $\varphi_K(j_K u)$ does not converge to the correct answer. The same holds restricted to the class where $\mathrm{Ann}(u)$ is principal or zero; hence no learner identifies $\mathrm{Ann}(u)$ on that class. Dim. CONV-T2, Catalogue p. 74

Intuition

Build u as a limit of polynomials. Whenever the learner says "law" on the current polynomial (it must, eventually), add Taylor coefficients, chosen outside a proper algebraic set, that destroy every law of degree $\leq k$. The limit function is differentially transcendental yet agrees with a polynomial where the learner said "law", infinitely often.

Takeaway. "$\mathrm{Ann}(u) \neq 0$" is *verifiable in the limit* (a rule that is eventually right when it holds) but not *decidable in the limit*.

13.3 Noisy data

Theorem 13.4 Noise: discontinuity, conditional stability, discrepancy rule (CONV-T3, T4, T5) [proved]

(T3) In spaces where polynomials are dense and $e^{|z|^2}$ lies, the sets $\{\mathrm{Ann} = 0\}$ and $\{\mathrm{Ann} \neq 0\}$ are both dense: $u \mapsto \mathrm{Ann}(u)$ is discontinuous everywhere. (T4) At a fixed level n_0 with a margin $\rho = \sigma_2(\Phi(u)) > 0$, the minimal singular vector of $\Phi(\tilde{u})$ is within $\sin\angle \leq 2\delta/\rho$ of the true law, an order that is optimal; under an L^2 error and an a priori H^{n_0+s} bound the rate is Hölder, and $\sigma_1(\Phi(\tilde{u})) > \delta$ certifies a misfit. (T5) The discrepancy rule $m(\delta) = \min\{m : \sigma_1(\Phi_m(\tilde{u})) \leq \delta_m\}$ returns the correct degree d and converges at rate δ once $\delta < \delta_0 = \frac{1}{2}\min_{m<d} s_m$; if $\mathrm{Ann}(u) = 0$, $m(\delta) \to \infty$. The threshold δ_0 cannot be bounded below from the data. Dim. CONV-T4, Catalogue p. 76

Example 13.5 The heat kernel from noisy data

For $u = (t+1)^{-1/2} e^{-x^2/(4(t+1))}$ (minimal degree 2, $\mathrm{Ann} = (\tau - \xi^2)$), the rule returned $m(\delta) = 2$ for $\delta = 10^{-3}$ and 10^{-4} in a numerical check.

13.4 Other results of this phase

- **Locality**: when the law is a non-local operator it is invisible to $\mathrm{Ann}(u)$; an irrational fractional power or a delay gives $\mathrm{Ann} = 0$ (a delay is in the local class through an auxiliary transport) (CONV-T6, CONV-P6, CONV-P7).
- **Dimension** $1 + d$: the ambiguity of the factor decomposition is infinite unless the factors are pairwise co-maximal (CONV-D1); on a convex region $\ker(Q_1 Q_2) = \ker Q_1 + \ker Q_2$ in C^∞ (CONV-D2).
- **Boundary conditions** are invisible to $\mathrm{Ann}(u)$; stability holds for a dictionary of traces (CONV-B1–B3). Splitting with a Dirichlet boundary has global order 1 (Lie) and $\Theta(h^{5/4})$ (Strang) in L^2 (CONV-T8).
- **Mori–Zwanzig** for constant-coefficient 2×2 systems: exact memory form and eliminated law (CONV-MZ1–3); the criterion "$E_{\mathrm{int}} = 0 \iff [\mathcal{P}, L] = 0$" is false (CONV-MZ3-F).
- **A convergence contract** for chains of steps: classes (A) exact, (B) convergent with explicit error, (C) no guarantee, (Lim) in the limit; the error of a chain of Lipschitz steps is $\leq \sum_i (\prod_{j>i} L_j)\varepsilon_i$ (CONV-T7).

> **Why it matters.** Learning an operator from growing data is possible only in the limit and without a certificate; deciding whether any law exists is impossible even in the limit; and noise can be handled only relative to a finite level of the dictionary and a margin that the data cannot supply.

CHAPTER 14

The analytic encoder

In this chapter	
Starting from	a Schrödinger-type equation with a variable potential $q(x)$ *known*, written $P(\partial_t, L_q)u = 0$ with $L_q = -\partial_x^2 + q$ and $L_0 = -\partial_x^2$, on a symmetric interval J.
Why	the variable potential is a nuisance: the theory of Phases 1–5 applies to L_0. Can one change the unknown by an explicit, invertible, stable map so that the potential disappears, without machine learning? Such a map is an *encoder*.
Role	supplies Step 5 of the extended algorithm (Chapter 17): the encoder is a *translation* to constant coefficients, not a source of information. The proofs are Dim. A44–A47 and B79.

14.1 What an encoder is

A *structured encoder* is a linear map W acting in x only, with (E1) finite-dimensional kernel on its own domain, (E2′) $WL_q = (L_0 + c)W$ for a constant c, and (E3) explicit. The conditions are all needed: without (E1) the zero map qualifies, and without (E3) the interaction-picture map $U_0(t)U_V(t)^{-1}$ does.

Proposition 14.1 A constant potential has no strict differential intertwiner (ENC-K0) [proved]

For $q \equiv c \neq 0$ no differential operator of any order satisfies $WL_c = L_0W$. With the relaxed requirement $WL_c = (L_0 + c)W$ the identity works. Dim. ENC-K0, Catalogue p. 155

14.2 The Volterra transformation operator

Let K solve the Goursat problem $K_{xx} - K_{ss} = q(x)K$ on the triangle $|s| \le |x| \le a$, with $K(x,x) = \frac{1}{2}\int_0^x q$ and $K(x,-x) = 0$, and put

$$Tf(x) = f(x) + \int_{-x}^{x} K(x,s)f(s)\,\mathrm{d}s, \qquad W = T^{-1}.$$

Theorem 14.2 The symmetric Volterra encoder (ENC-1, ENC-U1) [proved]

For $q \in C^1$ the problem has a unique solution, and for every $f \in C^2(J)$, without any boundary condition, $L_qTf = TL_0f$ with $(Tf)(0) = f(0)$, $(Tf)'(0) = f'(0)$. T is bijective on C, C^2 and (for $q \in C^\infty$) C^∞, with $\|K\| \le \frac{aQ}{2}e^{a\sqrt{Q}}$ and a Lipschitz dependence on

q. Among all Volterra intertwiners the one preserving Cauchy data at the base point is unique; it depends on the base point already for constant q. Dim. ENC-1, Catalogue p. 80

Intuition

In characteristic coordinates the kernel solves a Volterra integral equation by successive approximations, with the bound $(Q|\xi\eta|/4)^n/(n!)^2$. Volterra operators with a triangular kernel have no spectrum, so T is invertible, and the Goursat conditions are exactly the ones that make every boundary term in the Leibniz expansion of $L_qTf - TL_0f$ cancel.

Corollary 14.3 One encoder for every equation (ENC-C1) [proved]

For $q \in C^\infty$ and every polynomial P: $P(\partial_t, L_q)u = 0 \iff P(\partial_t, L_0)(Wu) = 0$. The same W serves Schrödinger, heat, wave and Klein–Gordon. Dim. ENC-C1, Catalogue p. 82

Example 14.4 Constant potential

For $q \equiv c$ the transformation maps the exponential $e^{ik\cdot}$ to $e^{ikx_0}[\cos\mu(x-x_0) + ik\sin\mu(x-x_0)/\mu]$ with $\mu^2 = k^2 - c$: the eigenfunction of L_c with the Cauchy data of e^{ikx} at x_0.

14.3 What no encoder can do

Theorem 14.5 No time-independent encoder for a time-dependent potential (ENC-P0) [proof written, not audited]

Let W be linear with finite kernel and $W(-\partial_x^2 + V(t,\cdot))f = (-\partial_x^2 + c(t))Wf$ for all t. Then $V(t,x) - V(s,x) = c(t) - c(s)$: the potential must be autonomous up to a function of time. The reason is that otherwise infinitely many independent functions $m\varphi_k$ would lie in $\ker W$. Encoders for a genuinely time-dependent potential exist but depend on time (interaction pictures), and a pointwise representative with the identity time change exists exactly for $V = \beta(t)x + \gamma(t)$. Dim. ENC-P0b, Catalogue p. 83

Theorem 14.6 Blind encoders and identifiability (ENC-N1′) [proved]

A blind encoder, one whose output does not depend on which potential produced the data, exists iff the coherence condition (C) holds for the chosen family; it is implied by identifiability and is strictly weaker (a witness: the data model that observes only a phase). The correct statement is: blind encoder $\iff$ identifiability modulo the gauge group of the data model. Dim. ENC-N1p, Catalogue p. 84

An earlier claim that turned out false

The synthesis of the first version, "the exceptions change the time", was false as written: Avron–Herbst and the whole class $V = \beta(t)x + \gamma(t)$ have a pointwise representative with the identity time change. Likewise "a blind encoder exists iff $W_V^{-1}v$ is injective" is false (ENC-N1).

14.4 Other results of this phase

- **Darboux encoders**: a differential W with $WL_q = (L_0 + c)W$ is a Darboux chain on each component of a complement of a countable closed set; for rational $V \to 0$ it exists iff V is an iterated Darboux transform of 0 (Adler–Moser) (ENC-K1′, ENC-K1). They are unbounded with a non-trivial kernel, so they lose modes.
- **The interaction term with a potential**: $\|E^{\mathrm{mult}} f\| \le \frac{t^2}{2}(\|V''\|\|f\| + 2\|V'\|\|f'\|) + \frac{2t^3}{3}\|V'\|^2\|f\|$ with the optimal constant $2/3$, and $E^{\mathrm{mult}} \equiv 0$ iff V is constant (ENC-A9).
- **Faddeev decomposition** on an interval and **Foldy–Wouthuysen** for the free Dirac $1+1$ are decompositions, not encoders (FAD-1, FW-1).

Why it matters. Removing a variable potential is possible by an explicit stable Volterra transformation when the potential is known, and the same map works for every equation in (∂_t, L_q). When the potential is unknown the encoder gives no information, and for a time-dependent potential no time-independent encoder exists.

CHAPTER 15

The universal dictionary and the misfit certificate

In this chapter	
Starting from	a filtered dictionary $\mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots$ of finite sets of operators (the *atoms*), enumerated by an explicit algorithm; a datum u *fits* a level if some non-zero combination of its atoms annihilates u.
Why	the project was asked: can a dictionary be built that "takes everything into account"? The honest answer has three parts: in the sense of approximation, yes; as exact identification of every operator, no; and in between, a misfit can always be certified by finite data, while a fit never can.
Role	supplies Steps 2 and 4 of the extended algorithm (Chapter 17). The proofs are Dim. A48–A51.

15.1 What "universal" can mean

Theorem 15.1 Every local linear operator is a limit of dictionary elements (DICT-U1) [proved]

By Peetre's theorem a linear operator on $C^\infty(\Omega)$ that does not enlarge supports is a differential operator with smooth coefficients, of locally finite order. With a partition of unity and polynomial approximation of the coefficients it is a limit, in the topology of locally bounded order, of elements of the explicit dictionary $\{\chi_j(y)y^\beta\partial^\alpha\}$. It is a pointwise limit of polynomial-coefficient operators iff its order is globally finite. Dim. DICT-U1, Catalogue p. 85

The same holds for classes beyond differential operators: regular non-linear operators (differential polynomials, by a local version of a Peetre–Slovák argument), symbols with loss ε, completely monotone memory kernels, delays (as an augmented transport system) and systems, each with its own explicit atoms (DICT-U2, U3, RIT1, U4).

15.2 What it cannot mean

Theorem 15.2 No countable dictionary identifies everything (DICT-N1) [proved]

(i) For any countable dictionary the exactly representable operators form a meagre set (Baire). (ii) No finite data certifies the fit or the identification. (iii) If $u \in \mathrm{EP}$, at every high level $\mathrm{Ann}(u) \cap R_{\le n} \supsetneq (L) \cap R_{\le n}$ for any single law L: the kernel is strictly larger than the consequences of one law. (iv) A procedure with finite outputs emits only countably many operators. (v) For a filtered family of differential polynomials the set of u that satisfy some equation of the dictionary is meagre *and* shy (prevalence): the generic u satisfies none. Dim. DICT-N1, Catalogue p. 87

Intuition

A compact set in an infinite-dimensional space has empty interior, so a countable family of finite-parameter families covers a "thin" set. For the measure statement one moves u along a direction on which the determinant of the dictionary applied to $u + sw$ is a non-zero polynomial in the single real parameter s: only finitely many s lie in the bad set.

What this theorem does not say

"Everything" holds in the sense of approximation (Theorem 15.1) plus a misfit certificate, not as exact identification. The first form of the statement for coefficients that are C^∞ functions ("no operator is identified") is false and was replaced by a rank theorem: an affine family is identified iff $\ker \Phi_u = 0$ (DICT-RANGO).

15.3 Fit and misfit

Theorem 15.3 Asymmetry of fit and misfit (DICT-A, MF1–MF3) [proved]

Let $\{D_0, \dots, D_N\}$ be independent continuous operators. The set of u with a *misfit* (the $D_i u$ independent) is open, and each such u has a certificate: a non-vanishing determinant $\det\langle D_i u, \psi_j\rangle$ for finitely many test functions, valid in a neighbourhood. The set of u that *fit* is closed with empty interior, and for differential atoms a fit cannot be certified even from finitely many *exact* measurements. Dim. DICT-A, Catalogue p. 89

Example 15.4 A certified misfit, an uncertifiable fit

For $u = e^{-t}\sin x$ the level $\{\partial_t, \partial_x\}$ shows a misfit: the Gram matrix of (u_t, u_x) is non-singular and $\varphi = \langle \cdot, \mathrm{res}\rangle / \|\mathrm{res}\|$ certifies that no first-order law holds. The level $\{\partial_t, \partial_x^2\}$ fits ($u_t = u_{xx} = -u$), and no finite data can certify that.

Misfit certificates exist in three forms: a Hahn–Banach functional (distance to the span), a Gram quotient in a Hilbert space, and a weak form that needs no derivative of u (only integrals against test functions) (MF1, MF2, MF3).

15.4 Stopping rules and the limit

Theorem 15.5 Stopping rule and identification in the limit (DICT-GR, I-LIM) [proved]

If n_0 is the least level with an affine fit and the kernel at every higher level equals the consequences of that law, then the test stops exactly at n_0. With a compact class of coefficients and uniqueness, small residual means close coefficients: $\|L_n - L\| \le \omega(C_u \varepsilon_n)$, and with data of error δ one gets $\omega(C_u \varepsilon_n + 2C_{R'}\delta)$, a Hölder rate in the one-coefficient case.

Dim. DICT-GR, Catalogue p. 90

15.5 Other results of this phase

- **Exact gain of an atom** (DICT-R3); **change of frame** by the Volterra encoder, with exact intertwining and constants $\|W\|, \|W^{-1}\| \le 1 + a^2 Q\, e^{a\sqrt{Q}}$ (DICT-R6).
- **Exponential families**: exact polynomialisation, kinks and the 2-kink (DICT-NS).
- **Minimax rate** $\Theta(\varepsilon^{1/3})$ for the potential reconstruction under $M > \delta_0$ (DM-δ-MINIMAX).

Why it matters. A dictionary can be universal for approximation, never for exact identification. Its negative side is both topological and measure-theoretic; its positive side is a one-way certificate: misfit is witnessed by finitely many integrals, fit by none.

CHAPTER 16

What Phase 10 changed

In this chapter	
Starting from	the open problems and conditional statements of all earlier phases, reviewed in Phase 10 by three research sessions (phases 1–4, phase 5, phase 8), each followed by an independent audit and, where needed, a second round.
Why	an honest edition has to say which earlier statements were refuted, which were closed, and which remain open. Phase 10 refuted five of them and closed many.
Role	in case of a difference with Chapters 8, 7 and 9 this chapter prevails. The proofs of the two corrections that change statements are Dim. A52 and A53; the other items are list cards (Part IV).

16.1 Statements that turned out false

Proposition 16.1 The interaction no-go with constant terms (S10A-T1, P1-Q6) [proved]

Chapter 8 left open (P1-Q6) whether $u_0 \mapsto E_{\mathrm{int}}$ fails to be continuous in H^m for every operator with at least two non-proportional terms. The statement as written is **false**: for reaction–diffusion $u_t = Du_{xx} + ru$ (two non-proportional terms) E_{int} is continuous from H^1, with

$$E_{\mathrm{int}}(u) = (e^{sr} - 1)\big(v(t) - v_0\big), \qquad s = t - t_0.$$

The correct form is: for monomial terms of distinct degrees, E_{int} is unbounded $H^m \to L^2$ for every m and base time iff there are *at least two terms of degree* ≥ 1; with at most one the interaction is bounded for every $m \geq 1$. Constant terms only multiply the solution by a bounded exponential and neither create nor destroy the no-go. Dim. S10A-T1, Catalogue p. 93

Proposition 16.2 Holes: the sum of kernels is closed but not dense (S10A-P1) [proved]

On the annulus $A = \{1 < |z| < 2\}$ the Laplacian is a product of the two coprime factors $\partial_t \pm i\partial_x$. The harmonic function $\log|z|$ is not in the closure of the sum of holomorphic and anti-holomorphic functions in any H^m, and for $m \geq 1$

$$\ker_{H^m(A)} \Delta = (\mathcal{H}_1 + \mathcal{H}_2) \oplus \mathbb{C}\log|z| :$$

the sum is closed, of codimension $b_1 = 1$, and not dense. Hence the density question of Chapter 7 has a negative answer on non-convex regions in general (P2-Q1 is false as a universal statement), and this is a *differential* example of "closed but not dense". Dim. S10A-P1, Catalogue p. 94

Intuition

A hole gives a period: the flux of a harmonic function through a loop around the hole. The flux is a continuous functional that vanishes on holomorphic and anti-holomorphic pieces (their derivative has a single-valued primitive), but not on $\log|z|$.

Three more statements were refuted: the extension of the convex density theorem to non-convex regions (S10B-C1, by a hole with an interior); P2-Q1 as a universal statement (for Δ on every Lipschitz region with $b_1 \geq 1$); and, in the variable-coefficient phase, "$P11(e)$ with B merely continuous" and "the pole depends only on $A(\xi_0)$ and the factorisation".

16.2 Problems closed

- **Density of the sum of kernels** on every simply connected Lipschitz region, and for $\delta(P) = 0$ on every Lipschitz region, modulo two cited facts [F1], [F2]; the source of the collar field was replaced by a proof (S10A-T3, S10COL).
- **C12-P7**: "closed but not dense" realised by the annulus.
- **C3-Q1**, the decidability question, closed as a *classification by data model* (composition of black-box impossibility, bound, analytic and jet models); no new mathematics.
- **OPF5C-T5, INT-T1(B), S10B-T2**: the multi-soliton is a Hirota solution; the reflectionless Jost computation holds.
- **Uniform shock rate with several births** (OPA-Q1): modulo the equivalence Oleinik $\Leftrightarrow$ Kružkov [F1].
- **Variable coefficients**: VAR-P11(e) by Volterra and Gronwall; a solution with $V = 0$, $|Z| > 0$ and $\operatorname{int} Z = \emptyset$ (Tychonoff and a Cantor-type set); Floquet in the quadratic class ($\operatorname{tr} M = -26/5$); a monic criterion in the Weyl algebra.
- **Systems**: Maxwell has an identifying field; at fixed real time no finite product of Dirac flows with polynomial times; the pole criterion $E_0 \cap F_0 \neq 0$.
- **Dictionary**: the L^1 rate for general completely monotone memory kernels; the minimax rate $\Theta(\varepsilon^{1/3})$.
- **Computer-assisted**: $N \leq 4$, and, after the triage, $N = 5$ and $\delta_6(12) = 7$ for the syzygy degrees (modulo a novelty hypothesis (Nov)); $\delta_6(11) = 8$ with a written proof.

16.3 Textbook examples and formalisation

Five textbook examples were checked against the whole theory: free Dirac $1+1$, the harmonic oscillator, the Pöschl–Teller potential, the fractional heat equation and Maxwell in vacuum; 80 of 80 checks pass, and three new arguments were proved. The formalisation in Lean 4 with Mathlib reached 37 entries: 27 are faithful to the ledger statement, 7 are weaker and 3 are different, none is vacuous, and all 85 `#print axioms` use only the standard axioms. Mathlib has no Weyl algebra and no C_0-semigroups, so the analytic parts are not formalised.

Why it matters. The corrections of Phase 10 all push in one direction: the negative results of Phases 1–3 are sharper than first stated (constants do not break the no-go; holes do break density), and the closures are explicit about what they depend on.

CHAPTER 17

The extended algorithm

In this chapter	
Starting from	the results of Chapters 11–15 and of Chapter 10; a datum in one of the models below.
Why	Chapter 10 turned the theorems of Phases 1–5 into a seven-step procedure for a scalar constant-coefficient equation. This chapter extends it to systems, variable coefficients, noise, and operators outside the catalogue, and states, for every step, what is promised and what is not.
Role	the operational summary of the whole edition. It introduces *no new claim*: it composes rows of the ledger.

17.1 Which senses of "identify"

The word "identify" has five senses, and every step declares which one it uses.

Sense	Meaning	Unique when
ID-ann	$\mathrm{Ann}(u) = (P)$: the annihilator ideal is principal and generated by the law	always if proved; never for $u \in \mathrm{EP}$ (the point spectrum is non-empty)
ID-ker	$\ker \Phi_u = 0$ in a fixed parametric class	only if a law of the class exists (H-ex)
ID-res	coefficients determined by the residue map, e.g. $V = (iu_t + u_{xx})/u$	on $\{u \neq 0\}$, in the classes of Chapter 12
ID-fam	identification modulo a family: $V + c$, a gauge class, an encoder modulo $c(t)$	–
ID-lim	identification in the limit, never certified at finite time	under (H-ID)

17.2 Data models and the contract

Model	Datum	Senses allowed
DM1	a closed form that is an exponential-polynomial	only ID-ker relative to a finite class, never ID-ann
M3-hol	a closed-form holonomic non-EP function (Bessel, Airy, sech, Gaussians)	ID-ann, ID-res, ID-ker, with a symbolic proof
DM2	the jet at a point	only ID-lim: no certificate at finite time
M2$^+$	jet plus a zero test, or a declared holonomic ideal	exact decidability modulo (AC)
DM3	u known exactly in $H^m(\Omega)$ or $C^\infty(\Omega)$	ID-ann, ID-res, ID-ker, ID-fam
DM-δ	$\tilde{u}$ with $\|\tilde{u} - u\|_Y \le \delta$, norm Y declared	no identification: only the moduli and conditional stability

The contract has four classes: (A) finite and exact, (B) convergent with an explicit error ε_i, (C) no guarantee, (Lim) in the limit without a certificate. The pipeline has **two layers**.

1. *Conditioning* (Steps 1–3) fixes the model, the level and the class. Its outputs are discrete, and since $\mathrm{Ann}(\cdot)$ is discontinuous everywhere (Theorem 13.3) no Lipschitz bound applies to them. Completeness ("the right law is in the chosen level") is always (C).
2. *Conditional correctness* (Steps 4–6) has metric outputs. *At the declared level and in the declared class* the error of the chain is bounded by $\sum_i (\prod_{j>i} L_j)\varepsilon_i$, provided every step is (A) or (B) and Lipschitz in its computed inputs.

17.3 The steps

Step	Task	Result relied upon	Output
1	model	Thms. 5.3, 5.3, 13.1; Prop. 11.5; Thm. 12.4	model, ring of coefficients, senses reachable
2	level of $\mathcal{D}_\infty$ (L0–L10)	Thm. 15.2	level n and finite dictionary, or `NON-DECISO`
3	identification (scalar, system, variable)	Chs. 4, 11, 12	$\mathrm{Ann}(u) \cap R_{\leq n}$, G_u, σ_p, sense
4	misfit certificate or best approximation	Thms. 15.3, 15.4, 13.3	`MISFIT`, `FIT-COMPATIBLE` or stop
5	analytic encoder for V	Thm. 14.2, Cor. 14.2	translated datum and certificates, constant C_W
6	factors, terms and E_{int}	Chs. 6–8, 16	components, ambiguity $\delta(P)$, E_{int}
7	convergence guarantees	Thm. 13.1–13.3	double label and chain bound

The levels L0–L10

L0–L2 scalar constant coefficients; L3 non-monic or degree ≥ 2 in τ; L4 polynomial non-linear; L5 memory and delay (augmented system); L6 constant-coefficient systems (Dirac, Lamé, Maxwell); L7 variable coefficients in x (Schrödinger, Airy); L8 coefficients in (t, x) (Stark, oscillator, Floquet); L9 non-local (fractional, multipliers); L10 general dictionary with misfit and stopping. The choice of level is a *heuristic* (completeness (C)): Theorem 13.2 excludes any a priori criterion.

Step 3 by case

Scalar (L0–L5): the tests of Chapter 4; in DM1 never ID-ann. **Systems** (L6): the existence test (socles ≤ 1, Theorem 11.2), then, for a given u, ID-ann or, in Hamiltonian form, ID-ker with a pointwise linear system of rank 8 (Dirac $1+1$) provided at least four generic modes. **Variable** (L7–L8): the residue map ρ_u (two derivatives and a division per point) or the affine set $\theta_0 + \ker \Phi_u$.

What the pipeline never outputs

It *never* emits "law identified" from finite numerical data. Its outputs are: `MISFIT(certificate)` at level n; `FIT-COMPATIBLE(`n, r_n`, bound)`; `IDENTIFIED[sense, model]`, only with an exact symbolic proof in DM3, M3-hol or M2$^+$ (modulo (AC)), or relative to a finite class in DM1; `ID-LIM(`n`)`, a current candidate without certificate; and `NON-DECISO`, when the budget runs out.

The encoder is a translation

Step 5 is only usable when q is *given*. If q is estimated at Step 3c the Lipschitz constant of $q \mapsto W_q$ is not explicit and the transport is (C). The only proved bound is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$, which can overestimate by about four orders of magnitude at $a\sqrt{Q} = 5$; a sharper bound (a cosh or an exact Gronwall estimate) is open. A Darboux encoder is unbounded with a non-trivial kernel: it loses modes, and an identifying datum can become exponential-polynomial.

Why it matters. The extended algorithm is two things at once: a procedure and a map of its own limits. What it can promise is correctness *at a declared level and class*; what it can never promise is that the level is the right one. That asymmetry, not a technical gap, is the content of the negative theorems of Phases 1–8.

CHAPTER 18

Limits, open problems and status

In this chapter	
Starting from	no new assumptions.
Why	to state what is impossible, which statements turned out false, what remains open, and how reliable each result is, for the whole project (Phases 0–10).
Role	delimits every claim of this edition. The analytic part of the project is *closed* by decision of the user (28 October 2026); the unresolved items are collected in the project file `PROBLEMI_APERTI.md`, none of them bearing a central result.

18.1 What is impossible

Each item below is a proved negative theorem, not a gap.

- a finite certificate of identification, and identification from a black box (Theorems 5.3 and 5.3); in the limit, no learner decides whether a law exists (Theorem 13.2);
- identification by restricting variable-coefficient operators to one solution (Proposition 5.6); a huge class of coefficients identifies nothing (Theorem 12.1);
- decomposition of a solution by additive terms (Proposition 6.1.1); factor decomposition on regions with holes (Proposition 6.4);
- components in H^m in general: one derivative can be lost (Counterexample 7.1); density of the sum of kernels on regions with holes (Proposition 16.1);
- a continuous or stably regularised interaction term in H^m *when at least two terms have degree* ≥ 1 (Theorems 8.5, 8.5, 16.1);
- a finite product of single-term flows reproducing non-commuting dynamics when the adjoint orbit is infinite (Theorem 8.7(c)); a uniform finite product for Dirac $1 + 1$ with transport and mass times independent of the frequency (Proposition 11.5);
- an identifying solution when a module has a socle of dimension ≥ 2 (Theorem 11.2); one field identifying (A, V) in the magnetic Schrödinger equation (Proposition 11.5);
- a time-independent encoder for a genuinely time-dependent potential (Theorem 14.3);
- exact identification of every operator by a countable dictionary (Theorem 15.2), and any finite certificate of a fit (Theorem 15.3).

18.2 Statements that turned out false

Statement	Counterexample or correction	Ledger
the spectrum of an expansion decides identification on a bounded domain	e^{x-t} on the unit square	C3-C1
components keep the regularity of u	parallel transports on the square	C12-C1
$u \mapsto E_{\text{int}}$ is continuous in H^m	real exponentials in advection–diffusion	P1-P1(e)
the companion matrix is normal iff $p = s^m - c$	$s^2 + s - 1$	R6-T3
splitting by terms lies in $\ker(L_1, L_2)$	$\frac{1}{2}x^2 + Dt$ for the heat equation	R2-T3 (corrected)
exact interaction impossible for infinite Lie algebras	$e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$	R5-Q5, P1-C1
NLS identification by scattering data without multiplicity	explicit counterexample	OPF5D
no continuity for *any* operator with two non-proportional terms	reaction–diffusion $u_t = Du_{xx} + ru$ (Theorem 16.1)	P1-Q6
density of the sum of kernels on every Lipschitz region	the Laplacian on the annulus (Proposition 16.1)	P2-Q1, S10B-C1
$\Phi(V, v) = W_V^{-1}v$ injective $\iff$ a blind encoder exists	the phase-only data model	ENC-N1
an eigenvalue crossing implies a pole of the projector (derogatory A)	$A = \operatorname{diag}(\xi, -\xi)$ has constant projectors	SYS-II-T2-C
every $u \not\equiv 0$ identifies $V(t, x)$	Tychonoff silent strips	VAR-T1(c)-F
the chirp–free–chirp word breaks at the lens time	explicit continuation	VAR-T10-F
Dyson–Phillips with B merely continuous	a rank-2 counterexample	VAR-Q18-CE

18.3 Open or conditional

These are not closed and are not declared closed. For each the ledger records what is proved.

With a proved reduction or condition.

- the Lie-flow question for algebras with (H) but without a weighted dilation [open] (P1-Q7) (candidate $(x^3 + x)\partial, \partial$ proved);
- the closedness of the sum of kernels for $N \geq 3$ differential factors, reduced to triples with at least two non-compact pairs, outside real transports [open] (C12-Q2);
- the graded case of the syzygy degrees [open] (OPF5C-Q1); the realisability of atoms [open] (OPR2-Q3); the sufficient condition (L-v) at every vertex [open] (OPR2-Q2′);

- identification for analytic KdV data with nonzero reflection, equivalent to (R), modulo [GT-u$_m$] and (WP) [open] (INT-Q1);
- boundedness of the Hölder constants in the Gelfand–Levitan analysis [open] ([GT-u$_0$]); the justification of a differentiation in a cited analysis [open] ([GT-u$_m$]).

Without a known route. Non-Lipschitz domains (P2-Q1-NL: only a model case is proved); the exact codimension for a general P (P2-Q1-COD); closedness of the sum for $(\tau - \xi^2)(\tau + c\xi)$ (C12-Q1); multiplications and a general Floquet logarithm (VAR-Q18, VAR-P12(d)); the non-monic case in the Weyl algebra (VAR-C3Q2-open, VAR-Q13); complex-time Dirac with polynomial times (SYS-DIRAC-POL); the DM3 reading of ENC-Q4 and the real residue ENC-Q10; the optimality of the L^1 rate for general kernels (DICT-U3d-v); the C^∞ division reformulation of DICT-Q3(c).

Conditional on sources not read in full. Oberst 1993 (read only as an abstract; it concerns novelty, not a load-bearing step); the equivalence Oleinik ⇔ Kružkov [F1] (a plan exists: vanishing viscosity on Cole–Hopf, no audited proof); Malgrange 1966; the existence of global solutions in $\mathcal{S}$ (WP); Björk. The project forbids downloading files, so these items are *blocked on sources unavailable offline*.

18.4 How reliable the results are

Proofs and audits. Every statement labelled [proved] has a complete proof that passed an independent review. The reviewers were AI agents with no access to the author's reasoning, not human referees. Statements labelled [proof written, not audited] have complete proofs that were not audited. For Phases 6–10 the labels follow the ledger: a result is [proved] if the ledger records it as proved after the audits of Gate 8 and of Phase 10 (S-CHECK-F8, S-CHECK-F10). **The proofs in this edition, however, were rewritten from the research packages for this volume (cards A32–A53 and B73–B79) and were not themselves re-audited**; the label refers to the result, not to the rewriting. Several packages cite theorems (Peetre, Malgrange, Oberst, Marchenko) whose exact statements were verified only on abstracts or secondary sources.

Formal verification. Fifteen theorems of the algebraic core of Phases 0–5 and, in Phase 10, 37 entries (27 faithful, 7 weaker, 3 different, none vacuous, 85 `#print axioms` with standard axioms only) are formalised in Lean 4 with Mathlib. The analytic parts are not.

Citations. Many ingredients are classical. The primary sources of the load-bearing citations of Phases 0–5 were checked; for Phases 6–10 a number of citations were checked only on abstracts or secondary sources (Oberst, the transformation-operator theorem numbers, Casas for Magnus). A result relied upon should be traced to its primary source.

Novelty. The contribution is the package: sharp characterisations, negative theorems and counterexamples, each with a verified status. Parts of the identification theory are folklore in commutative algebra (Oberst duality, socle criteria, Peetre's theorem, Volterra transformation operators); parts of the finite-data theory are known in substance (Prony, Mourrain).

18.5 What this edition does not contain

- The proofs of every result of Phases 6–10: the catalogue contains proofs of 22 results in Part II and of 25 more in Part IV; the other 86 results appear as list cards with their trust label and the research file (Chapter 15 of the catalogue onwards).
- The catalogue of about 60 equation types and the tropical reading of scaling limits.
- The Lean sources.
- Phase 11 as a research phase: this edition *is* its deliverable. The text written for it (the rewritten proofs of Parts II and IV, the corrections added to Chapters 7 and 8, and the reorganisation of the extended algorithm in Chapter 17) has not had the proof-checker audit, citation audit and kill-argument review that the plan required before closing.

CHAPTER 19

The complete list of open problems

In this chapter	
Starting from	the registry `PROBLEMI_APERTI.md` of 28 October 2026, the closing report of Phase 10, the final report of Phases 0–5, the open-question files of every research package, and Section 11 of the extended algorithm.
Why	Chapter 18 states what is open in a few lines. This chapter is the full registry: every open or conditional item of the project, with what is already proved about it and what would be needed to close it. Nothing here is a central result.
Role	a starting point for any later research phase. The status of every item in the ledger is unchanged: a label is raised only after a proof and an audit.

19.1 Why the analytic part can be closed

None of the items below carries a central result of the project (the linear characterisations, the algebraic theorems ALG-*, the dictionary criterion S4ID-T1, the sum-of-kernels theorems C12-T3 and P2-T3, the negative theorems and the counterexamples). Each item is of one of three kinds: (i) an extension to more general classes of a result already proved in a model case; (ii) a new question born during the work; (iii) a dependence on a published source that was not available offline.

The only items from which statements of the report depend are listed with their dependence, and each is already marked "modulo" or "conditional" where it is used:

- **[F1] Lax–Oleinik = Kružkov.** The source is classical and was read (Kružkov 1970, Thm 1, §3); what is missing is an autonomous proof, for which the triage gives a plan (vanishing viscosity on Cole–Hopf). Burgers beyond the shock depends on it (S4T-T3(iii), OPA-T1, OPF5A-T1/T2/C1). This is *not* an open problem.
- **[L-loc]**: the localisation lemma on which OPA-Q1 depends for several shock births; the case of a single shock birth (for example $\sin x$) is closed modulo [F1].
- **[GT-u]**, that is [GT-u_0] and [GT-u_m]: S4ID-P8 and OPR2-P7 (asymptotics with derivatives).
- **[F4$^-$]**: INT-T1(B) (reflectionless N-solitons), verified for $N = 2$.
- **Malgrange 1966**: C3-P1 for general data on convex sets; the jet case is proved without it and the cited theorem is verified.

19.2 Research problems with no known route

Item	Question	Already obtained	What is needed
C12-Q2 = OPF-Q2	closedness of the sum of kernels for $N \geq 3$ differential factors	false in abstract Hilbert space (counterexample); $N = 2$ proved	a differential family with ≥ 2 non-compact interactions
C12-Q1	closedness for $(\tau - \xi^2)(\tau + c\xi)$	—	boundary-layer estimates (route OPR2-L1, heat $\times$ transport)
OPF5C-Q1	syzygy degrees $\delta_N(K)$ for general N	nine exact values, computer-assisted; $\delta_6(12) = 7$ new; conjectures $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$	a structure for general N; for fixed N it is a finite computation; $\delta_6(11) = 8$ has a written proof, Hirota $N = 5$ is not concluded (a run of about 17 minutes was interrupted)
OPR2-Q3	repeated socle, "no finite test"	proved under (I) and (II)	a lifting theory of atoms
OPR2-Q2′	wedges with degenerate directions	tangent model closed and direct (OPF5B-P8); numerical check 7/7	(L-int): an explicit Hilbert–Schmidt estimate, a few hours of work; (L-v): analysis at the vertex of the cone
INT-Q1	KdV, $r \not\equiv 0$, u analytic: is there an autonomous ODE in x?	OPF5C-T6: order ≤ 3 implies 0 or a 1-soliton	singular relations at the origin and sub-dominant terms of the radiative asymptotics (upstream: [GT-u$_m$])
P1-Q7, general	Lie algebras with (H) but without a weighted dilation (Dil)	the candidate $(x^3 + x)\partial, \partial$ is proved	no route without a dilation
P2-Q1-NL	non-Lipschitz domains	Δ, $m \geq 1$, $L^{1,2} \subset L^2$ proved	general P and $m = 0$
P2-Q1-COD	exact codimension for general P	—	continuous period functionals in H^m
VAR-Q18	multiplications	a rank-2 counterexample with B merely continuous	a counterexample with rough W, v merely continuous, and control of the whole Dyson series
VAR-P12(d)	non-quadratic Floquet logarithms	partial observations (parity, dilations)	an obstruction for non-quadratic logarithms
VAR-C3Q2-open, VAR-Q13	non-monic case in the Weyl algebra	monic case (a′) proved (VAR-C3Q2-R)	a decomposition without monicity
SYS-DIRAC-POL	polynomiality for complex t, m	real t, m proved	the triage sketches a lemma E′ "by growth"; not audited
ENC-Q4	the DM3 reading	—	the class $W(t)$ must be defined first

Item	Question	Already obtained	What is needed
DICT-Q3(c)	C^∞ division at variable rank	—	a general division criterion
AM-CL	closure AM	no longer needed (ledger F10-TRIAGE)	—
OPA-Q2	density of the Morse+(SB) class	a sketch (Sard with finitely many parameters)	write and audit the sketch
non-convex regions	density of OPF5B-T1 outside convex sets	convex case proved; the extension is false on regions with holes (A53); simply connected Lipschitz modulo F1, F2	—

19.3 Items blocked on sources unavailable offline

The project forbids downloading files. Each item below is unblocked by placing the indicated text in the project folder.

Item	Source needed	After reading
[GT-u_0]	Grunert–Teschl §5, Krüger–Teschl	rewriting with explicit constants; a few days of work
[GT-u_m]	Deift–Zhou, CMP 165 (1994) 175–191; Buslaev–Sukhanov 1986	term-by-term derivation; not short
VAR-Q17(b)	Baudouin–Puel; Lasiecka–Triggiani–Zhang; Mercado–Osses–Rosier (Carleman for Schrödinger with $V(t,x)$)	an adaptation
ENC-Q10	Matveev–Salle; Bagrov–Samsonov (pairwise Darboux)	an adaptation, plus a decision on the class $\mathfrak{E}$
DICT-U3d-v	Braess; Braess–Hackbusch	for $t^{-\beta}$ probably the τ_k are not needed; the uniform limit on the class stays a hard problem
Malgrange 1966	*Ideals of differentiable functions*	only to have an autonomous proof: the theorem is already cited
(WP)	Bona–Smith; Kato 1983; Lax 1975	alternatively an autonomous proof of several days
[F4$^-$]	the primary source for reflectionless N-solitons	verified for $N = 2$
Oberst 1993	MCSS 6 (read only as an abstract)	*priority and novelty only*: SYS-EX is proved on an arbitrary Ω without it

19.4 Open questions of each research package

Systems (Phase 8).

- **Q-S8SYS-1** (closed for real times; complex times open): finite words of Dirac flows with times polynomial in ξ.
- **Q-S8SYS-3**: for fixed Ω, does (i$\Leftarrow$) of SYS-S1 hold for every pair iff $\mathrm{Ext}^1(N, C^\infty(\Omega)) = 0$ for every N that occurs, i.e. iff Ω is P-convex for all P ($C^\infty(\Omega)$ injective)?
- **Q-S8SYS-4** (derogatory A): the candidate "the projector has a pole at ξ_0 iff $A(\xi_0)$ has a non-trivial Jordan block mixing eigenvalues of coprime factors"; not proved, high cost; the pole criterion $E_0 \cap F_0 \neq 0$ was proved in Phase 10.
- **Q-S8SYS-5** (novelty of SYS-EX): read Oberst 1990 §2–4, Oberst 1995 (*Variations on the fundamental principle*) and Pillai–Shankar 1998 to decide whether SYS-EX on $\mathbb{R}^n$ is already stated.
- **Q-S8SYS-6** (Maxwell): socle of vacuum Maxwell in (E, B), an over-determined 8×6 system; closed in Phase 10 ($\mathrm{pd} \leq 2 \Rightarrow \mathrm{soc} = 0$) in the symmetric class; in the general Hamiltonian class ID-ker is impossible (EX10-e-K).
- **Q-S8SYS-7** (SYS-N2): the determinantal reduction for massless Weyl needs the class group of the light cone (Knörrer/Eisenbud); not touched.

Variable coefficients (Phase 8).

- **VAR-Q10** (closed in Phase 10 by an example). **VAR-Q6′**: the Galois group of $-\partial^2 + x^2 - E$ over $\mathbb{C}(x)$ (SL_2 for $E \notin \pm(2\mathbb{N}+1)$, the Borel group otherwise); not needed for identification; tool: Kovacic or a direct argument.
- **VAR-Q12**: for L monic of order $m \geq 2$ in ∂_t, an intrinsic characterisation of B-independence (holonomy of $A_2 u$, characteristic dimension); is there a non-D-finite u that does not identify? **VAR-Q13**: for non-constant leading coefficient in every direction, when is $\mathrm{Ann}(u)$ equal to the saturation $A_2 \cap \mathbb{C}(t,x)\langle\partial\rangle L$? A decision procedure needs factorisation of D-modules (Grigoriev–Schwarz).
- **VAR-Q11**: for generic periodic $V(t,x)$, does $U(T,0)$ have a differential (unbounded) logarithm? Known for $V = v(t)$ and quadratic cases.
- **VAR-Q8, Q8′**: convergence of the Magnus series for unbounded operators; when $Y(t;\varepsilon)$ is normal for complex ε, as Casas 2007, Thm 3.4 requires in infinite dimension.
- **VAR-Q14**: whether Saut–Scheurer (1987) or Isakov (1993) contain VAR-L1 for $V \in L^\infty(t,x)$ in 1D (probably VAR-L2 and VAR-T1(c)(ii)); if so, cite them. **VAR-Q15**: read Schindelar–Levandovskyy–Zerz (2011) to confirm the delta of the Weyl-algebra comparison.
- **VAR-Q16**: for continuous matrix V, does "V determined on an open U" imply $\mathrm{span}_t \psi(t,x) = \mathbb{C}^2$ on a dense subset of U? **VAR-Q17**: lateral observability for mild solutions ($\psi_0 \in L^2$, $V \in L^\infty$). **VAR-Q18**: see above. **VAR-Q19**: robustness in DM-δ: construct an estimator of V with guaranteed error, or prove that the Hölder exponent $1/3$ of VAR-P3 is also optimal for the estimation problem (settled with $M > \delta_0$ in Phase 10).

Encoder (Phase 8).

- **Q1**: ENC-U1 in wider classes (kernel $K \in L^1$ for $q \in L^1$, non-triangular kernels). **Q2**: boundedness of W on $H^{2k}(J)$ with explicit constants (the induction with $(Wf)'' = W(f'') - W(qf)$ is only sketched; it is needed for rule R6 with Sobolev norms).
- **Q3** (ENC-S1): transport of the factor decomposition for coprime factors in $S = \mathbb{C}[\tau, \sigma]$ through φ and C12-T1′; to be written out, and to check whether $\operatorname{lcm}(Q, \tilde{Q})$ is the generator.
- **Q4**: a natural class of potentials with $\mathcal{S}_V \cap \mathcal{S}_{V'} \neq \{0\}$ for $V \neq V'$ and a structured encoder for each, so that separation (S) is non-empty. **Q14**: the same modulo the gauge of the data model.
- **Q5**: E^{mult} for Schrödinger on a bounded Ω with Dirichlet conditions (leading term $t^2/2\,[H_0, V]$ presumably, to be proved). **Q6**: ENC-1 in dimension $d \geq 2$: no triangular Volterra operator with local kernel (the characteristic problem is not Goursat).
- **Q7**: does the local Darboux chain extend to all of J across the zeros of the seeds, and is Z discrete (it is only shown closed and countable)? Known for rational potentials (Duistermaat–Grünbaum).
- **Q8**: a quantitative bound for the truncated Foldy–Wouthuysen series with $V \in W^{k,\infty}$ (error $O(m^{-k-1})$?). **Q10**: non-local but explicit encoders for non-quadratic V (Bluman–Shtelen classify only pointwise encoders).
- **Q11**: W commuting with ∂_t with non-constant $c(t)$ and W non-local in t (for example convolutions in t). **Q12**: lateral continuation and (E1) on solutions. **Q13**: regularity of the propagator with V merely norm-continuous (Kato asks for $t \mapsto H(t)$ strongly C^1). **Q15**: ENC-T2(c′) in the class of principal annihilators (proved for the specific family in Phase 10).
- **Q9**: theorem numbers to verify (Marchenko ch. 1, van der Put–Singer, Duistermaat–Grünbaum, Casas).

Convergence (Phase 8).

- **CONV-Q1**: without an upper bound, is $G_u = \gcd \operatorname{Ann}(u)$ identifiable in the limit when $\operatorname{Ann}(u)$ is not principal? (the rule converges to g_d, which can be a proper multiple of G_u). **CONV-Q2**: an explicit jet threshold K^* for classes wider than EP (finite growth order). **CONV-Q3**: a quantitative version of T2 in terms of the complexity of changes of mind.
- **CONV-Q4**: characterise $\mathcal{N}_{\mathcal{K}}$ when $\mathcal{K}$ is the class of augmented systems with finitely many auxiliary variables. **CONV-Q5**: the decomposition $\ker(Q_1 Q_2) = \ker Q_1 + \ker Q_2$ on non-convex domains in $1 + d$ (P-convexity, Hartogs–Ehrenpreis phenomenon). **CONV-Q17**: injectivity of $\mathcal{D}'(\Omega)$ on convex Ω (Malgrange calls it probable; to check in Palamodov 1970).
- **CONV-Q7**: T4 and T5 with stochastic noise, rates in probability. **CONV-Q8**: explicit margin ρ_d for concrete families (heat, waves) as a function of Ω. **CONV-Q14**: T4(b) and T5 for non-integer s (complex or real interpolation; withdrawn from the statement).
- **CONV-Q11**: Mori–Zwanzig with variable coefficients ("closed local law $\Longleftrightarrow$ finite elimination" proved only in the direction $\Leftarrow$ with constant coefficients); look for a Weyl-algebra / D-module version. **CONV-Q12**: the L^∞ order of Strang in the P3-T3 frame-

work (numerically 1, an example, not proved). **CONV-Q13**: higher orders in ε for CONV-T8 and non-modal data in H^1_0 or H^2. **CONV-Q6′**: the norm of the orders 1 (Dirichlet) and 1.5 (Neumann/Robin) in Dang–Einkemmer–Ostermann 2025. **CONV-Q16**: identification of a boundary condition with positive margin through a quotient dictionary for tangential consequences (CONV-B3).

Dictionary (Phase 8).

- **Q1**: a uniform-in-t rate for general completely monotone kernels (the explicit rate is known for $t^{-\beta}$; closed in Phase 10; optimality is open). **Q2**: Hölder for $s \geq 1$ in I-LIM needs a Gagliardo–Nirenberg interpolation $\|f\|_{C^s} \leq C\|f\|_{L^2}^{\theta}\|f\|_{C^{s,1}}^{1-\theta}$, $\theta = 2/(2s+2+N)$, on Lipschitz domains with a constant (the exponent is verified by scaling; the source is missing).
- **Q3**: the intermediate case of RANGO (the pointwise residue map $A_u(y)$ is neither injective on a dense set nor onto everywhere): is identifiability decided by division at variable rank, or are there counterexamples? **Q4**: Theorem A(iii) for non-local atoms (memory, symbols, fractional): the proof uses locality.
- **Q5**: order optimality of polynomials via Kolmogorov n-widths (Pinkus 1985, source not read). **Q6**: the 2-kink has kernel 1 (only checked on examples; a proof needs exact independence of $u_{xt}, u_{xx}, u_{tt}, u$). **Q8**: prior art for N1(v)-ALG, "the generic u is differentially transcendental". **Q9**: R6 for $Y = H^{2k}$ and transport of the weak form MF3 with compactly supported test functions.
- **Q7**: theorem numbers and citations to verify: Hörmander III 18.1.13; Trefethen–Weideman (trapezoid rule); Boman 1967; Christensen 1972 and Hunt–Sauer–Yorke 1992 (prevalence); Ax 1971; Weiss–Tabor–Carnevale 1983; Whitney extension.

19.5 Open questions of the extended algorithm

The twelve questions of Section 11 of the extended algorithm:

1. robustness in DM-$\delta(Y)$ (VAR-Q19);
2. an example at level L9 with infinitely many modes and an identified non-local atom, and DICT-A(iii) for non-local atoms;
3. DICT-ILIM-Hs (closed in Phase 10 with an elementary cone lemma);
4. CONV-Q1: G_u in the limit without an upper bound;
5. VAR-T1(c)-Q10 (closed by an example in Phase 10);
6. C3-Q2 in general;
7. CONV-Q9: unbounded V;
8. the cost of rule R1 per level; the audit of C3-A1 (elimination at polynomial cost, a written proof without a ledger row); a rigorous inclusion for the misfit (A);
9. rule R6 in H^{2k} and with the weak form MF3; a bound on $\|W_q\|$ tighter than C_W (at $a\sqrt{Q} = 5$ the proved $C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2(a\sqrt{Q}) \approx 5.5 \cdot 10^3$);
10. an explicit Lipschitz constant of $q \mapsto W_q$ (ENC-1) for Step 5 with an estimated q;

11. ENC-T2(c′) in the principal class; VAR-P12(d); II-T2(b′) and Maxwell (Q6);
12. a ledger row for Trotter–Kato; the closure of the citations of the next section.

Citations to close. so that the steps "conditional on verification" become fully grounded: van der Put–Singer Prop. 1.34/Thm 1.27 (VAR-T6); DICT-B3; Whitney–Stein (DICT-ILIM, (H2)); Pazy ch. 5 (VAR-P11(e), VAR-L2-cor); Thaller ch. 5 (FW-1 with V); Lax–Oleinik = Kružkov [F1] (S4T-T3); Humphreys §4.1 Cor. B (P1-T4); Wei–Norman 1964, content (VAR-P8(b) "$\Leftarrow$"); Casas 2007 Thm 3.4 (VAR-T10(4)); Oberst 1993, full text; and the Phase 0–5 numbers not checked: Krantz–Parks Thm 2.3.5 (OP-T1), Cohen GTM 138 for (AC), Hörmander SCV 1.4.4, Runge in Rudin.

19.6 Statuses of the claims used by the algorithm

For completeness, the statuses recorded in the table "claims used $\to$ status" of the extended algorithm that are *not* fully proved: C3-A1 (written proof, not audited, no ledger row: every output that depends on it is "(A) conditional on C3-A1"); VAR-T6, DICT-B3, DICT-ILIM-Hs (candidate, later closed), VAR-L2-cor (not used), FW-1 with V, VAR-P8(b) "$\Leftarrow$", VAR-P11(e), VAR-T10(2)(4)(5), P1-T4 (Lie from a cited source), S4T-L2 and S4T-T3 (modulo [F1]), ENC-A9(b3) (cited source and example), and the five textbook examples (checked on examples only).

CHAPTER 20

Directions for improvement and routes not taken

In this chapter	
Starting from	the route registry `ROUTES.md` (decisions at Gates 1, 4a, 6 and 8), the research notes on the learning pipeline (`NOTE_ML`), the open questions of the extended algorithm, and the honest limits of the final report.
Why	a record of what could make the work better: sharper constants, missing audits, whole directions that were scouted and parked, and the statistical wrapper that was proposed and postponed. It also records which routes were eliminated and why, so that nobody repeats them.
Role	companion to Chapter 19. That chapter lists open *questions*; this one lists *improvements and directions*.

20.1 Improving what is already there

Sharper encoder constants. The only proved bound on the Volterra encoder is $C_W = 1 + a^2 Q\, e^{a\sqrt{Q}}$; it overestimates by about four orders of magnitude at $a\sqrt{Q} = 5$ ($C_W^2 \approx 1.4 \cdot 10^7$ against $\cosh^2 \approx 5.5 \cdot 10^3$). A cosh bound or an exact Gronwall estimate would make the transported certificates of Step 5 useful in practice. An *explicit* Lipschitz constant of $q \mapsto W_q$ would allow Step 5 with an estimated potential.

Rigorous misfit. The misfit certificate is class (A) only with a rigorous inclusion of r_n (interval arithmetic or an error bound for the quadrature); with floating-point quadrature it is class (B). The numerically stable computation is the least-squares residual, not the Gram determinant ratio.

Cost model. No general cost estimate is proved for the dictionary levels other than the linear family ($(n+1)(n+2)/2$ monomials in $1+1$); the cost of rule R1 per level, and of the Goursat solve ($O(M^2)$ evaluations of q by iterated quadrature, indicative only), should be proved.

Audits not yet run. The elimination procedure C3-A1 has a written proof, no audit and no ledger row; every output that depends on it is "(A) conditional on C3-A1". Trotter–Kato convergence has no ledger row. The final text of this edition has had no proof-checker audit, no citation audit and no kill-argument review (the Phase 11 plan required all three).

Citations. Close the list of Chapter 19 (van der Put–Singer, Whitney–Stein, Pazy, Thaller, Humphreys, Wei–Norman, Casas, Oberst) and read the sources that unblock the "blocked" items; read Saut–Scheurer and Isakov to decide whether the unique-

continuation lemma is already known; read Schindelar–Levandovskyy–Zerz in full to confirm the stated delta.

Rates under stronger hypotheses. Explicit jet thresholds beyond exponential-polynomials; explicit margins ρ_d for the heat and wave equations as functions of the domain; stochastic noise with rates in probability; non-integer smoothness in the Hölder estimates.

Dimension. The Volterra encoder is one-dimensional: in dimension $d \geq 2$ the characteristic problem is not of Goursat type. A genuine multidimensional analogue, or a proof that none exists with a local kernel, is open.

A first-principles proof of the Burgers closure. A written proof of the equivalence Oleinik $\Leftrightarrow$ Kružkov by vanishing viscosity on Cole–Hopf would remove the "modulo [F1]" from the shock results; the triage gives the plan.

20.2 Routes scouted and parked

During the project each candidate approach (a *route*) was scouted, then promoted, merged, parked or eliminated with a stated reason. The parked routes are the natural next directions.

Route	Content / why parked	Reopen when
RT-5 (symbol tensor)	the tensor keeps only the principal symbol (R6-P1); useful as a geometric invariant; reopened for systems (RT-SYS) and parked again	a use as an invariant "at infinity"
RT-R1b	needs a sign hypothesis on the datum	such a hypothesis is acceptable
Koopman / DMD	"dynamic mode decomposition is regression", excluded by the no-learning constraint; reopen with the normal form: eigenfunctions carry the nonlinear term	the no-ML rule is lifted
Painlevé, NLSPEC, HAANTJES	triggers in the verdicts of the nonlinear scouting; Painlevé kept as a *tool* (DICT-R4)	a nonlinear exact family needs it
RT-NA (non-autonomous propagators)	parked after Phase 8	triggers VAR-Q8, Q11, Q18
RT-ENC (the encoder as a route)	parked; lemmas moved to the dictionary and variable-coefficient routes	triggers ENC-Q4, ENC-Q10
RIGID, STOKES, CONV-BORDO	not touched in Phase 8	boundary identification with margin (CONV-Q16)
Weyl algebra for PDE	scouted only for ODE and A_2	a D-module factorisation tool is available
Inverse Sturm–Liouville (spectral data)	parked	spectral data become available as a datum
U3(a) kernels, U5	universality classes not covered	a use case

Route	Content / why parked	Reopen when
Homogenisation / gamblets	approximate, a tool (S) per scale, not an encoder	oscillating or rough V treated as noise
Wave operators $\Omega_{\pm}^{*}$	outside the bounded-domain model	scattering data are the datum
Hartogs $1+d$	P-convexity on non-convex domains in several variables (CONV-Q5)	non-convex $1+d$ decomposition needed
MON, NAT, MOB	minor routes, remain parked	—
Mori–Zwanzig, RT-R2a, POLE, gamblets as *methods*	eliminated: the criteria are false or only cited; the lemmas remain (Chapter 13)	—

Eliminated for good. Statistical consistency as a method (excluded by the no-learning rule), SPLIT-f, free probability, ICA, quiver representations, microlocal analysis beyond the characteristic variety, and Lie/Olver symmetry methods were eliminated or merged. RT-S1g was eliminated for the core (it gives no information for elliptic or parabolic operators) and kept for the hyperbolic part.

20.3 The statistical wrapper that was proposed and postponed

At the end of Phase 5 a learning pipeline was drafted around the exact analytic core (the note *Reading the structure of a differential equation from data*). It was *excluded* from the project by decision of the user (no machine learning of any kind, including the autoencoder/mixture idea, "Idea I", postponed to a later phase). It is recorded here as a direction.

1. **Interpolate.** A learning algorithm fits a smooth function to noisy data (t_i, x_i, y_i) and returns not one function but a weighted family $(\hat{u}_s, \pi_s)$, with $\pi_s \geq 0$, $\sum \pi_s = 1$.
2. **Analytic core.** Each plausible function is passed to the exact algorithm, which returns its factors and the component each carries.
3. **Aggregate.** Weighted by the probabilities, this gives a distribution over factors (for example "transport 0.7 ± 0.1, diffusion 0.3 ± 0.1").
4. **Group.** The distribution sets the parameters of a standard clustering (how many groups, how spread out), cutting the data into groups that ideally contain one factor each, with a small "tangle number" (the Friedrichs cosine between factors; the size of E_{int} between terms).
5. **Repeat inside each group** until all groups are clean; the factors then rebuild the equation. The dashed loop of the pipeline: each group becomes a new data set.

The exact parts are the analytic core and the two tangle measures; the interpolation, the aggregation and the clustering were left as a proposal. The note asked two questions: whether the basis functions of the first step must be chosen in advance (and therefore only discover equations whose building blocks were anticipated), and how to avoid that choice. The answer built

in Phase 8 is the filtered dictionary with a misfit certificate and a stopping rule (Chapter 15); the statistical aggregation itself remains open.

20.4 What was never done

- the catalogue of about 60 equation types (ODE, transport, diffusion, hyperbolic, integral and memory, beams and elasticity, quantum, non-linear) used as scouting material, and the tropical reading of scaling limits (only the dissipative-face estimate TROP-3′ and its corollary for advection–diffusion are proved);
- Lean sources and a Lean statement for the analytic parts (the Mathlib libraries are missing: no Weyl algebra, no C_0-semigroups);
- an independent human review: all reviewers were AI agents;
- the decision on global installation of the project skills and on Mathpix (Step 0 of the original plan, left to the user);
- the Phase 11 audits on the final text of this edition.

20.5 Honest limits of the whole project

The overall novelty is modest: many building blocks are classical (Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering, Peetre, Volterra transformation operators, Oberst duality). The contribution is the package of characterisations, the negative theorems and the counterexamples, with verified statuses. The source policy (decision of the user, 26 October 2026) allowed results to rest on published theorems provided the source is verified: of 112 sources, 34 had the theorem number checked, 75 had the content checked (45 only in the metadata), none was not found, and three wrong citations were corrected (Humphreys Cor. B, Gel'fand–Dikii 1976, DLMF 9.7.10). Several load-bearing numbers were still unchecked at the end (Chapter 19). Work interrupted by API usage limits was resumed from the files with no loss recorded.

APPENDIX A

Glossary of classical tools

Each entry states a classical tool in one sentence and points to where this edition uses it.

Name	Statement in one sentence	Used in
Unique factorisation	Polynomials in two variables factor uniquely into irreducibles, up to constants.	Ch. 3
Bézout's theorem	Two curves without a common component meet in at most the product of their degrees points [2].	Ch. 3, 5
Hilbert's Nullstellensatz	A polynomial vanishing on the zero set of an ideal has a power in the ideal.	Ch. 4
Chinese remainder theorem	Coprime moduli split a quotient ring into a product.	Ch. 6
Malgrange–Ehrenpreis	Every nonzero constant-coefficient operator has a fundamental solution, and is onto on smooth functions on a convex set [3, 4].	Ch. 6, 5
Oberst duality	Solution spaces of constant-coefficient systems correspond to modules over the operator ring [6].	Ch. 4
Poincaré lemma	On a simply connected region every closed 1-form is exact.	Ch. 6
Friedrichs angle	$M + N$ is closed iff the cosine of the angle between M and N, after removing $M \cap N$, is smaller than 1.	Ch. 7
Miyadera–Voigt	Small relatively bounded perturbations of a generator generate a semigroup [9].	Ch. 8
Baker–Campbell–Hausdorff	$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$.	Ch. 8
Wei–Norman	Linear time-dependent systems on a solvable Lie algebra reduce to ODEs for scalar coefficients [11].	Ch. 8
Tychonoff	The heat equation has nonzero smooth solutions with zero initial datum [16].	Ch. 8
Cole–Hopf	$u = -2\nu\varphi_x/\varphi$ maps Burgers to the heat equation [13].	Ch. 9
Kružkov	Entropy solutions of scalar conservation laws are unique and L^1-stable [14].	Ch. 9
Inverse scattering	KdV and NLS are linearised by a nonlinear Fourier transform with trivial time evolution.	Ch. 9
Novikov	The stationary KdV hierarchy characterises finite-gap and soliton potentials [15].	Ch. 9

APPENDIX B

Bibliography of the project

The 112 sources of the project bibliography (file `BIBLIOGRAFIA.bib`), with the verification note recorded for each ("VERIFICATA, the checked theorem numbers are in `BIBLIOGRAFIA.md`). Only the sources whose theorem number or content was checked are listed here.

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